MILITARY SPECIFICATION

Control, Orderwire C-10717()/TRC

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The Control, Orderwire C-10717()/TRC (Orderwire Control Unit (OCU)) serves as an orderwire termination for TRI-TAC equipments. Thirteen Digital Voice Orderwires (DVOW), six Analog Voice Orderwires (AVOW), and fourteen Data Orderwires (DOW) can be handled by this unit. The C-10717()/TRC is also known as OCU Type 2 (OCU-2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents form a part of this specification to the extent specified herein. The effective issue or revision of specifications and standards shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (a) specific issues are set forth herein, or (b) different issues than those specified in this specification are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract. The issue in effect for drawings shall be as stated in the solicitation.

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet and Strip, Thin Gauge, Ployolefin
NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: DRSEL-PED-EC-2, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Federal

PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Non-metallic (and Connectors)
PPP-T-97	Tape, Packaging, Industrial, Filament Reinforced

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-Q-9859	Quality Program Requirements
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-S-19500	Microelectronics Devices, General Specifications for
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-I-45208	Inspection System Requirements
MIL-C-45662	Calibration System Requirements
MIL-P-55110	Printed Wiring Boards
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of
MIL-C-55583	Cable Assembly, Special Purpose CX-11230()/G, Cable Assembly Adapter CS-10734()/G
MIL-E-55585	Electronics Equipment and Parts, Packaging of
MIL-P-55640	Printed Wiring Boards, Multilayer (Plated through Holes)

TRI-TAC Office

TT-B1-2102-0080	Appendix I, Nuclear Hardness Criteria for Type 1 and Type 2 Orderwire Control Units
TT-B1-2102-0080	Appendix II, TEMPEST Provisions for TRI-TAC Family of Orderwire Units

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STANDARDS

Federal

FED-STD-101	Preservation, Packaging, and Packing Materials, Test Procedures

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners or Pallet-Type Base
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 Notice 4	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family

DRAWINGS

Communications Research and Development Command

SM-D-876554	Power Supply Assembly, LGM
SM-E-876642	Circuit Card Assembly, Group Framing - GF
DL-SM-B-959045	Control, Orderwire C-10717()/TRC
SM-D-959064	Circuit Card Assembly, Control Panel Logic
SM-D-959069	Circuit Card Assembly, Analog Transmit Amplifier
SM-D-959072	Circuit Card Assembly, Analog Signaling
SM-D-959075	Circuit Card Assembly, Analog Bridge

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SM-D-959078	Circuit Card Assembly, Data Channel Mux
SM-D-959081	Circuit Card Assembly, Level Converter
SM-D-959084	Circuit Card Assembly, Telemetry Combiner Generator
SM-D-959087	Circuit Card Assembly, Mux/Demux
SM-D-959090	Circuit Card Assembly, Interface
SM-D-959093	Circuit Card Assembly, VINSON Buffer Ring Detector
SM-D-959096	Circuit Card Assembly, Digital Bridge
SM-D-959099	Circuit Card Assembly, Timing Distributor
SM-D-959105	Circuit Card Assembly, BITE
SM-D-959108	Circuit Card Assembly, Alarm Interface

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number should be stipulated when requesting copies).

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3. REQUIREMENTS

3.1 Production, acceptance, testing and evaluation (PATE). When specified in the contract or purchase order, the contractor shall furnish PATE units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-SM-B-959045, and the requirements specified herein.

3.2.2 Serial numbers. The Control Unit, Orderwire C-10717()/TRC Type 2 shall have consecutively assigned serial numbers unless specified otherwise by the contracting officer.

3.2.3 Printed wiring. The printed wiring shall meet the requirements of MIL-P-55110 and MIL-P-55640.

3.2.4 Semiconductor devices. The semiconductor devices shall meet the requirements of MIL-S-19500.

3.3 Parts, materials, and processes - general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and drawings specified in 3.2.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. Serial numbers are required on each nomenclatured item, except cords and cables.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled C-10717()/TRC shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.14 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Balanced equipment-side signal characteristics. (See 4.7.1) The equipment shall accept and deliver balanced data and timing at 2, 16, 256 and 512 kilobits per second (kb/s). Timing signals also shall be delivered at 16 and 32 kilohertz (kHz).

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3.7.1.1 Line-to-line voltage levels. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1 and between -0.1 and -4.0 for a digital 0, measured into a 100-ohm dummy load.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.3 Clock/data phasing. (See 4.7.1.3) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be no more than 240 nanoseconds (ns) for 256 and 512 kb/s signals and no more than 1500 (ns) for 2 and 16 kb/s signals. Times shall be measured at the zero voltage point of the signals.

3.7.1.4 Offset. (See 4.7.1.4) The magnitude of the offset voltage measured between the center tap of a 100-ohm dummy load and ground shall not be greater than 3 volts.

3.7.1.5 Signal balance. (See 4.7.1.5) The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be no greater than 10% of the larger of the two.

3.7.1.6 Slew rate. (See 4.7.1.6) When 100 feet of RG-108 A/U cable terminated in a 78-ohm dummy load is connected to the driver output, the transition between ±40 mV (either direction) shall take no more than 10 ns for the 256 and 512 kb/s signals and the 16 and 32 kHz timing distributor signals and 100 ns for all other 2 and 16 kb/s signals.

3.7.1.7 Receiver sensitivity. (See 4.7.1.7) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 millivolts (mV) for a digital 1 or -50 mV for a digital 0 is applied to the receiver input terminals.

3.7.1.8 Input load impedance. (See 4.7.1.8) The input impedance shall be 100 ohms ±10%.

3.7.1.9 Clock asymmetry. (See 4.7.1.9) The positive half of the clock waveform shall be equal to half the period ±120 ns for 256 and 512 kHz timing signals and the 16 and 32 kHz timing distributor signals and 1500 ns for other 2 kHz and 16 kHz timing signals.

3.7.2 Unbalanced dedicated loop encryption device (DLED) interface. (See 4.7.2)

3.7.2.1 Driver output voltage. (See 4.7.2.1) The voltage for a digital 1 shall be +6 ±1 volts, and the voltage for a digital 0 shall be -6 ±1 volt. These voltage levels shall be met when measured into an open circuit and into a 6 kilohms load.

3.7.2.2 Receiver operating range. (See 4.7.2.2) The receiver shall deliver a digital 1 for an input voltage from +0.6 to +7.0 volts. The receiver shall deliver a digital 0 for an input voltage from -7.0 to -0.6 volts.

3.7.2.3 Transmitter output impedance. (See 4.7.2.3) Transmitter output impedance shall be no greater than 150 ohms.

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3.7.2.4 Receiver input impedance. (See 4.7.2.4) Receiver input impedance shall be no less than 6 kilohms.

3.7.2.5 Rise and fall times. (See 4.7.2.5) Rise and fall times shall be no greater than 3 microseconds (3µs).

3.7.2.6 Data/timing phase. (See 4.7.2.6) With a 2 kHz timing wave applied to both TC DATA IN and TC TIM IN RCV in parallel, the displacement between zero crossings shall be less than 244.5µs, measured at ERXC2 and ERCT2. With a 2 kHz timing wave applied to both TD DATA IN and TD TIM IN RCV in parallel, the displacement between zero crossings shall be less than 244.5µs measured at ERXC1 and ERCT1. The time between the falling edge of the TX CLOCK and the data transitions of TXDPT shall be at least 2-1/2µs measured at the ten percent or the 90 percent point, whichever is closest.

3.7.2.7 Timing asymmetry. (See 4.7.2.7) The width of the positive half of the timing wave shall be equal to half the clock period ±62-1/2µs.

3.7.2.8 Sync signal. (See 4.7.2.8) The sync signal shall be a positive going pulse, having a width at zero crossing of 30 to 200µs.

3.7.3 Unbalanced VINSON interface.

3.7.3.1 Driver output voltage. (See 4.7.3.1) The voltage for a digital 1 shall be +6 ±1 volt, and the voltage for a digital 0 shall be -6 ±1 volt. These voltage levels shall be met when measured into an open circuit and into a 6 kilohm load.

3.7.3.2 Receiver operating range. (See 4.7.3.2) The receiver shall deliver a digital 1 for an input voltage of +0.6 to +7.0 volts. The receiver shall deliver a digital 0 for an input voltage of -7.0 to -0.6 volts.

3.7.3.3 Rise and fall times. (See 4.7.3.3) Rise and fall times shall be 3 to 15µs.

3.7.4 Analog voice orderwire (AVOW) channel interface. (See 4.7.4)

3.7.4.1 Input and output impedance. (See 4.7.4.1) Input and output impedance shall be 600 ohms ±10%.

3.7.4.2 Input test tone level. Input test tone level shall be -18.2 dBm.

3.7.4.3 Output test tone level. Output test tone level shall be +21.3 dBm ±1 dB.

3.7.4.4 Frequency response. Frequency response shall be ±1/2 dB from 300 to 3400 Hz, referred to 1000 Hz.

3.7.4.5 Signaling frequency. Signal frequency shall be 1600 Hz ±2%.

3.7.4.6 Signal detector operating range. Signal detector operating range shall be -29 to -15 dBm for frequencies of 1600 Hz ±2%.

3.7.4.7 Signal generator output. Signal generator output shall be +21.3 dBm ±1 dB.

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3.7.5 AVOW headset interface. (See 4.7.5)

3.7.5.1 Output. (See 4.7.5.1) With the test tone level applied at the AVOW channel input, the output shall be +4 to +7 dBm into a 150 ohm load.

3.7.5.2 Input impedance. (See 4.7.5.2) Input impedance shall be 430 to 600 ohms at 1 kHz.

3.7.5.3 Input level. (See 4.7.5.3) With a 150 mV rms input applied, the output at the AVOW channel shall be +21.3 dBm ±1 dB.

3.7.5.4 Frequency response. (See 4.7.5.4) Frequency response shall be ±1-1/2 dB from 300 to 3400 Hz, referenced to 1 kHz.

3.7.6 Alarm interface. (See 4.7.6)

3.7.6.1 Relay contact alarms. (See 4.7.6.1) A closure shall cause the appropriate alarm. Forty eight (48) alarms shall be provided as shown in Table XXII.

3.7.6.2 Logic driver alarms. (See 4.7.6.2) An input voltage of +2 to +6 volts shall cause the appropriate alarm. Seven (7) alarms shall be provided as shown in Table XXIII. An input voltage of -6 to +0.8 volts shall not cause an alarm. The input impedance shall be no less than 6 kilohms.

3.7.7 Bit error rate (BER) interface. (See 4.7.7) The signal characteristics of the strobe and BCD data lines are balanced as described in 3.7.1 with the additional characteristics of 3.7.7.1. Offset requirement of 3.7.1.4 does not apply.

3.7.7.1 Strobe pulsewidth. (See 4.7.7.1) Strobe pulsewidth shall be 25 to 70 µsec.

3.7.7.2 Slew time. (See 4.7.7.2) When the strobe is connected to a 100-ohm dummy load, the transition between ±40 mV (either direction) shall take no more than 100 ns.

3.7.7.3 Strobe voltage. (See 4.7.7.3) Strobe voltage shall be:

Idle: -4.0 to -0.1 volts

Active: +0.1 to +4.0 volts

Both idle and active strobe voltages shall be measured line-to-line into a load of 50 ohms from both lines to ground.

3.7.8 Digital voice orderwire (DVOW) performance. (See 4.7.8)

3.7.8.1 Signaling code generation. (See 4.7.8.1) Upon operator selection, the equipment shall be able to generate any one of the 16 codes listed in 4.7.8. A single operation of the signal button shall generate a transmission of two to three seconds duration.

3.7.8.2 Signaling detection. (See 4.7.8.2) The equipment shall be able to detect upon selection by internal switches any three of the 16 codes listed in 4.7.8.

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3.7.8.2.1 Detection with errors. (See 4.7.8.2.1) Ring detection of any of the three selecxed signaling codes shall be accomplished with a 90 percent probability or greater in an error environment of 20 percent BER random bursts interspersed with 0.1 percent BER random error intervals. The time between successive bursts shall be random between 50 ms and one second and the burst length shall be five percent of the time between the start of the burst and the start of the next burst.

3.7.8.2.2 False ring detection. (See 4.7.8.2.2) The equipment shall have a probability of false ring detection of 10^{-5} or less when in the presence of an idle DVOW channel (all ONES) in the burst BER of 3.7.8.2.1. The mean time between false ring detection with random digital data on all incoming lines shall be 24 hours or more.

3.7.8.3 Operator access and bridging. (See 4.7.8.3) The equipment shall provide for a half duplex connection between any channel and the operator in isolation from the bridge. The equipment shall provide for half duplex connections between any number of channels and the operator through the bridge.

3.7.8.4 Bridge seizure and release. (See 4.7.8.4) The bridge shall be seized by channel activity within 50 ms. The bridge shall be released by cessation of channel activity within 20 ms.

3.7.8.5 VINSON buffer. (See 4.7.8.5) The VINSON buffer shall provide for at least 2.2 minutes of error free operation with a frequency difference between the data source and the equipment timing distributor of $\pm$30 parts per million (ppm).

3.7.8.6 DVOW multiplexing. (See 4.7.8.6) The DVOW multiplexer shall multiplex up to 12 DVOW channels into a 256 or 512 kb/s group. The frame format shall be as shown in Table I. When the 16 kb/s channels are offset in frequency by $\pm$1 ppm referred to the DVOW mux transmit timing, channel BCI shall be maintained for at least 1 minute.

TABLE I. DVOW frame format.

Main frame: O D D D C_1 C_2 C_3 C_4 C_5 C_6 C_7 C_8 C_9 C_{10} C_{11} C_P
O = Overhead channel
D = Dummy bit set to 1
C_1 - C_{11} = The 11 DVOW channels
C_P = The dedicated DVOW channel
Overhead channel: F 1 1 1 1 1 1 1
F = Framing bit. 0101 ... when in frame, 1111 when out of frame.

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3.7.8.7 DVOW demux framing. (See 4.7.8.7) The DVOW demux shall acquire frame synchronization within 100 ms in at least 90% of the tests in a random error environment of 0.001 BER. The total time for the detection of loss of frame synchronization and reacquisition shall not exceed 200 ms in at least 90% of the tests in a random error environment of 0.001 BER and with random data on all channels. While BCI is maintained, the demux shall not declare out of frame more than once per 24 hours on the average in a random error environment of 0.001 BER.

3.7.9 Timing distribution. (See 4.7.9) The equipment shall supply three timing signals at 16 kHz and three timing signals at 32 kHz. The equipment shall derive these signals from an input signal having a frequency accuracy of up to ±150 ppm.

3.7.10 AVOW performance. (See 4.7.10)

3.7.10.1 Signaling generation. (See 4.7.10.1) A single operation of the signal pushbutton shall generate a transmission of two to three seconds duration.

3.7.10.2 Signaling detection. (See 4.7.10.2) Under the following conditions, the equipment shall detect a tone burst no less than 90 percent of the time:

a. Tone level from -29 to -15 dBm.

b. Tone to random noise level ratio of 15 dB or better.

c. Tone frequency between 1568 and 1632 Hz.

d. Tone burst two seconds minimum.

3.7.10.3 False signaling detection. (See 4.7.10.3) For a period of one hour, the equipment shall not detect a tone with either Gaussian noise in the band 300 to 3400 Hz at -18.2 dBm or a voice tape at normal conversational level.

3.7.10.4 Operator access and bridging. (See 4.7.10.4) The equipment shall provide a full duplex connection between any channel and the operator in isolation from the bridge. The equipment shall provide a full duplex connection between any number of channels and the operator through the bridge.

3.7.11 Data orderwire (DOW) performance. (See 4.7.11)

3.7.11.1 Format. (See 4.7.11.1) The equipment shall generate a 20 character message block as shown in 4.7.11.1. The equipment shall provide 13 channels, any one of which can be used for the message block. For one or two alarms, one message block shall be transmitted. For three or four alarms, two message blocks shall be transmitted. For five or six alarms, three message blocks shall be transmitted. For seven or eight alarms, four message blocks shall be transmitted. For more than eight alarms, one message block shall be transmitted with the message "MAJOR MAJOR".

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3.7.11.2 Telemetry decombiner framing. (See 4.7.11.2) With a steady bit error rate of 10^{-3}, the decombiner shall lock-in within 0.4 seconds for no less than 90 percent of the trials. Once framed-in, the decombiner shall declare out of frame within 0.6 second of removal of data for no less than 90 percent of the trials. Once framed, and with 10^{-3} bit error rate, the decombiner shall not declare loss of frame more often than once per 24 hours on the average.

3.7.11.3 Resync command pulse rate. (See 4.7.11.3) When a decombiner number 1 is out of frame, it shall issue a resync command pulse to the DLED every 1.6 to 5.0 seconds.

3.7.11.4 DOW BCI. (See 4.7.11.4) When the decombiner timing is offset ±1 ppm with respect to the combiner timing, bit count integrity (BCI) loss on the channels shall occur no more often than once in 15 minutes (100 ppm = 9.0 seconds).

3.7.11.5 DCM multiplexing. (See 4.7.11.5) Each DCM shall multiplex up to seven 2 kb/s channels into a 16 kb/s group. The frame format shall be as shown in Table II. Unused channels shall be set to 1. When the 2 kb/s channel inputs are offset in frequency by ±1 ppm referred to the DCM transmit timing, channel BCI shall be maintained for at least three minutes.

TABLE II. DCM frame format.

$O_1S_1S_2S_3S_4S_5S_6S_7O_2S_1S_2S_3S_4S_5S_6S_7O_3S_1S_2S_3S_4S_5S_6S_7O_4S_1S_2S_3S_4S_5S_6S_7$
$O_1 = O_2 = O_3$ = Digital 0
O_4 = Digital 1
$S_1 - S_7$ = the seven 2 kb/s channels

3.7.11.6 DCM framing. (See 4.7.11.6) The DCM demux shall acquire frame synchronization within 100 ms for at least 90% of the tests in a random error environment of 0.001 BER. While BCI is maintained, the DCM demux shall not declare out of frame no more than once in 240 hours on the average in a random error environment of 0.001 BER. The total time for the DCM demux to detect loss of frame synchronization and reacquire it shall be less than 200 ms for at least 90% of the tests in a random error environment of 0.001 BER.

3.7.12 Prime power variation. (See 4.7.12) The ac primary power source shall be 115V ±10% at frequencies of 50 Hz, 60 Hz, and 400 Hz ±5% single phase. The power input to the unit shall not exceed 80 watts. The ac primary power shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Mode, I, II, and III generator sets.

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3.7.13 BITE and alarms. (See 4.7.13) The built-in test equipment (BITE) shall monitor signals in and out of the unit and internal circuits and power supply voltages. The BITE shall operate without interrupting traffic or interfering with normal system operation. Alarm conditions are removed automatically when proper operation is restored. Failures detected by the BITE circuitry shall cause the following:

a. S Alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm shall provide a visible indication on the front panel except for input power failure.

b. SA Alarm. The SA alarm shall indicate that the malfunction is within the unit. The SA alarm shall cause the status light to go red and the alarm word OCU to be transmitted.

3.7.13.1 Alarm operation. (See 4.7.13.1) Means shall be provided in the unit to check the operation of the alarms without interrupting the equipment operation.

3.7.14 Operational check. (See 4.7.14)

3.7.14.1 Alarm monitor check. (See 4.7.14.1) There shall be no alarm lights lighted.

3.7.14.2 DVOW error rate test. (See 4.7.14.2) There shall be no more than one error over a period of one minute.

3.7.14.3 DVOW ring test. (See 4.7.14.3) The FVOW shall be able to cause itself to ring when looped back.

3.7.14.4 AVOW levels. (See 4.7.14.4) With a channel input of -18.2 dBm at 1000 Hz, the channel output shall be between +20.3 and +22.3 dBm; the headset output shall be between +4 and +7 dBm.

3.7.14.5 AVOW ringing. (See 4.7.14.5) The unit shall generate a ring signal of two to three seconds duration. The unit shall respond to a ring tone of two seconds duration at -29 dBm input and 1568 to 1632 kHz frequency.

3.7.14.6 DCM error rate test. (See 4.7.14.6) There shall be no errors for a period of 30 seconds.

3.8 Service conditions. (See 4.8) The C-10717()/TRC shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified in the following environmental conditions.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) During and after testing as specified in 4.8.1.1, the equipment shall meet the requirements specified in 3.7.14.

3.8.1.2 Low temperature. (See 4.8.1.2) During and after testing as specified in 4.8.1.2, the equipment shall meet the requirements specified in 3.7.14.

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3.8.2 Humidity. (See 4.8.2) During and after testing as specified in 4.8.2, the equipment shall meet the requirements specified in 3.7.14. At the conclusion of this test, the equipment shall exhibit no physical damage.

3.8.3 Low pressure (altitude). (See 4.8.3) During and after testing as specified in 4.8.3, the equipment shall meet the requirements specified in 3.7.14.

3.8.4 Dust. (See 4.8.4) After testing as specified in 4.8.4, and in addition to the constraints of 3.8, the equipment shall exhibit no evidence of clogging or binding when an attempt is made is made to operate any mechanical part through its full range of movement.

3.8.5 Salt fog. (See 4.8.5) After testing as specified in 4.8.5, the equipment when visually examined, shall show no evidence of degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered or otherwise loosened finish or metal surface or exfoliation (see 6.4.2) of metal. In addition, the equipment shall exhibit no evidence of clogging or binding when an attempt is made to operate any mechanical part through its full range of movement.

3.8.6 Fungus. (See 4.8.6) After testing as specified in 4.8.6, the equipment shall be visually examined with the aid of a 10-power magnifier. The equipment shall show no more than six unrelated spots, each no greater than 0.015 square inch in area of sparse microbial growth (see 6.4.4) as evidenced by growth colonization (see 6.4.3), which includes branching (see 6.4.1) and sporulation (see 6.4.5) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.4.6) shall not be included in this evaluation.

3.8.7 Vibration and shock. (See 4.8.7)

3.8.7.1 Vibration. (See 4.8.7.1) After testing as specified in 4.8.7.1, the equipment shall meet the requirements specified in 3.7.14.

3.8.7.2 Shock. (See 4.8.7.2) After testing as specified in 4.8.7.2, the equipment shall meet the requirements specified in 3.7.14.

3.9 Electromagnetic interference. (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4.

CE02	CS02	RE02	RS02
CE03	CS06	RE02.1	RS03
CE04			
CE05			

Notes:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

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3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The C-10717()/TRC shall have no less than 3500 hours Mean-Time-Between-Failures (MTBF), θ_o, rate when operating in the normal modes of operation, 100 percent duty cycle fixed ground environment at an ambient temperature (T_A) of 50°C (122°F) plus the incremental temperature rise (Δt) inside the unit.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements of their respective test requirement drawings and specifications described in 4.5.1:

Circuit Card Assembly (CCA) Control Panel Logic	SM-D-959064
Circuit Card Assembly (CCA) Analog Transmit Amplifier	SM-D-959069
Circuit Card Assembly (CCA) Analog Signaling	SM-D-959072
Circuit Card Assembly (CCA) Analog Bridge	SM-D-959075
Circuit Card Assembly (CCA) Data Channel Mux	SM-D-959078
Circuit Card Assembly (CCA) Level Converter	SM-D-959081
Circuit Card Assembly (CCA) Telemetry Combiner/Generator	SM-D-959084
Circuit Card Assembly (CCA) Mux/Demux	SM-D-959087
Circuit Card Assembly (CCA) Interface	SM-D-959090
Circuit Card Assembly (CCA) VINSON Buffer/Ring Detector	SM-D-959093
Circuit Card Assembly (CCA) Digital Bridge	SM-D-959096
Circuit Card Assembly (CCA) Timing Distributor	SM-D-959099
Circuit Card Assembly (CCA) BITE	SM-D-959105
Circuit Card Assembly (CCA) Alarm Interface	SM-D-959108
Circuit Card Assembly (CCA) Group Framing	SM-E-876642
Power Supply Assembly LGM	SM-D-876554

3.13 Operational tests. (See 4.7.14) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.14 after all other inspections and tests are completed to determine that the unit is operating prior to packaging. The operational test of Group A satisfies this requirement for those units for which that is the last inspection or test.

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3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives.

b. Cables, cords and wires, electrical.

c. Wire (Harnessing).

d. Plastic material and parts.

e. Rubber.

f. Wiring and cabling.

g. Connectors (electrical).

h. Printed wiring assemblies.

3.16 Nuclear survivability. (See 4.17) The requirements of Appendix I to TT-B1-2102-0080 shall apply.

3.17 TEMPEST. (See 4.18) The requirements of Appendix II to TT-B1-2102-0080 shall apply.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. PATE inspection. (See 4.3) Does not include preparation for delivery.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.14)

4.3 PATE. Unless otherwise specified in the contract or purchase order, the PATE inspections shall be performend by the contractor.

4.3.1 PATE units. The contractor shall furnish a minimum of 15 PATE units of the complete Control Unit, Orderwire C-10717( )/TRC Type 2. The PATE quantity shall be as specified in the contract or purchase order.

4.3.2 PATE inspection. The PATE inspections shall consist of inspections specified in Table III and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in-process Group A inspection prior to burn-in.

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TABLE III. PATE inspection.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8-15
1. Inspections by Subsidiary documents.	N/A	4.4	Inspections to be performed on all units.							
2. Burn-in	3.6	4.6	Inspections to be performed on all units.							
3. Group A Inspection	(See Table II)		Inspections to be performed on all units.							
4. Group B Inspection	(See Table III)		Inspections to be performed on all units.							
5. Group C Inspection										
Subgroup 1										
High Temperature	3.8.1	4.8.1.1	1							
Low Temperature	3.8.1	4.8.1.2	2							
Subgroup 2										
EMI/EMC	3.9	4.9		1						
Vibration/Shock	3.8.7	4.8.7			1					
Subgroup 3										
Humidity (Note C)	3.8.2	4.8.2				1				
Subgroup 4										
Altitude	3.8.3	4.8.3					1			
Sand/Dust (Note C)	3.8.4	4.8.4					2			
Salt Fog (Note C)	3.8.5	4.8.5					3			
Fungus (Note C)	3.8.6	4.8.6						1		
Subgroup 5										
Balanced Equipment-Side Signal Characteristics	3.7.1	4.7.1								
Clock/Data Phasing	3.7.1.3	4.7.1.3							1	
Offset	3.7.1.4	4.7.1.4							2	
Signal Balance	3.7.1.5	4.7.1.5							3	
Slew Rate	3.7.1.6	4.7.1.6							4	
Balanced and Unbalanced Input Impedance	3.7.1.8	4.7.1.8							5	
Clock Asymmetry	3.7.1.9	4.7.1.9							6	
Unbalanced Dedicated Loop Encryption (DLED) Interface	3.7.2	4.7.2								
Transmitter Output Impedance	3.7.2.3	4.7.2.3							7	
Receiver Input Impedance	3.7.2.4	4.7.2.4							8	
Rise and Fall Times	3.7.2.5	4.7.2.5							9	
Data/Timing Phasing	3.7.2.6	4.7.2.6							10	
Timing Asymmetry	3.7.2.7	4.7.2.7							11	
Sync Signal	3.7.2.8	4.7.2.8							12	

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TABLE III. PATE inspection. (Cont'd)

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8-15
Subgroup 5										
Unbalanced VINSON Interface	3.7.3	4.7.3								
Rise and Fall Times	3.7.3.3	4.7.3.3							13	
Analog Voice Orderwire (AVOW) Channel Interface	3.7.4	4.7.4								
Input and Output Impedance	3.7.4.1	4.7.4.1							14	
AVOW Headset Interface	3.7.5	4.7.5								
Input Impedance	3.7.5.2	4.7.5.2							15	
Bit Error Rate Interface	3.7.7	4.7.7								
Slew Time	3.7.7.2	4.7.7.2							16	
Digital Voice Orderwire (DVOW) Performance	3.7.8	4.7.8								
Signaling Detection	3.7.8.2	4.7.8.2							17	
Detection With Errors	3.7.8.2.1	4.7.8.2.1							18	
False Ring Detection	3.7.8.2.2	4.7.8.2.2							19	
Bridge Seizure and Release	3.7.8.4	4.7.8.4							20	
VINSON Buffer	3.7.8.5	4.7.8.5							21	
DVOW Multiplexing	3.7.8.6	4.7.8.6							22	
DVOW Demux Framing	3.7.8.7	4.7.8.7							23	
Timing Distribution	3.7.9	4.7.9								
AVOW Performance	3.7.10	4.7.10								
Signaling Detection	3.7.10.2	4.7.10.2							24	
False Signaling Detection	3.7.10.3	4.7.10.3							25	
Data Orderwire (DOW) Performance	3.7.11	4.7.11								
Format	3.7.11.1	4.7.11.1							26	
Telemetry Decombiner Framing	3.7.11.2	4.7.11.2							27	
Resync Command Pulse Rate	3.7.11.3	4.7.11.3							28	
DOW BCI	3.7.11.4	4.7.11.4							29	
DCM Multiplexing	3.7.11.5	4.7.11.5							30	
DCM Framing	3.7.11.6	4.7.11.6							31	
6. Group D Inspection										
Reliability	3.10	4.10								8 units

NOTES:

a. The Inspections 1 through 4, in the order shown, shall be performed on all PATE units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by Inspections 5 and 6 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. The quantity of units shall be as specified in the contract or purchase order. All PATE units shall be subjected to Group A inspection. Units 1 through 15 shall be subjected to Group B inpsection.

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4.3.3 PATE data. The PATE test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where Required
Printed Wiring	3.2.3
Semiconductor Devices	3.2.4
Parts, Materials, and Processes	3.3
Finish	3.4
Marking	3.5
Workmanship	3.15

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in Table IV shall be inspected for conformance to the inspections specified in Table IV.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in Table IV. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table IV shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table IV.

NOTE: Assemblies and subassemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's Group A inspection of the end equipment, except for spares that are not installed into complete units.

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TABLE IV. Group A inspection.

Inspection	Requirement Paragraph	Inspection Paragraph	AQL	
			Major	Minor
VISUAL/MECHANICAL (Including Spare Assemblies)				
CCA Control Panel Logic	3.15	4.13	2.5%	4.0%
CCA Analog Transmit Amplifier	3.15	4.13	2.5%	4.0%
CCA Analog Signaling	3.15	4.13	2.5%	4.0%
CCA Analog Bridge	3.15	4.13	2.5%	4.0%
CCA Data Channel Mux	3.15	4.13	2.5%	4.0%
CCA Level Converter	3.15	4.13	2.5%	4.0%
CCA Telemetry Combiner/Generator	3.15	4.13	2.5%	4.0%
CCA Mux/Demux	3.15	4.13	2.5%	4.0%
CCA Interface	3.15	4.13	2.5%	4.0%
CCA VINSON Buffer/Ring Detection	3.15	4.13	2.5%	4.0%
CCA Digital Bridge	3.15	4.13	2.5%	4.0%
CCA Timing Distributor	3.15	4.13	2.5%	4.0%
CCA BITE	3.15	4.13	2.5%	4.0%
CCA Alarm Interface	3.15	4.13	2.5%	4.0%
Power Supply	3.15	4.13	2.5%	4.0%
CCA Group Framing	3.15	4.13	2.5%	4.0%
Case and Covers	3.15	4.13	2.5%	10%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
CCA Control Panel Logic	3.12	4.12	1.0%	*
CCA Analog Transmit Amplifier	3.12	4.12	1.0%	*
CCA Analog Signaling	3.12	4.12	1.0%	*
CCA Analog Bridge	3.12	4.12	1.0%	*
CCA Data Channel Mux	3.12	4.12	1.0%	*
CCA Level Converter	3.12	4.12	1.0%	*
CCA Telemetry Combiner/Generator	3.12	4.12	1.0%	*
CCA Mux/Demux	3.12	4.12	1.0%	*
CCA Interface	3.12	4.12	1.0%	*
CCA VINSON Buffer/Ring Detector	3.12	4.12	1.0%	*
CCA Digital Bridge	3.12	4.12	1.0%	*
CCA Timing Distributor	3.12	4.12	1.0%	*
CCA BITE	3.12	4.12	1.0%	*
CCA Alarm Interface	3.12	4.12	1.0%	*
Power Supply	3.12	4.12	1.0%	*
CCA Group Framing	3.12	4.12	1.0%	*
ELECTRICAL				
BITE and Alarm Tests	3.7.13	4.7.13	1.0%	*
Operational Check	3.7.14	4.7.14	1.0%	*
DVOW Operator Access and Bridging	3.7.8.3	4.7.8.3	1.0%	*
AVOW Operator Access and Bridging	3.7.10.4	4.7.10.4	1.0%	*

*All Electrical Defects are Major.

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b. Only for spares which are not installed into complete units, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in Table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the Quality Assurance Verification Inspection of step a above shall be assembled into the end equipments, which shall than be subjected by the contractor to the burn-in test requirements of 4.6 and a 100% visual/mechanical and electrical inspection in accordance with Table IV in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in Table II. Group A electrical inspection shall be performed in the order shown in Table IV.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection, except that the gauging for interchangeability is to be done prior to assembly of the end equipment. This inspection shall conform to Table V and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table V and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in Table V, except that the gauging for interchangeability is to be done prior to assembly of the end equipment.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table VI. Samples shall be selected in accordance with 4.5.3.1

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing with the unit subjected to testing in that order. The unit selected every eighth month shall be subjected to EMI/EMC testing.

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TABLE V. Group B inspection.

Inspection	Requirement Paragraph	Inspection Paragraph	AQL
Dimensional Interchangeability	3.11	4.11	6.5%
Balanced Equipment-Side Signal Characteristics	3.7.1	4.7.1	6.5%
Line-to-Line Voltage Levels	3.7.1.1	4.7.1.1	6.5%
Line-to-Ground Voltage Levels	3.7.1.2	4.7.1.2	6.5%
Receiver Sensitivity	3.7.1.7	4.7.1.7	6.5%
Unbalanced Dedicated Loop Encryption Device (DLED) Interface	3.7.2	4.7.2	6.5%
Driver Output Voltage	3.7.2.1	4.7.2.1	6.5%
Receiver Operating Range	3.7.2.2	4.7.2.2	6.5%
Unbalanced VINSON Interface	3.7.3	4.7.3	6.5%
Driver Output Voltage	3.7.3.1	4.7.3.1	6.5%
Receiver Operating Range	3.7.3.2	4.7.3.2	6.5%
Analog Voice Orderwire (AVOW) Channel Interface	3.7.4	4.7.4	6.5%
Input Test Tone Level	3.7.4.2	4.7.4.2	6.5%
Output Test Tone Level	3.7.4.3	4.7.4.3	6.5%
Frequency Response	3.7.4.4	4.7.4.4	6.5%
Signaling Frequency	3.7.4.5	4.7.4.5	6.5%
Signal Detector Operating Range	3.7.4.6	4.7.4.6	6.5%
Signal Generator Output	3.7.4.7	4.7.4.7	6.5%
AVOW Headset Interface	3.7.5	4.7.5	6.5%
Output	3.7.5.1	4.7.5.1	6.5%
Input Level	3.7.5.3	4.7.5.3	6.5%
Frequency Response	3.7.5.4	4.7.5.4	6.5%
Alarm Interface	3.7.6	4.7.6	6.5%
Relay Contact Alarms	3.7.6.1	4.7.6.1	6.5%
Logic Driver Alarms	3.7.6.2	4.7.6.2	6.5%
Bit Error Rate Interface	3.7.7	4.7.7	6.5%
Strobe Pulse Width	3.7.7.1	4.7.7.1	6.5%
Strobe Voltage	3.7.7.3	4.7.7.3	6.5%
Digital Voice Orderwire (DVOW) Performance	3.7.8	4.7.8	6.5%
Signaling Code Generation	3.7.8.1	4.7.8.1	6.5%
AVOW Performance	3.7.10	4.7.10	6.5%
Signaling Generation	3.7.10.1	4.7.10.1	6.5%
Prime Power Variation	3.7.12	4.7.12	6.5%

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4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to Altitude, Sand/Dust and Salt Fog testing and one unit shall be subjected to Fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in Table VI.

4.5.3.2 Order of inspection within Group C. Group C inspections shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

TABLE VI. Group C inspections.

Inspection	Requirement Paragraph	Inspection Paragraph
Subgroup 1		
High Temperature	3.8.1.1	4.8.1.1
Low Temperature	3.8.1.2	4.8.1.2
Subgroup 2		
EMI/EMC	3.9	4.9
Vibration	3.8.7.1	4.8.7.1
Shock	3.8.7.2	4.8.7.2
Basic Design		4.8.7.2.1
Bench Handling		4.8.7.2.2
Subgroup 3		
Humidity	3.8.2	4.8.2
Subgroup 4		
Sand/Dust	3.8.4	4.8.4
Salt Fog	3.8.5	4.8.5 (Note 1)
Altitude	3.8.3	4.8.3
Fungus	3.8.6	4.8.6 (Note 1)

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TABLE VI. Group C inspections. (Cont'd)

Inspection	Requirement Paragraph	Inspection Paragraph
Subgroup 5		
Balanced Equipment-Side Signal Characteristics	3.7.1	4.7.1
Clock/Data Phasing	3.7.1.3	4.7.1.3
Offset	3.7.1.4	4.7.1.4
Signal Balance	3.7.1.5	4.7.1.5
Slew Rate	3.7.1.6	4.7.1.6
Balanced and Unbalanced Input Impedance	3.7.1.8	4.7.1.8
Clock Asymmetry	3.7.1.9	4.7.1.9
Unbalanced Dedicated Loop Encryption Device (DLED) Interface	3.7.2	4.7.2
Transmitter Output Impedance	3.7.2.3	4.7.2.3
Receiver Input Impedance	3.7.2.4	4.7.2.4
Rise and Fall Times	3.7.2.5	4.7.2.5
Data/Timing Phasing	3.7.2.6	4.7.2.6
Timing Asymmetry	3.7.2.7	4.7.2.7
Sync Signal	3.7.2.8	4.7.2.8
Unbalanced VINSON Interface	3.7.3	4.7.3
Rise and Fall Times	3.7.3.3	4.7.3.3
Analog Voice Orderwire (AVOW) Channel Interface	3.7.4	4.7.4
Input and Output Impedance	3.7.4.1	4.7.4.1
AVOW Headset Interface	3.7.5	4.7.5
Input Impedance	3.7.5.2	4.7.5.2
Bit Error Rate Interface	3.7.7	4.7.7
Slew Time	3.7.7.2	4.7.7.2
Digital Voice Orderwire (DVOW) Performance	3.7.8	4.7.8
Signaling Detection	3.7.8.2	4.7.8.2
Detection with Errors	3.7.8.2.1	4.7.8.2.1
False Ring Detection	3.7.8.2.2	4.7.8.2.2
Bridge Seizure and Release	3.7.8.4	4.7.8.4
VINSON Buffer	3.7.8.5	4.7.8.5
DVOW Multiplexing	3.7.8.6	4.7.8.6
DVOW Demux Framing	3.7.8.7	4.7.8.7
Timing Distribution	3.7.9	4.7.9
AVOW Performance	3.7.10	4.7.10
Signaling Detection	3.7.10.2	4.7.10.2
False Signaling Detection	3.7.10.3	4.7.10.3
Data Orderwire (DOW) Performance	3.7.11	4.7.11
Format	3.7.11.1	4.7.11.1
Telemetry Decombiner Framing	3.7.11.2	4.7.11.2
Resync Command Pulse Rate	3.7.11.3	4.7.11.3
DOW BCI	3.7.11.4	4.7.11.4
DCM Multiplexing	3.7.11.5	4.7.11.5
DCM Framing	3.7.11.6	4.7.11.6

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table VII and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspections. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE VII. Group D inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight units shall be selected from the first month's production and subjected to the test of 4.10. For subsequent Group D inspections, a maximum of eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected per Figure 1. Test Level B of MIL-STD-781 shall apply except that duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. The requirements of 3.6 shall be met.

4.6.1 Burn-in. Set control panel switches per Table VIII. Set CCA switches per Table IX. Set audio oscillator to provide a 1 kHz signal.

Loop Back Data From Channel 1 to All Other Channels

1 Out to 2 In
2 Out to 3 In
3 Out to 4 In
4 Out to 5 In
5 Out to 6 In
6 Out to 7 In
7 Out to 8 In
8 Out to 9 In
9 Out to 10 In
10 Out to 11 In
11 Out to MG In
MG Out to P In

4.6.2 Setup. Secure unit to test bed capable of supplying 1.5G for 10 minutes each hour, located in a chamber maintained at a temperature of 50°C ±5°C. Connect per Burn-in diagram (Figure 1).

4.6.3 Test. Turn unit power ON. Run the unit continuously for 48 hours, using BITE to indicate failure. If a failure is detected, it is to be repaired and recorded. The last 24 hours of burn-in must be failure free as indicated by BITE.

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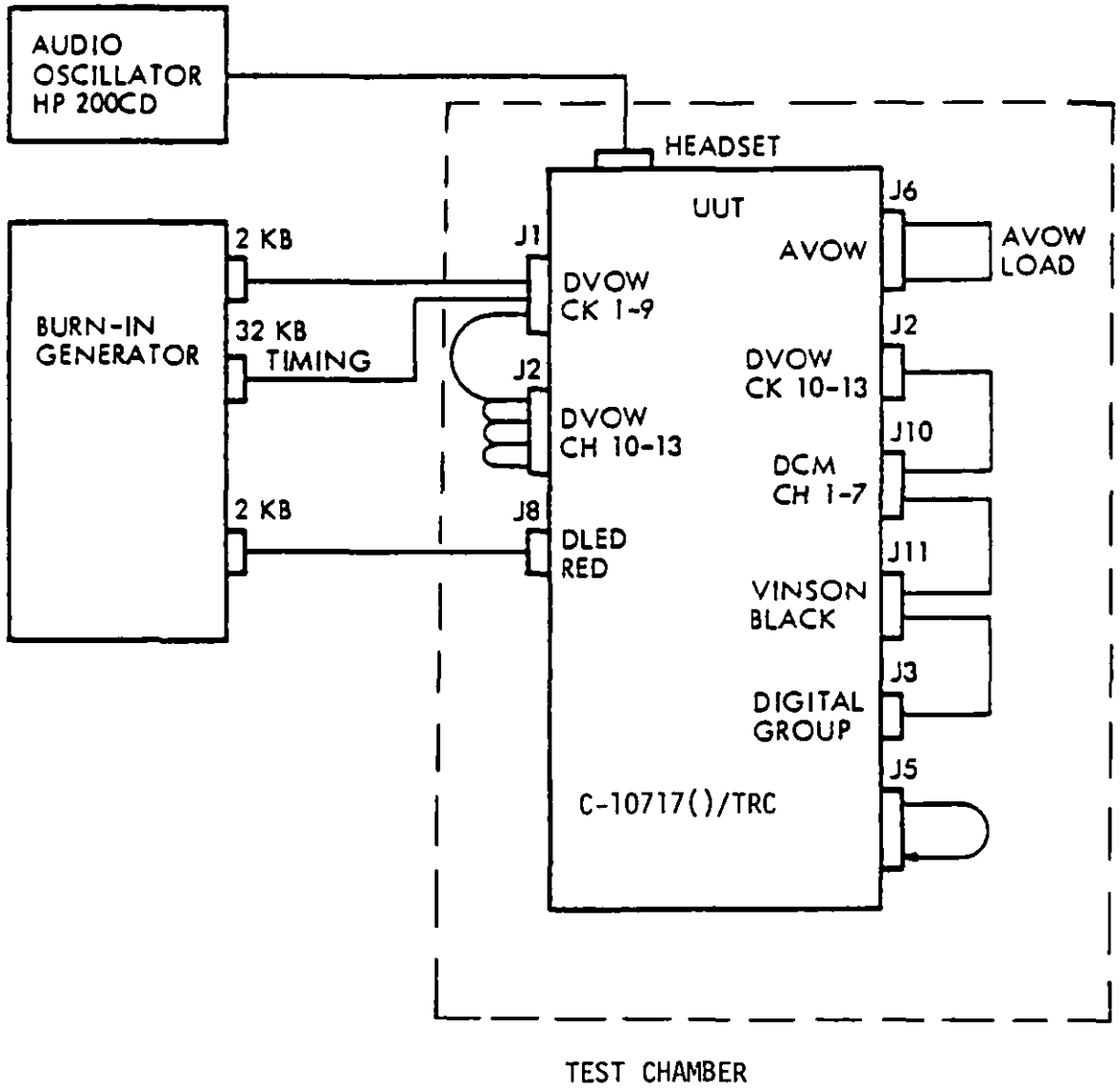


FIGURE 1. Burn-in diagram.

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TABLE VIII. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1	X	
	2	X	
	3	X	
	4	X	
	5	X	
	6	X	

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1	X							
	2	X							
	3	X							
	4	X							
	5	X							
	6	X							
	7	X							
	8	X							
	9	X							
	10	X							
	11	X							
	MG	X							
	P	X							

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S1	S2	S1
											X

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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TABLE IX. OCU CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	ON
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

*See Front Cover for code switch positions.

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4.7 Electrical. (See 3.7) Unless otherwise specified, all tests shall be performed at 115 volts +5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in Table X or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-C-45662A. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see Table XI) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test listed in Table XII.

TABLE X. Standard test equipment.

Description	Mfg./Model
DC Power Supply	Harrison Labs 510A
Oscilloscope	Tektronix 7904
Attenuator	Kay 30-0
AC Voltmeter	HP 403B
Audio Oscillator	HP 200CD
Distortion Analyzer	HP 333A
RMS Voltmeter	HP 406A
Selective Voltmeter	HP 302A
Counter	HP 5345A
Frequency Synthesizer	GRC 1162A
Oscilloscope	Tektronix
Digital Volt-Ohmmeter	Fluke 8040A
Tape Player	
Serial Data Analyzer	Spectron D-502B
Time Interval Counter	HP 5233L
Logic Analyzer	HP 1067A
Wattmeter	Weston 432
AC Power Source	Invertron 1501A

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Table XI. Special test equipment.

Equipment Name	Description
Error Inserter	Allows errors to be added to signals to the UUT.
BER Number Generator	Simulates the BER radio signals.
Alarm Box	Allows the application of single or multiple alarms to the UUT.
Level Converter	Converts balanced signals to VINSON in both directions.
Level Converter	Converts and gates balanced - TTL - balanced.
Noise Mixer and Gate	Gates and allows for noise measurements.

Table XII. Government furnished equipment (GFE).

Equipment Name	Quantity	Description
DSVT Simulator	1	The DSVT Simulator test set simulates the 16 kb/s DVOW and 2 kb/s DOW signals.
Station Clock Timing Source	1	978890 provides related timing signals to the UUT.
Group Modem	1	Supplied as GFE for test purposes.

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4.7.1 Balanced equipment-side signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage levels. (See 3.7.1.1)

STEP 1 Connect the equipment as shown in Figure 2.

STEP 2 Ensure that the following connections are made.

DSVT Simulator #1	UUT Connections			
	Connector	Pins		
		Low	High	Ground
DVOW DATA IN CH 1	J1	B	C	A
STATION 16 kHz	J2	w	x	A
CLOCK	J7	Y	Z	A
DSVT Simulator #2 As Required				
DCM DATA IN CH1	J10	B	C	A
DCM TIM OUT CH1		g	h	A
DCM DATA IN CH2		D	E	A
DCM TIM OUT CH2		i	j	A
DCM DATA IN CH3		F	G	A
DCM TIM OUT CH3		k	m	A
DCM DATA IN CH4		H	J	A
DCM TIM OUT CH4		n	p	A
DCM DATA IN CH5		K	L	A
DCM TIM OUT CH5		q	r	A
DCM DATA IN CH6		M	N	A
DCM TIM OUT CH6		s	t	A
DCM DATA IN CH7		P	R	A
DCM TIM OUT CH7	J10	u	v	A
DCM DATA IN CH8	J11	B	C	A
DCM TIM OUT CH8		g	h	A
DCM DATA IN CH9		D	E	A
DCM TIM OUT CH9	J11	i	j	A

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DSVT Simulator #2 (As required)	UUT Connections Pins			
	Connector	Low	High	Ground
DCM DATA IN CH10	J11	F	G	A
DCM TIM OUT CH10		k	m	A
DCM DATA IN CH11		H	J	A
DCM TIM OUT CH11		n	p	A
DCM DATA IN CH12		K	L	A
DCM TIM OUT CH12		q	r	A
DCM DATA IN CH13		M	N	A
DCM TIM OUT CH13		s	t	A
DCM Data IN CH14		P	R	A
DCM TIM OUT CH14	J11	u	v	A
ETXC2 TO TXCLOCK	J9-J8	Jumper J9-J to J8 G		
TXDPT TO ETCT2	J8-J9	J8-F to J9 K		
TXCPT TO ETCT1	J8-J9	J8-F to J9 L		

STEP 3 Set up the Station clock and DSVT Simulators for proper operation at 2 kb/s and 16 kb/s respectively.

STEP 4 For each of the signals listed in Table XIII, measure the positive and negative voltage levels line-to-line (i.e., between high and low pins) at the pin numbers indicated.

NOTE: When measuring DCM channels connect the DSVT Simulator to each channel one at a time, i.e., DATA IN and TIM OUT and then measure the DATA OUT. Measure the TIM OUT with the DSVT disconnected. The DCM Group must be looped back when measuring DCM channels.

STEP 5 Verify that the line-to-line voltage levels are within the required limits of 3.7.1.1.

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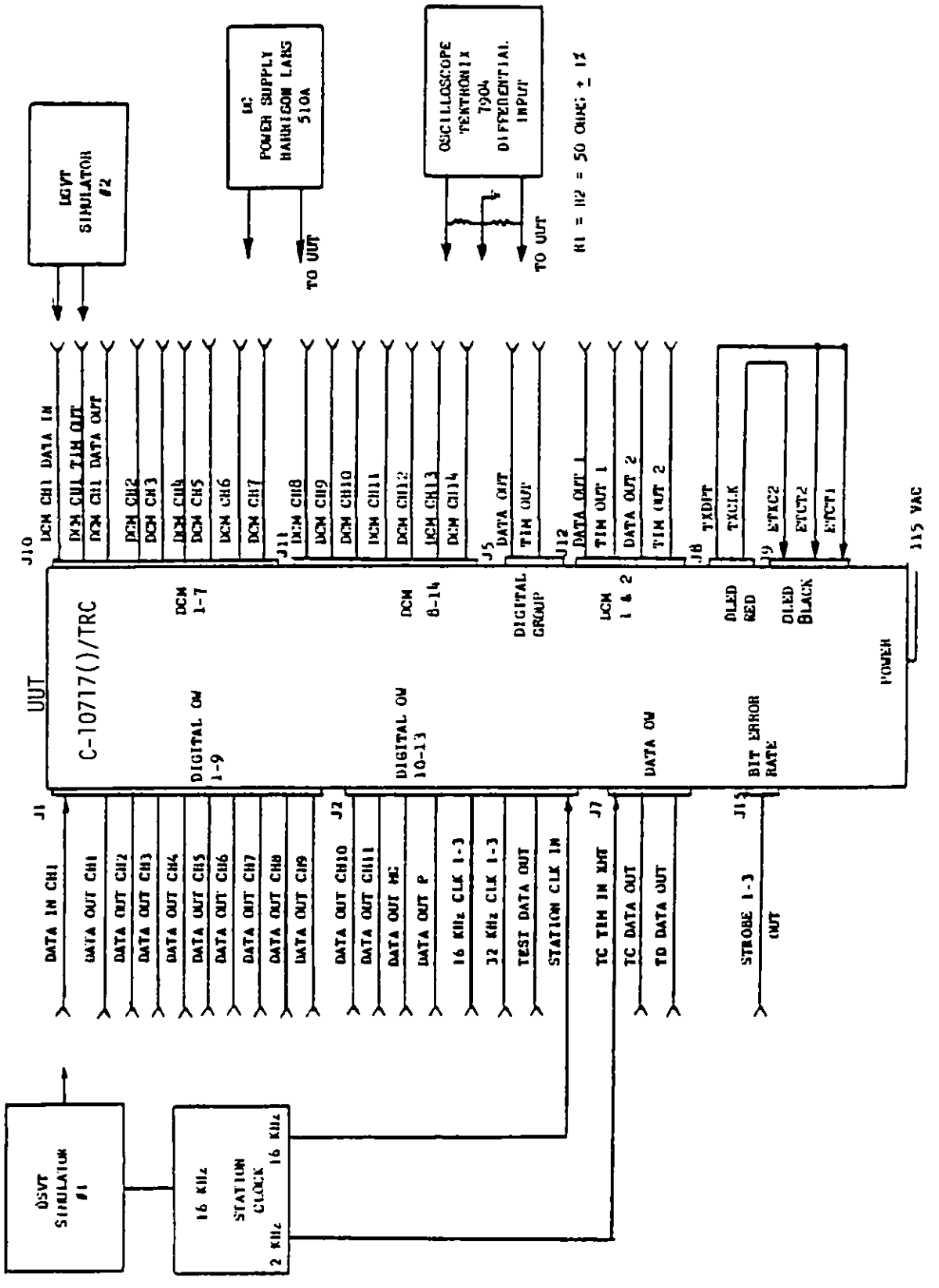


FIGURE 2. Line-to-line voltage levels.

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TABLE XIII. Line-to-line voltage level signals.

Signal	Connector	UUT Point of Measurement		
		Pins		
		Low	High	Ground
DVOW DATA OUT CH 1	J1	q	r	A
DVOW DATA OUT CH 2		s	t	A
DVOW DATA OUT CH 3		u	v	A
DVOW DATA OUT CH 4		w	x	A
DVOW DATA OUT CH 5		y	z	A
DVOW DATA OUT CH 6		AA	BB	A
DVOW DATA OUT CH 7		CC	DD	A
DVOW DATA OUT CH 8		EE	FF	A
DVOW DATA OUT CH 9	J1	GG	HH	A
DVOW DATA OUT CH 10	J2	n	p	A
DVOW DATA OUT CH 11		q	r	A
DVOW DATA OUT CH MG		s	t	A
DVOW DATA OUT CH P		u	v	A
TEST DATA OUT		y	z	A
16 kHz CLOCK 1 OUT		H	J	A
16 kHz CLOCK 2 OUT		K	L	A
16 kHz CLOCK 3 OUT		M	N	A
32 kHz CLOCK 1 OUT		P	R	A
32 kHz CLOCK 2 OUT		S	T	A
32 kHz CLOCK 3 OUT	J2	U	V	A
TC DATA OUT	J7	U	V	A
TD DATA OUT	J7	P	R	A
STROBE 1 OUT	J15	R	S	A
STROBE 2 OUT	J15	K	L	A
STROBE 3 OUT	J15	M	N	A

NOTE: Polarity is measured on the pin with the later letter of the alphabet. The pin with the earlier letter is used as reference.

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TABLE XIII. Line-to-line voltage level signals. (Cont'd)

Signal	Connector	UUT Point of Measurement		
		Pins		
		Low	High	Ground
DCM CH1 DATA OUT	J10	S	T	A
DCM CH2 DATA OUT		U	V	A
DCM CH3 DATA OUT		W	X	A
DCM CH4 DATA OUT		Y	Z	A
DCM CH5 DATA OUT		a	b	A
DCM CH6 DATA OUT		c	d	A
DCM CH7 DATA OUT	J10	e	f	A
DCM CH8 DATA OUT	J11	S	T	A
DCM CH9 DATA OUT		U	V	A
DCM CH10 DATA OUT		W	X	A
DCM CH11 DATA OUT		Y	Z	A
DCM CH12 DATA OUT		a	b	A
DCM CH13 DATA OUT		c	d	A
DCM CH14 DATA OUT	J11	e	f	A
GROUP DATA OUT	J5	E	F	P
GROUP TIM OUT	J5	G	H	R
DCM 1 DATA OUT	J12	F	G	A
DCM 1 TIM OUT		H	J	A
DCM 2 DATA OUT		P	R	A
DCM 2 TIM OUT	J12	S	T	A

NOTE: Polarity is measured on the pin with the later letter of the alphabet. The pin with the earlier letter is used as reference.

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4.7.1.2 Line-to-ground voltage. (See 3.7.1.2)

STEP 1 The equipment setup is the same as in 4.7.1.1 except that the scope input must be unbalanced.

STEP 2 For each of the signals given below, measure the voltage peaks on both lines to ground.

STEP 3 Verify that the voltage levels are within the required limits.

Unbalanced Voltage Measurement	Requirement
DVOW Data Out CH 1	±3V maximum
DVOW Data Out CH 2	
DVOW Data Out CH 3	
DVOW Data Out CH 4	
DVOW Data Out CH 5	
DVOW Data Out CH 6	
DVOW Data Out CH 7	
DVOW Data Out CH 8	
DVOW Data Out CH 9	
DVOW Data Out CH 10	
DVOW Data Out CH 11	
DVOW Data Out CH MG	
DVOW Data Out CH P	
16 kHz Clock 1 Out	
16 kHz Clock 2 Out	
16 kHz Clock 3 Out	
32 kHz Clock 1 Out	
32 kHz Clock 2 Out	
32 kHz Clock 3 Out	
TEST DATA OUT	
TC DATA OUT	
TD DATA OUT	±3V maximum

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Unbalanced Voltage Measurement	Requirement
STROBE 1 OUT	±3V maximum
STROBE 2 OUT	
STROBE 3 OUT	
DCM CH 1 DATA OUT	
DCM CH 2 DATA OUT	
DCM CH 3 DATA OUT	
DCM CH 4 DATA OUT	
DCM CH 5 DATA OUT	
DCM CH 6 DATA OUT	
DCM CH 7 DATA OUT	
DCM CH 8 DATA OUT	
DCM CH 9 DATA OUT	
DCM CH 10 DATA OUT	
DCM CH 11 DATA OUT	
DCM CH 12 DATA OUT	
DCM CH 13 DATA OUT	
DCM CH 14 DATA OUT	
DCM CH 1 TIM OUT	
DCM CH 2 TIM OUT	
DCM CH 3 TIM OUT	
DCM CH 4 TIM OUT	
DCM CH 5 TIM OUT	
DCM CH 6 TIM OUT	
DCM CH 7 TIM OUT	
DCM CH 8 TIM OUT	
DCM CH 9 TIM OUT	
DCM CH 10 TIM OUT	
DCM CH 11 TIM OUT	
DCM CH 12 TIM OUT	
DCM CH 13 TIM OUT	
DCM CH 14 TIM OUT	±3V maximum

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Unbalanced Voltage Measurement	Requirement	Measured Value	
		High Pin to GND	Low Pin to GND
GROUP DATA OUT	±3V maximum		
GROUP TIM OUT			
DCM GROUP 1 DATA OUT			
DCM GROUP 2 DATA OUT			
DCM GROUP 2 TIM OUT			
DCM GROUP 1 TIM OUT	±3V maximum		

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4.7.1.3 Clock/data phasing. (See 3.7.1.3)

STEP 1 Connect the equipment as shown in Figure 3. Ensure that the following connections are made.

Signal	Connector	UUT Connections		
		Pins		
		Low	High	Ground
DSVT Simulator #1 Data Out	J1	B	C	A
2 kHz Station Clock	J2	w	x	A
Vinson Data Out to Vinson Data In	J3	Jumper J3B to J3C		A
PTT	J3	J3D to J3E		A
DSVT Simulator #2 (As Required)	J10			
DCM Data In CH 1	J10	B	C	A
DCM Data Out CH 1	J10	g	h	A
DCM Data In CH 2	J10	D	E	A
DCM Data Out CH 2	J10	i	j	A
DCM Data In CH 3	J10	F	G	A
DCM Data Out CH 3	J10	k	m	A
DCM Data In CH 4	J10	H	J	A
DCM Data Out CH 4	J10	m	p	A
DCM Data In CH 5	J10	K	L	A
DCM Data Out CH 5	J10	q	r	A
DCM Data In CH 6	J10	N	M	A
DCM Data Out CH 6	J10	s	t	A
DCM Data In CH 7	J10	P	R	A
DCM Data Out CH 7	J10	u	v	A
DCM Data In CH 8	J11	B	C	A
DCM Data Out CH 8	J11	g	h	A
DCM Data In CH 9	J11	D	E	A
DCM Data Out CH 9	J11	i	j	A
DCM Data In CH 10	J11	F	G	A
DCM Data Out CH 10	J11	k	m	A

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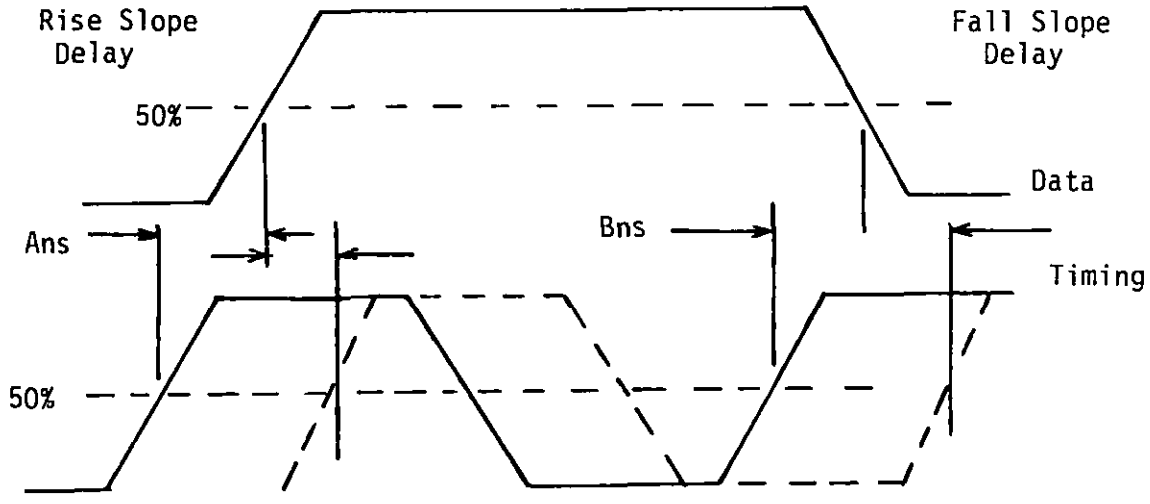
Signal	Connector	UUT Connections Pins		
		Low	High	Ground
DCM Data In CH11	J11	H	J	A
DCM Data Out CH11	J11	m	p	A
DCM Data In CH12	J11	K	L	A
DCM Data Out CH12	J11	q	r	A
DCM Data In CH13	J11	M	N	A
DCM Data Out CH13	J11	s	t	A
DCM Data In CH14	J11	P	R	A
DCM Data Out CH14	J11	u	v	A

STEP 2 Set the UUT circuit board switches and the control panel switches as shown in Tables XIV and XV.

STEP 3 Push the DVOW Channel 1 pushbutton and close the PTT switch.

STEP 4 For each of the pairs of signals in Table XVI at the pin numbers indicated, measure the delay between the 50% points on both the data rise and fall time slopes to the 50% points of the timing rise slopes as indicated below.

STEP 5 Verify that the delays meet the requirements of 3.7.1.3.



Typical Measurement

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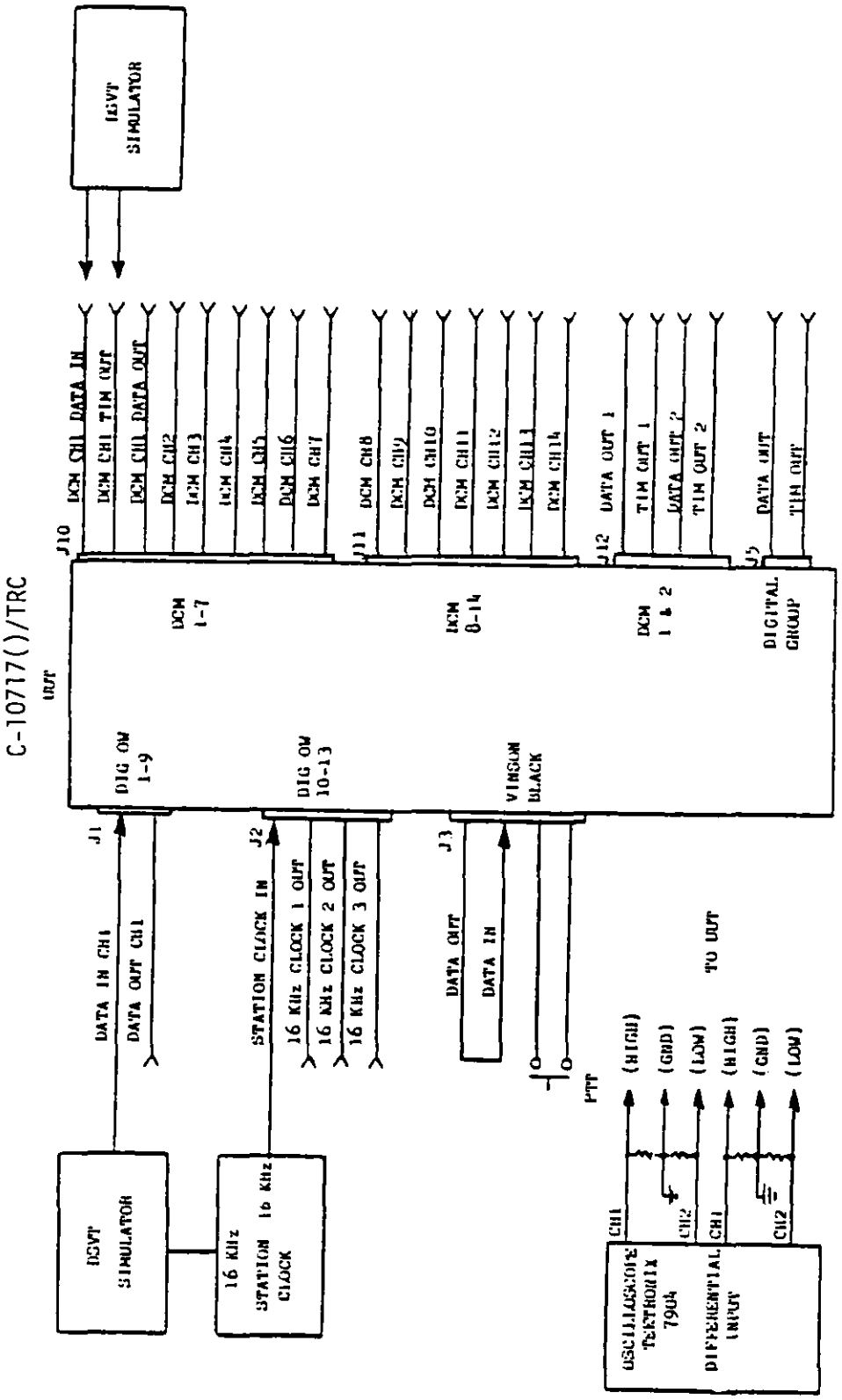


FIGURE 3. Clock/data phasing.

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TABLE XIV. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s / 16 kb/s	As Used
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4* Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8* Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4* Codes 0 thru 9	0
	Channel Select S2	5 thru 8* Codes 1 thru 13	1
	BER Threshold	5 thru 8* Codes 1 thru 9	8
A8	VBRD Ring S1	Codes 1 thru 16*	1
	Codes S2		1
	S3		1

*See Front Cover for code switch positions.

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TABLE XV. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1		X						
	2		X						
	3		X						
	4		X						
	5		X						
	6		X						
	7		X						
	8		X						
	9		X						
	10		X						
	11		X						
	MG		X						
	P		X						

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
											X

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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TABLE XVI. Clock/data phasing signals.

Signal	Connector	UUT Point of Measurement		
		Pins		
		Low	High	Ground
DVOW Data Out CH 1 (Data)	J1	q	r	A
16 kHz Clock 1 Out (Tim)	J2	H	J	A
DVOW Data Out CH 1 (Data)	J1	q	r	A
16 kHz Clock 2 Out (Tim)	J2	K	L	A
DVOW Data Out CH 1 (Data)	J1	q	r	A
16 kHz Clock 3 Out (Tim)	J2	M	N	A
DCM CH 1 Data Out	J10	S	T	A
DCM CH 1 Tim Out	J10	g	h	A
DCM CH 2 Data Out	J10	U	V	A
DCM CH 2 Tim Out	J10	i	j	A
DCM CH 3 Data Out	J10	W	X	A
DCM CH 3 Tim Out	J10	k	m	A
DCM CH 4 Data Out	J10	Y	Z	A
DCM CH 4 Tim Out	J10	n	p	A
DCM CH 5 Data Out	J10	a	b	A
DCM CH 5 Tim Out	J10	q	r	A
DCM CH 6 Data Out	J10	c	d	A
DCM CH 6 Tim Out	J10	s	t	A
DCM CH 7 Data Out	J10	e	f	A
DCM CH 7 Tim Out	J10	u	v	A
DCM CH 8 Data Out	J11	S	T	A
DCM CH 8 Tim Out	J11	g	h	A
DCM CH 9 Data Out	J11	U	V	A
DCM CH 9 Tim Out	J11	i	j	A
DCM CH 10 Data Out	J11	W	X	A
DCM CH 10 Tim Out	J11	k	m	A
DCM CH 11 Data Out	J11	V	Z	A
DCM CH 11 Tim Out	J11	n	p	A
DCM CH 12 Data Out	J11	a	b	A
DCM CH 12 Tim Out	J11	q	r	A

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TABLE XVI. Clock/data phasing signals. (Cont'd)

Signal	Connector	UUT Point of Measurement		
		Pins		
		Low	High	Ground
DCM CH 13 Data Out	J11	c	d	A
DCM CH 13 Tim Out	J11	s	t	A
DCM CH 14 Data Out	J11	e	f	A
DCM CH 14 Tim Out	J11	u	v	A
DCM Data Out 1	J12	F	G	A
DCM Tim Out 1	J12	H	J	A
DCM Data Out 2	J12	P	R	A
DCM Tim Out 2	J12	S	T	A
Digital Group Data Out	J5	E	F	P
Digital Group Tim Out	J5	G	H	R

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4.7.1.4 Offset. (See 3.7.1.4)

STEP 1 Change the scope connections as shown in Figure 4.

STEP 2 For each of the signals given below, measure the dc offset voltage between the high and low pins.

NOTE: Polarity is measured on the pin with the later letter in the alphabet. The pin with the earlier letter is used as the reference.

DC Offset Measurement	Requirement
DVOW Data Out CH 1	±3V maximum
DVOW Data Out CH 2	
DVOW Data Out CH 3	
DVOW Data Out CH 4	
DVOW Data Out CH 5	
DVOW Data Out CH 6	
DVOW Data Out CH 7	
DVOW Data Out CH 8	
DVOW Data Out CH 9	
DVOW Data Out CH 10	
DVOW Data Out CH 11	
DVOW Data Out CH MG	
DVOW Data Out CH P	
DCM CH 1 Data Out	
DCM CH 2 Data Out	
DCM CH 3 Data Out	
DCM CH 4 Data Out	
DCM CH 5 Data Out	
DCM CH 6 Data Out	
DCM CH 7 Data Out	
DCM CH 8 Data Out	
DCM CH 9 Data Out	
DCM CH 10 Data Out	
DCM CH 11 Data Out	
DCM CH 12 Data Out	
DCM CH 13 Data Out	±3V maximum

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DC Offset Measurement	Requirement
DCM CH 14 Data Out	±3V maximum
DCM CH 1 Tim Out	
DCM CH 2 Tim Out	
DCM CH 3 Tim Out	
DCM CH 4 Tim Out	
DCM CH 5 Tim Out	
DCM CH 6 Tim Out	
DCM CH 7 Tim Out	
DCM CH 8 Tim Out	
DCM CH 9 Tim Out	
DCM CH 10 Tim Out	
DCM CH 11 Tim Out	
DCM CH 12 Tim Out	
DCM CH 13 Tim Out	
DCM CH 14 Tim Out	
DCM Data Out 1	
DCM Tim Out 1	
DCM Data Out 2	
DCM Tim Out 2	
Group Data Out	
Group Tim Out	
Test Data Out	
32 kHz Clock Out 1	
32 kHz Clock Out 2	
32 kHz Clock Out 3	
16 kHz Clock Out 1	
16 kHz Clock Out 2	
16 kHz Clock Out 3	
TC Data Out	
TD Data Out	±3V maximum

STEP 3 Verify that the dc offset voltages are within the required limits of 3.7.1.4.

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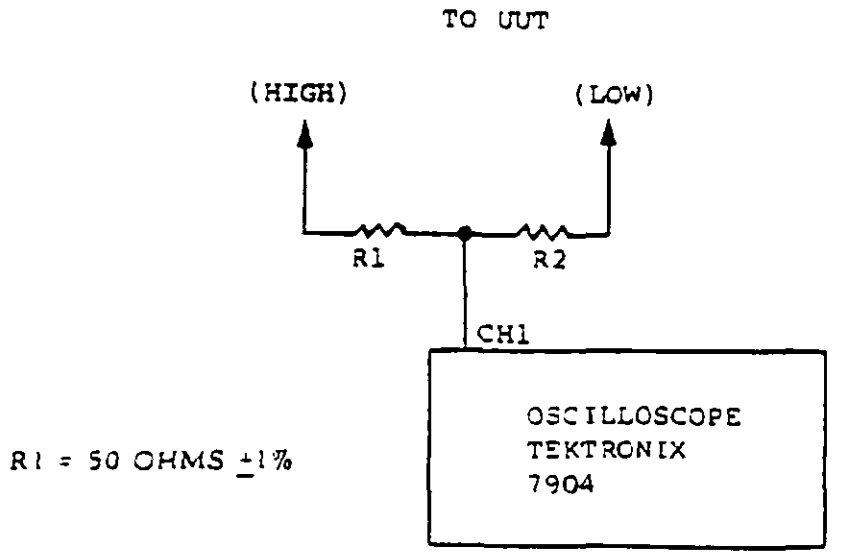


FIGURE 4. Offset.

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4.7.1.5 Signal balance. (See 3.7.1.5)

STEP 1 Compare the value of the magnitudes of the positive and negative voltage levels measured in 4.7.1.1. Determine that the larger of the two values is negative or positive. Then, compare the difference between the positive and negative to 10% of the larger value.

STEP 2 Verify that the differences are within the required limits given in 3.7.1.5.

Balance Measurement	Requirement
TEST DATA OUT	The magnitude of the difference between the magnitudes of the digital "1" level and the digital "0" level shall be less than 10% of the larger of the two.
TC DATA OUT	
TD DATA OUT	
STROBE 1 OUT	
STROBE 2 OUT	
STROBE 3 OUT	
DCM CH 1 DATA OUT	
DCM CH 2 DATA OUT	
DCM CH 3 DATA OUT	
DCM CH 4 DATA OUT	
DCM CH 5 DATA OUT	
DCM CH 6 DATA OUT	
DCM CH 7 DATA OUT	
DCM CH 8 DATA OUT	
DCM CH 9 DATA OUT	
DCM CH 10 DATA OUT	
DCM CH 11 DATA OUT	
DCM CH 12 DATA OUT	
DCM CH 13 DATA OUT	
DCM CH 14 DATA OUT	
DCM CH 1 TIM OUT	
DCM CH 2 TIM OUT	
DCM CH 3 TIM OUT	

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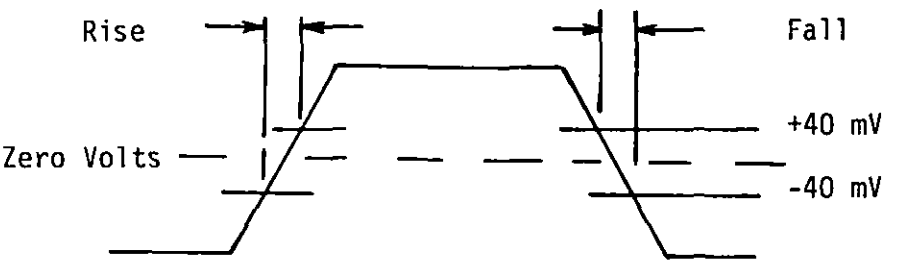
Balance Measurement	Requirement
DCM CH 4 TIM OUT	The magnitude of the difference between the magnitudes of the digital "1" level and the digital "0" level shall be less than 10% of the larger of the two.
DCM CH 5 TIM OUT	
DCM CH 6 TIM OUT	
DCM CH 7 TIM OUT	
DCM CH 8 TIM OUT	
DCM CH 9 TIM OUT	
DCM CH 10 TIM OUT	
DCM CH 11 TIM OUT	
DCM CH 12 TIM OUT	
DCM CH 13 TIM OUT	
DCM CH 14 TIM OUT	
GROUP DATA OUT	
GROUP TIM OUT	
DCM GROUP 1 DATA OUT	
DCM GROUP 1 TIM OUT	
DCM GROUP 2 DATA OUT	
DCM GROUP 2 TIM OUT	

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4.7.1.6 Slew rate. (See 3.7.1.6)

STEP 1 The equipment setup is the same as in 4.7.1.1 except change the scope connections as shown in Figure 5 and measurements shall be made through 100 feet of RG-108 cable.

STEP 2 For each of the signals given in Table XVII, measure the time between the +40mV and -40mV points around the zero voltage levels of the rise and fall slopes.



STEP 3 Verify that the slew rates are within the required limits given in 3.7.1.6.

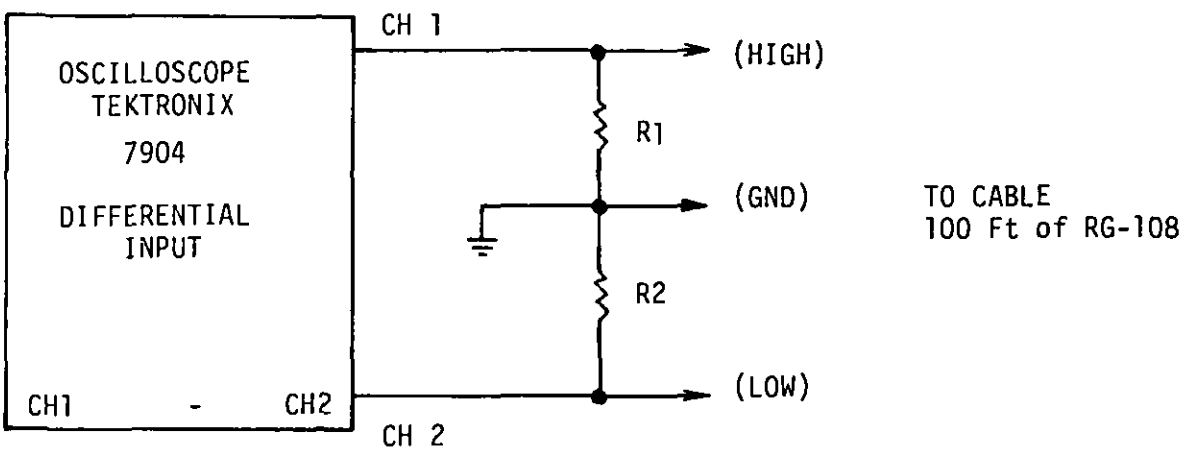


FIGURE 5. Balanced output data and timing: slew rate.

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TABLE XVII. Slew rate signals.

Measurement Slew Rate	Requirement
DVOW Data Out CH 1	100 nsec maximum
DVOW Data Out CH 2	
DVOW Data Out CH 3	
DVOW Data Out CH 4	
DVOW Data Out CH 5	
DVOW Data Out CH 6	
DVOW Data Out CH 7	
DVOW Data Out CH 8	
DVOW Data Out CH 9	
DVOW Data Out CH 10	
DVOW Data Out CH 11	
DVOW Data Out CH MG	
DVOW Data Out CH P	
DCM Data Out CH 1	
DCM Data Out CH 2	
DCM Data Out CH 3	
DCM Data Out CH 4	
DCM Data Out CH 5	
DCM Data Out CH 6	
DCM Data Out CH 7	
DCM Data Out CH 8	
DCM Data Out CH 9	
DCM Data Out CH 10	
DCM Data Out CH 11	
DCM Data Out CH 12	
DCM Data Out CH 13	
DCM Data Out CH 14	
DCM Tim Out CH 1	
DCM Tim Out CH 2	
DCM Tim Out CH 3	100 nsec maximum

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TABLE XVII. Slew rate signals. (Cont'd)

Measurement Slew Rate	Requirement
DCM Tim Out CH 4	100 nsec maximum
DCM Tim Out CH 5	
DCM Tim Out CH 6	
DCM Tim Out CH 7	
DCM Tim Out CH 8	
DCM Tim Out CH 9	
DCM Tim Out CH 10	
DCM Tim Out CH 11	
DCM Tim Out CH 12	
DCM Tim Out CH 13	
DCM Tim Out CH 14	100 nsec maximum
Group Data Out	10 nsec maximum
Group Tim Out	
DCM Data 1 Out	100 nsec maximum
DCM Data 2 Out	
DCM Tim 1 Out	
DCM Tim 2 Out	100 nsec maximum
Test Data Out	100 ns maximum
32 kHz Clock Out 1	10 ns maximum
32 kHz Clock Out 2	
32 kHz Clock Out 3	
16 kHz Clock Out 1	
16 kHz Clock Out 2	
16 kHz Clock Out 3	10 ns maximum
TC Data Out	100 ns maximum
TD Data Out	
Strobe Out 1	
Strobe Out 2	
Strobe Out 3	100 ns maximum

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4.7.1.7 Receiver sensitivity and receiver operating range. (See 3.7.1.7 and 3.7.2.2)

STEP 1 Connect the equipment as shown in Figure 6. Set the balanced attenuators (A) to give ±50mV input and the unbalanced attenuators (B) to give ±0.6 volt input. Measure the output levels using the oscilloscope.

STEP 2 Ensure that the following connections are made.

Signal	Connector	UUT Connections		
		Pins		
		Low	High	Ground
Station Clock				
16 kHz - DVOW CH 1 Tim in	J1	W	X	A
2 kHz - Buffer Clock 1	J2	B	C	A
2 kHz - TC Tim In Rcv	J7	W	X	A
2 kHz - TD 1 Tim In Rcv	J7	F	G	A
Data Out DVOW CH P	J2	u	v	A
to				
Test Data In	J2	AA	BB	A
TC Data Out	J7	U	V	A
to				
TD 1 Data In	J7	B	C	A
ETXC2 to TXCLK	Jumper	J9L to J8G		
ETCT2 to TXCPT	Jumper	J9K to J8F		
ERCT1 to RXDPT	Jumper	J9F to J8B		
ERXC1 to RXCLK	Jumper	J9E to J8C		
BER Test Set Data 1	J9	B	C	A
BER Test Set Data 2	J9	D	E	A
BER Test Set Data 3	J9	F	G	A
BER Test Set Data 4	J9	H	J	A
BER Test Set Strobe 1	J9	R	S	A
BER Test Set Strobe 2	J9	K	L	A
BER Test Set Strobe 3	J9	M	N	A

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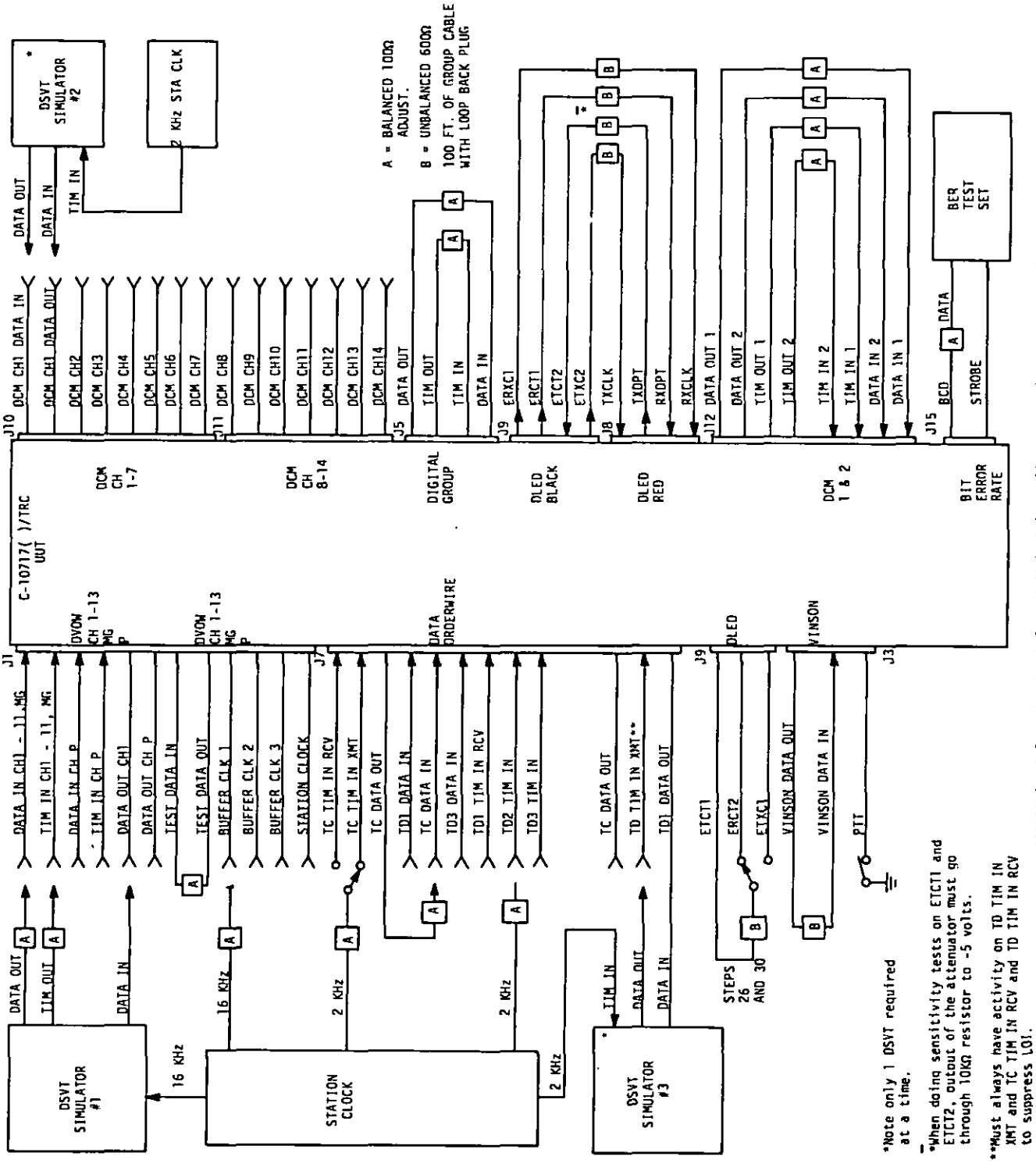


FIGURE 6. Balanced and unbalanced digital input line protection and sensitivity.

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Signal	Connector	UUT Connections		
		Pins		
		Low	High	Ground
DCM Data Out 1 to DCM Data In 1	J12 J12	F B	G C	A A
DCM Data Out 2 to DCM Data In 2	J12 J12	P K	R L	A A
DCM Tim Out 1 to DCM Tim In 1	J12 J12	H D	J E	A A
DCM Tim Out 2 to DCM Tim In 2	J12 J12	S M	T N	A A
DCM CH 1 Data Out to DCM CH 1 Data In	J10 J10	S B	T C	A A
DCM CH 2 Data Out to DCM CH 2 Data In	J10 J10	U D	V E	A A
DCM XH 3 Data Out to DCM CH 3 Data In	J10 J10	W F	X G	A A
DCM CH 4 Data Out to DCM CH 4 Data In	J10 J10	Y H	Z J	A A
DCM CH 5 Data Out to DCM CH 5 Data In	J10 J10	a K	b L	A A
DCM CH 6 Data Out to DCM CH 6 Data In	J10 J10	c M	d N	A A
DCM CH 7 Data Out to DCM CH 7 Data In	J10 J10	e P	f R	A A
DCM CH 8 Data Out to DCM CH 8 Data In	J11 J11	S B	T C	A A
DCM CH 9 Data Out to DCM CH 9 Data In	J11 J11	U D	V E	A A

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Signal	Connector	UUT Connections Pins Low	Pins High	Ground
DCM CH 10 Data Out	J11	W	X	A
to				
DCM CH 10 Data In	J11	F	G	A
DCM CH 11 Data Out	J11	Y	Z	A
to				
DCM CH 11 Data In	J11	H	J	A
DCM CH 12 Data Out	J11	a	b	A
to				
DCM CH 12 Data In	J11	K	L	A
DCM CH 13 Data Out	J11	c	d	A
to				
DCM CH 13 Data In	J11	M	N	A
DCM CH 14 Data Out	J11	e	f	A
to				
DCM CH 14 Data In	J11	P	R	A
Digital Group Data Out	J5	E	F	A
to				
Digital Group Data In	J5	A	B	A
Digital Group Tim Out	J5	G	H	A
to				
Digital Group Tim In	J5	C	D	A

STEP 3 Remove the front panel to gain access to the printed circuit boards.

STEP 4 Set the CCA switches as specified in Table XVIII.

STEP 5 Set the control panel switches as shown in Table XIX.

STEP 6 Measure the DVOW Channels 1 through 5 and MG BER as follows:

a. Set the STATION TIMING thumbwheel switch to the selected channel (MG uses Position 6).

b. Connect the DSVT Simulator data and timing input to the selected channel.

c. Monitor the DSVT Simulator for a period of 10 seconds. Verify that no errors are indicated.

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TABLE XVIII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	OFF
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	256
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	ON
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	RCV
	TLMY Decomb TMG	XMT/RCV	RCV
	TLMY Decomb 1	ON/OFF	ON
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

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TABLE XIX. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1		X						
	2		X						
	3		X						
	4		X						
	5		X						
	6		X						
	7		X						
	8		X						
	9		X						
	10		X						
	11		X						
	MG		X						
	P		X						

Switch Timing Buffer	Position			
	S	1	2	3
		As Required		

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
				As Required							

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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STEP 7 Repeat step 6 for channels 6 through 11, with the STATION TIMING thumbwheel switch set to S1. Connect timing from DSVT Simulator to Station Clock.

STEP 8 Measure the BER as in step 6. For Channels 6 through 11, verify that no errors are indicated.

STEP 9 Connect the DSVT DATA IN to the Channel 1 DATA OUT and the DSVT DATA OUT to the P Channel IN.

STEP 10 Measure the BER on the P Channel. Verify the no errors are indicated.

STEP 11 Reconnect the DSVT timing and DATA OUT to DVOW Channel 1 inputs. Set timing to Station 1.

STEP 12 Select Channel 1 by depressing the digital 1 pushbutton switch on the control montiro. Set the Channel 1 toggle switch to position (T), connect the 16 kHz output from the timing source to Buffer Timing 1 In. Set the BUFFER TIMING thumbwheel switch to 1 and ground the PTT switch. Measure the BER at Channel 1. Verify that no errors are indicated. Connect the VINSON DATA IN to VINSON DATA OUT.

STEP 13 Repeat step 12 with the timing source connected to Buffer Timing 2 and the BUFFER TIMING thumbwheel switch set to 2.

STEP 14 Repeat step 12 with the timing source connected to Buffer Timing 3 and the BUFFER TIMING thumbhwheel switch set to 3.

STEP 15 Drive the DSVT Simulator #2 at 2 kHz and connect the DSVT Simulator DATA OUT to DCM Channel 1 DATA IN and the DSVT Simulator DATA IN to DCM Channel 1 DATA OUT. Connect DSVT Simulator TIM IN to DCM CH 1 TIM OUT. Connect timing source 16 kHz to DVOW CH 1. Use Bal Atten on DCM GP 1 and 2.

STEP 16 Measure the BER of DCM Channel 1 for 80 seconds. Verify that no errors are indicated.

STEP 17 Repeat steps 15 and 16 for DCM Data Channels 2 through 14.

STEP 18 Set the BER selector on the BER test set to position 7. Verify that the Summary Status Light on the Control Monitor is amber.

STEP 19 Set the BER selector to position 8. Verify that the Summary Status Light is green.

STEP 20 Connect the TD data (from TC DATA OUT) to TD1. Connect a 2 kHz Station Clock Signal to TD TIM IN REC and TC TIM IN REC. Set the TD1 switch on CCA A12 to ON. Verify that TD OOF and LOI LED are OFF. (TCG card (1A15)).

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STEP 21 Check proper operation by inverting TD DATA IN to cause OOF.

STEP 22 Set the TC TIMING switch on the UUT to SMT and repeat step 21. Apply 2 kHz clock signal to TC TIM IN XMT.

STEP 23 Drive the DSVT Simulator #3 with 2 kHz.

STEP 24 Connect the DSVT Simulator DATA OUT to TC DATA IN and the DSVT Simulator DATA IN to TD1 DATA OUT.

STEP 25 Connect ERCT2 (OUT of OCU) to ETCT1 (IN to OCU) through a "B" attenuator. Connect output of attenuator through a 10KQ resistor to -5Vdc.

STEP 26 Monitor the DSVT Error Rate for 10 seconds.

STEP 27 Verify that there are no errors indicated.

STEP 28 Connect the DSVT Simulator DATA OUT to TD TIM IN XMT. Connect output of attenuator through 10K resistor to -5Vdc.

STEP 29 Connect ETCT1 (IN to OCU) to ETXC1 (OUT of OCU) through a "B" attenuator. Set the TD TMG switch to SMT.

STEP 30 Monitor the DSVT Error Rate for 10 seconds.

STEP 31 Verify that there are no errors indicated.

STEP 32 Check PRG operation at 2 and 16 kb/s (balanced).

STEP 33 Connect the DSVT Simulator DATA IN to DVOW CH1 DATA OUT, DATA OUT to DVOW CH 1 DATA IN, TIM OUT to DVOW CH 1 TIM IN, and TIM IN to 15 kHz clock of timing source.

STEP 34 Set all DVOW switches to TERMINATE/THROUGH.

STEP 35 Monitor the DSVT Error Rate for 10 seconds. Verify that no errors are indicated.

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4.7.1.8 Balanced and unbalanced input impedance. (See 3.7.1.8, Input load impedance and 3.7.2.4, Receiver input impedance).

STEP 1 Connect the equipment as shown in Figure 7.

STEP 2 Connect the test setup in sequence to the signal inputs listed below. Use the connectors and pin numbers indicated.

(Input Load Impedance)

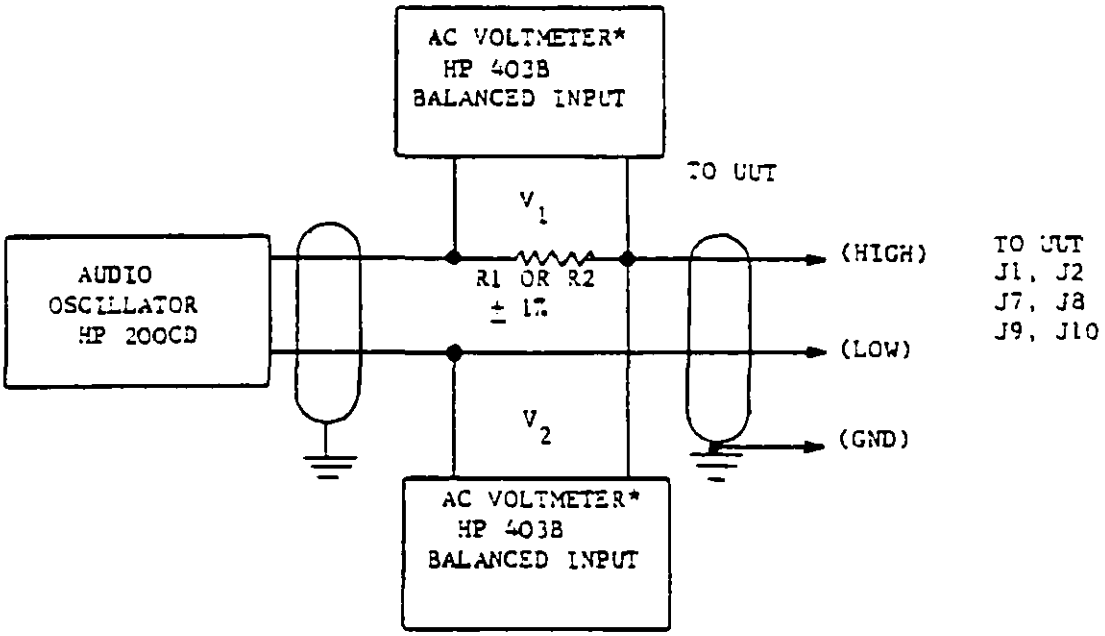
Signal	Connector	UUT Point of Measurement		
		Low	High	Ground
DVOW CH 1 Tim In	J1	W	X	A
DVOW CH 2 Tim In	J1	Y	Z	A
DVOW CH 3 Tim In	J1	a	b	A
DVOW CH 4 Tim In	J1	c	d	A
DVOW CH 5 Tim In	J1	e	f	A
DVOW CH 6 Tim In	J1	g	h	A
DVOW CH 7 Tim In	J1	i	j	A
DVOW CH 8 Tim In	J1	k	m	A
DVOW CH 9 Tim In	J1	n	p	A
DVOW CH 10 Tim In	J2	e	f	A
DVOW CH 11 Tim In	J2	g	h	A
DVOW CH MG Tim In	J2	i	j	A
DVOW CH P Tim In	J2	k	m	A
DVOW CH 1 Data In	J1	B	C	A
DVOW CH 2 Data In	J1	D	E	A
DVOW CH 3 Data In	J1	F	G	A
DVOW CH 4 Data In	J1	H	J	A
DVOW CH 5 Data In	J1	K	L	A
DVOW CH 6 Data In	J1	M	N	A
DVOW CH 7 Data In	J1	P	R	A
DVOW CH 8 Data In	J1	S	T	A
DVOW CH 9 Data In	J1	U	V	A
DVOW CH 10 Data In	J2	W	X	A
DVOW CH 11 Data In	J2	Y	Z	A
DVOW CH MG Data In	J2	a	b	A
DVOW CH P Data In	J2	c	d	A

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(Input load impedance)

Signal	Connector	UUT Point of Measurement		
		Low	High	Ground
Buffer Tim In 1	J2	B	C	A
Buffer Tim In 2	J2	D	E	A
Buffer Tim In 3	J2	F	G	A
TC Tim In Xmt	J7	Y	Z	A
TD Data In	J7	B	C	A
TD Tim In Rcv	J7	F	G	A
BER Data In Bit 1	J9	B	C	A
BER Data In Bit 2	J9	D	E	A
BER Data In Bit 3	J9	F	G	A
BER Data In Bit 4	J9	H	J	A
DCM CH 1 Data In	J10	B	C	A
DCM CH 2 Data In	J10	D	E	A
DCM CH 3 Data In	J10	F	G	A
DCM CH 4 Data In	J10	H	J	A
DCM CH 5 Data In	J10	K	L	A
DCM CH 6 Data In	J10	M	N	A
DCM CH 7 Data In	J10	P	R	A
DCM CH 8 Data In	J10	B	C	A
DCM CH 9 Data In	J10	D	E	A
DCM CH 10 Data In	J10	F	G	A
DCM CH 11 Data In	J10	H	J	A
DCM CH 12 Data In	J10	K	L	A
DCM CH 13 Data In	J10	M	N	A
DCM CH 14 Data In	J10	P	R	A
DCM Data In 1	J10	B	C	A
DCM Data In 2	J10	F	G	A
DCM Tim In 1	J10	D	E	A
DCM Tim In 2	J10	H	J	A
TD TIM IN XMT	J7	M	N	A
TC Data IN	J7	S	T	A
TC Tim IN RCV	J7	Y	Z	A
Station Clock	J2	w	x	A
DVOW Group Data IN	J5	A	B	R
DVOW Group Tim IN	J5	C	D	R

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R1 = 100 ohms ± 1%

R2 = 6000 ohms ± 1%

*Use battery powered voltmeters.

FIGURE 7. Balanced digital input impedance tests.

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STEP 3 At each signal input using R1 = 100 ohms ±1%, adjust the audio oscillator at a frequency of 1.0 kHz to produce an output voltage at V1 of 1.0 volts. Then measure the V2 voltage. Record the V2 voltage on the test data sheet.

STEP 4 Connect the test setup in sequence to the signal inputs listed below, using the connector and pin numbers indicated.

(Receiver input impedance)

Signal	Conn	Point of Measurement UUT Pin Numbers		
		High	Low	Ground
Vinson Data In	J3	B	E	A
TX Clock	J8	G	H,J	A
ETCT 1	J9	F	M,N	A
ETCT 2	J9	K	M,N	A
RX DPT	J8	B	H,J	A
RX Clock	J8	C	H,J	A

STEP 5 At each signal input using R2 = 6000 ohms ±1%, adjust the audio oscillator at a frequency of 1 kHz to produce an output voltage at V1 of 1.0 volts. Then record the V2 voltage on the test data sheet.

STEP 6 Connect the test setup in sequence to the signal inputs listed below. Use the connector and pin numbers indicated.

(Logic driver alarm input impedance)

Signal	Conn	UUT Point of Measurement Pin Numbers		
		High	Low	Ground
Logic Alarm 49	J14	W	X	A
Logic Alarm 50	J14	Y	Z	A
Logic Alarm 51	J14	AA	DD	A
Logic Alarm 52	J14	CC	FF	A
Logic Alarm 53	J14	EE	HH	A
Logic Alarm 54	J14	GG	FF	A
Logic Alarm 55	J14	BB	X	A

STEP 7 At each signal input using the R2 = 6000 ohm ±1% resistor, adjust the audio oscillator to a frequency of 10 Hz to produce an output voltage at V1 of 1.0 volts. Then measure and record the V2 voltage.

STEP 8 Calculate all the input impedances below using the formulas below:

$$R = (R1, R2)\ \frac{V2}{V1}\ \text{ohms}$$

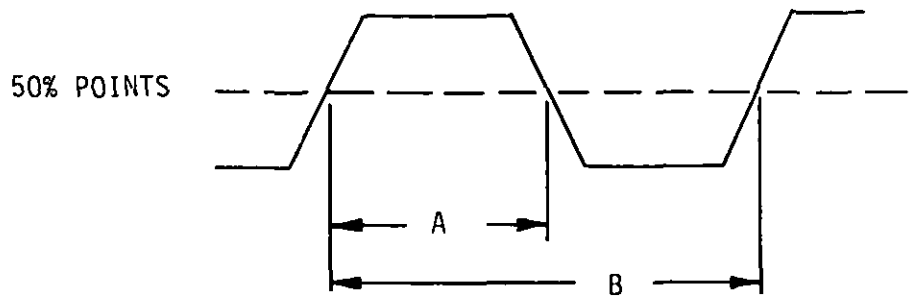
NOTE: Use R1 = 100 ohms or R2 = 6000 ohms as required.

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4.7.1.9 Clock asymmetry. (See 3.7.1.9)

STEP 1 The equipment test setup is the same as in 4.7.1.1.

STEP 2 For each of the signals given below, measure the time (A) between the rise and fall slopes at the 50% points; then, measure the time (B) between two successive rise slopes at the 50% points as indicated below.



Clockwidth Limits Measurements	Requirement
32 kHz Clock Out 1	±1500 ns
32 kHz Clock Out 2	
32 kHz Clock Out 3	
16 kHz Clock Out 1	
16 kHz Clock Out 2	
16 kHz Clock Out 3	
DCM CH 1 Tim Out	
DCM CH 2 Tim Out	
DCM CH 3 Tim Out	
DCM CH 4 Tim Out	
DCM CH 5 Tim Out	
DCM CH 6 Tim Out	
DCM CH 7 Tim Out	
DCM CH 8 Tim Out	
DCM CH 9 Tim Out	
DCM CH 10 Tim Out	
DCM CH 11 Tim Out	
DCM CH 12 Tim Out	
DCM CH 13 Tim Out	
DCM CH 14 Tim Out	
DCM Grp 1 Tim Out	
DCM Grp 2 Tim Out	
DVOW Group Tim Out	±120 ns

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4.7.2 Unbalanced dedicated loop encryption device (DLED) interface. (See 3.7.2)

4.7.2.1 Driver output voltage. (See 3.7.2.1)

STEP 1 Connect the equipment as shown in Figure 8. Ensure that the following connections are made.

Signal	Conn	UUT Connections Pin Numbers		
		High	Low	Ground
DVOW Data In CH 1	J1	B	C	A
DVOW Tim In CH 1	J1	W	X	A
TC Tim In Xmt	J7	Y	Z	A
TD Tim In Xmt	J7	M	N	A
TC Data In	J7	S	T	A
TC Tim In Rcv	J7	W	X	A
TD Data In	J7	B	C	A
TD Tim In Rcv	J7	F	G	A
ETXC2	J9	L		
to TXCLK (S1)	J8	G		

STEP 2 Set the UUT circuit board switches and the control panel switches as shown in Tables XX and XXI. Push the DVOW Channel 1 pushbutton on the control panel.

STEP 3 For each of the signals listed below, measure and record the loaded and unloaded voltage levels both positive and negative of each signal.

NOTE: Close S1 when measuring signal TXDPT and open S1 for all other measurements. (Put in jumper TXCLK to ETXC1).

Signal	Conn	UUT Point of Measurement	
		Pin	Ground
Vinson Data Out	J3	C	A
ERCT2	J9	K	A
DRCT1	J9	D	A
ERXC2	J9	J	A
ERXC1	J9	E	A
ETXC1	J9	G	A
ETXC2	J9	L	A
SYNC CMD TX	J8	D	A
TXDPT	J8	F	A

Measurement Impedance	Requirement
Vinson Data Out ERCT2 ERXC2 ERCT1 ETXC1 ETXC2 SYNC CMD TX TXDPT	Positive voltage levels +5 to +7V. Negative voltage levels -5 to -7V.

STEP 4 Verify that the requirements of 3.7.2.1 are met.

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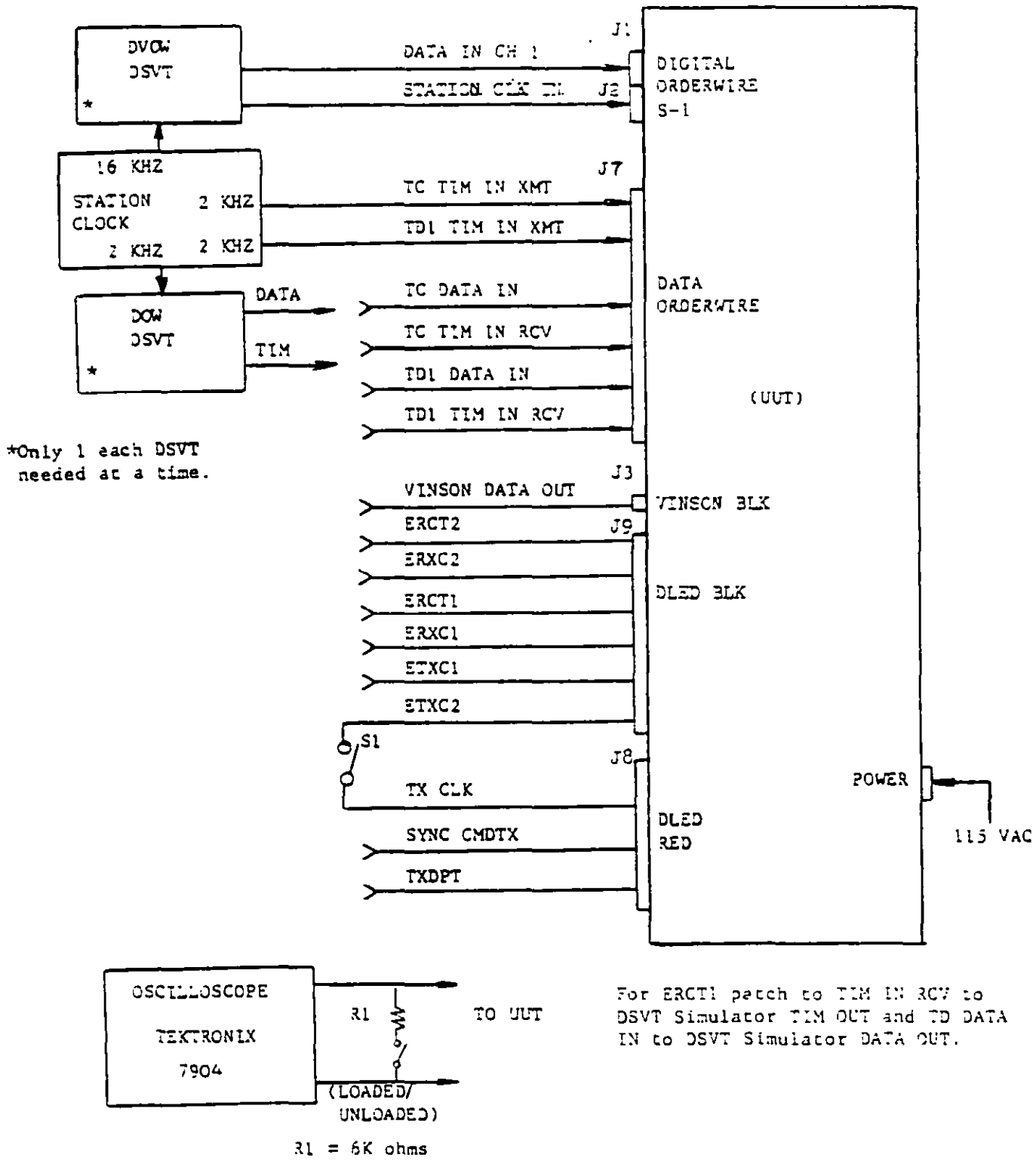


FIGURE 8. Driver output voltage and transmitter output impedance.

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TABLE XX. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s / 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	256
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	ON/OFF for ERCI-ETXC
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4* Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8* Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4* Codes 0 thru 9	0
	Channel Select S2	5 thru 8* Codes 1 thru 13	1
	BER Threshold S3	5 thru 8* Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16*	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

TD-OOF ON when TLMY Decombiner is on.

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TABLE XXI. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1	X							
	2	X							
	3	X							
	4	X							
	5	X							
	6	X							
	7	X							
	8	X							
	9	X							
	10	X							
	11	X							
	MG	X							
	P	X							

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
											X

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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4.7.2.2 Receiver operating range. (See 3.7.2.2) This test run as part of the receiver sensitivity tests in 4.7.1.7.

4.7.2.3 Transmitter output impedance. (See 3.7.2.3)

STEP 1 Using a digital voltmeter, measure the rms voltage into a 6000 ohm load on the outputs of VINSON DATA OUT, ERCT2, ERXC2, ERCT1, ERXC1, ETXC1, ETXC2.

STEP 2 Using an oscilloscope, measure the peak voltage into a 6000 ohm load on the output of SYNC CMD TX.

STEP 3 Using an oscilloscope, measure the peak voltage into a 6000 ohm load on the output of TXDPT.

STEP 4 Repeat steps 1, 2 and 3 using a 600 ohm load.

STEP 5 Calculate and record the output impedance using the following formula:

$$Z = \frac{6000\ (V_{6K} - V_{600})}{(10V_{600} - V_{6K})}$$

STEP 6 Verify that the requirements of 3.7.2.3 are met.

4.7.2.4 Receiver input impedance. (See 3.7.2.4) This test run as part of the Balanced and Unbalanced Input Impedance Tests of 4.7.1.8.

4.7.2.5 Rise and fall times. (See 3.7.2.5)

STEP 1 The equipment setup is the same as in 4.7.2.1.

STEP 2 The UUT circuit board switches and control panel switches are the same as shown in Tables XX and XXI.

STEP 3 For each of the signals listed below, measure the loaded rise and fall times at the 10% to 90% points.

NOTE: Close S1 when measuring signal TXDTP and open S1 for all other readings.

Signal	Conn	UUT Point of Measurement	
		Pin	Ground
Vinson Data Out	J3	C	A
ERCT2	J9	H	A
ERXC2	J9	J	A
ERCT1	J9	D	A
ERXC1	J9	E	A
ETXC1	J9	G	A
ETXC2	J9	L	A
SYNC CMD TX	J8	D	A
TXDPT	J8	F	A

NOTE: For DVOW tests jumper TC TIM RCV to TC TIM XMT-TD TIM RCV to TD TIM XMT to remove alarms.

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STEP 4 Verify that the rise and fall times are within the required limits.

Measurement Rise and Fall Times	Requirement
Vinson Data Out	3 to 15 μsec
ERCT2	3 μsec max
ERXC2	
ERCT1	
ERXC1	
ETXC1	
ETXC2	
SYNC CMD TX	
TXDPT	

4.7.2.6 Data/timing. This test demonstrates the data/timing phasing, timing asymmetry and the sync signal requirements of 3.7.2.6, 3.7.2.7 and 3.7.2.5. The asymmetry shall be computed by substracting the measured input asymmetry from the measured output asymmetry. The spec level is the permitted output asymmetry minus the permitted input asymmetry; 62.5 μs - 1.5 μs = 61 μs maximum.

STEP 1 The equipment setup, UUT switch positions and control panel settings are the same as in 4.7.2.1 (Tables XX and XXI).

STEP 2 Measure the loaded (100 ohms) positive half of the 2 kHz clock cycle of the DSVT Simulator timing output (input asymmetry). Record the results in column B.

STEP 3 Measure the loaded (6K ohms) positive half of the clock cycle of the following signals and record the results in column A. Connect the DSVT timing input to the UUT input association with each output.

Signal	Conn	UUT Point of Measurement		UUT DSVT Input	
		Pin	Ground	Signal	Pin No.
ETXC2	J9	M,N		TC Tim In Xmt	J7 YZ, A
ERXC2	J9	M,N		TC Tim In Rcv	J7 WX, A
ETXC1	J9	M,N		TD Tim In Xmt	J7 MN, A
ERXC1	J9	M,N		TD Tim In Rcv	J7 FG, A

Signal	A Pos Half of Clock Cycle	B Pos Half of Input Timing	A-B	Requirement
ETXC2				±62-1/2 μs
ERXC2				
ETXC1				
ERXC1				

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STEP 4 Compute the asymmetry and verify that the asymmetry is within the requirement.

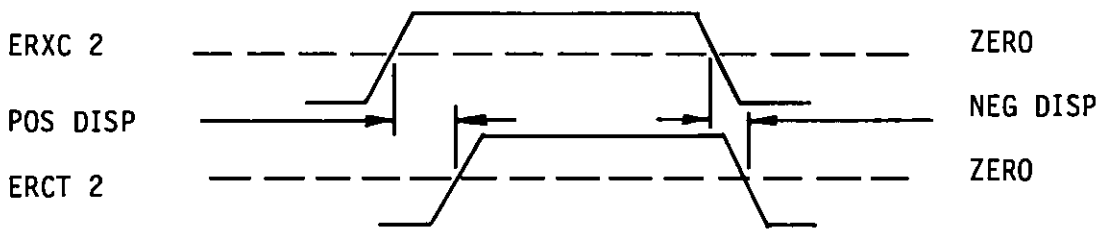
STEP 5 Connect the 2 kHz timing output from the DSVT Simulator simultaneously to the TC DATA IN and the TC TIM IN RCV inputs.

STEP 6 Measure the displacement between the positive zero crossing and between the negative zero crossing of ERXC2 and ERCT2 as shown.

STEP 7 Record the results.

STEP 8 Repeat steps 5 through7 using TD DATA IN and TD TIMING IN RCV (J5)

Measurement	Requirement
$TC_{ERXC2/ERCT2}$	Displacement shall not exceed 244.5 µs.
$TD_{ERXC2/ERCT2}$	



(ASYMMETRY)

STEP 9 Verify that the difference between the positive and negative zero displacements are within the required limits.

STEP 10 Measure the phasing between TX CLOCK and TXDPT.

STEP 11 Verify that the falling edgel of the TX CLOCK is at least 2.5 µs from the nearest data transition at the 10% or 90% points.

Measurement Phasing	Requirement
TXCLK to TXDPT	2.5 µsec minimum

STEP 12 Measurement the pulse width of the SYNC CMD TX signal. Verify that the pulse width is within the required limits.

Measurement Phasing Width	Requirement
SYNC CMD TX	30 to 200 µsec

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4.7.3 Unbalanced VINSON interface. (See 3.7.3) The unbalanced VINSON data requirements are demonstrated as part of the Unbalanced Dedicated Loop Encryption Device (DLED) Interface tests of 4.7.2.

4.7.3.1 Driver output voltage. See 4.7.2.1.

4.7.3.2 Receiver operating range. See 4.7.2.2.

4.7.3.3 Rise and fall times. See 4.7.2.5.

4.7.4 Analog voice orderwire (AVOW) channel interface tests. (See 3.7.4) The following tests demonstrate the input/output test tone levels, frequency response, signaling frequency, signal detector operating range and signal generator output requirements of 3.7.4.2, 3.7.4.3, 3.7.4.4, 3.7.4.5, 3.7.4.6 and 3.7.4.7. The input impedance of 3.7.4.1 is demonstrated in 4.7.4.1.

STEP 1 Connect the equipment as shown in Figure 9.

STEP 2 Ensure that the following connections are made.

Signal	Connector	UUT Point of Measurement		
		High	Low	Ground
AVOW CH1 OUT	J6	P	R	A,C
AVOW CH2 OUT	J6	S	T	A,C
AVOW CH3 OUT	J6	U	V	A,C
AVOW CH4 OUT	J6	W	X	A,C
AVOW CH5 OUT	J6	V	Z	A,C
AVOW CH6 OUT	J6	a	b	A,C

STEP 3 Set the control panel AVOW toggle switches (input and output channels) to the bridge position. Then push the AVOW B switch.

STEP 4 Set the oscillator frequency to 1 kHz and its output level (600 ohms termination) to -12.2 dBm. (This level will produce -18.2 dBm into a 600 ohm load.)

STEP 5 For each of the signals listed, measure and record the loaded output voltage in dBm. Use the connector and pin numbers specified.

NOTE: Measure AVOW Channels 2 through 6 with the input connected to Channel 1 and then measure Channel 1 with the input connected to Channel 2.

Item	Connector	Pins
OSC IN - CH1 OW IN	J6	B C
CH2 OW IN	J6	D E

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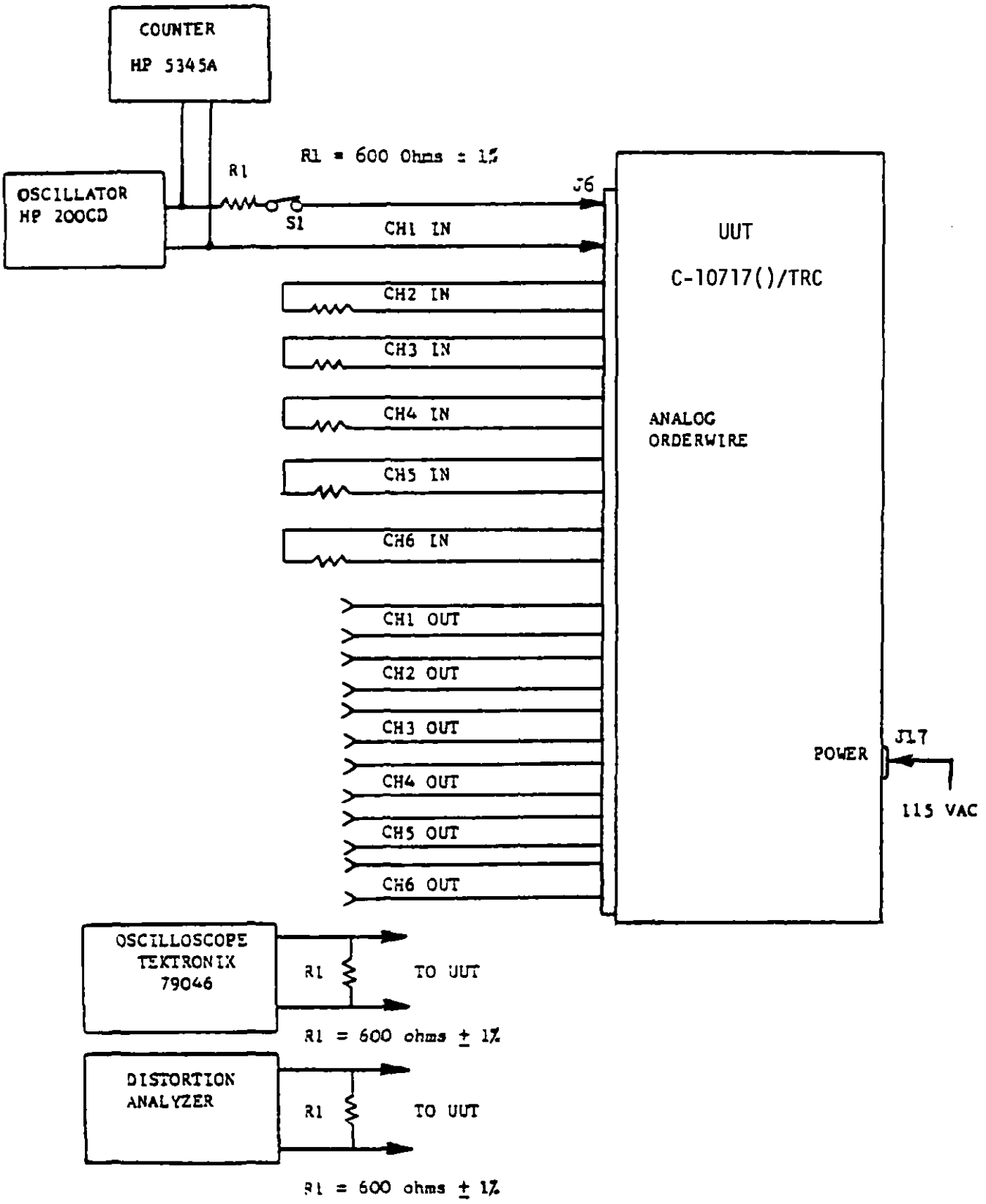


FIGURE 9. Analog voice orderwire (AVOW) channel interface test.

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STEP 6 Verify that the output test tone level is within the required limits.

Measurement Power	Requirement
AVOW CH1 OUT	+20.3 to 22.3 dBm
AVOW CH2 OUT	
AVOW CH3 OUT	
AVOW CH4 OUT	
AVOW CH5 OUT	
AVOW CH6 OUT	

STEP 7 Push the analog Channel 1 operator control switch on the control panel.

STEP 8 Push the RING button and measure and record the AVOW CH1 output voltage level. Repeat this procedure for Channels 2 through 6.

Measurement	Requirement
AVOW CH1 Voltage	+21.3 dBm ±1 dB
AVOW CH2 Voltage	+21.3 dBm ±1 dB
AVOW CH3 Voltage	+21.3 dBm ±1 dB
AVOW CH4 Voltage	+21.3 dBm ±1 dB
AVOW CH5 Voltage	+21.3 dBm ±1 dB
AVOW CH6 Voltage	+21.3 dBm ±1 dB
AVOW CH1 Frequency	1568 Hz to 1632 Hz

STEP 9 Measure the ring frequency. Connect the counter to pin 24 on the SIG A, CCA A2 assembly, for continuous ring frequency.

STEP 10 Verify that the above requirements are met.

STEP 11 For each of the signals listed below, measure and record the frequency response of each AVOW channel.

NOTE: Measure AVOW Channels 2 through 6 with the oscillator set to -12.2 dBm connected to Channel 1, then measure Channel 1 with the oscillator connected to Channel 2. This level will produce -18.2 dBm into the 600 ohms.

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Channel	Frequency				
	300 Hz	400 Hz	Ref 1000 Hz	3000 Hz	3400 Hz
AVOW CH1					
AVOW CH2					
AVOW CH3					
AVOW CH4					
AVOW CH5					
AVOW CH6					

STEP 12 Use HP 3555B transmission and noise measuring set. Compare the outputs to the value measured at 1 kHz. Verify that the frequency response is within the required limits.

Measurement Frequency Response	Requirement
AVOW CH1	±.5 dB max from 1000 Hz reference
AVOW CH2	
AVOW CH3	
AVOW CH4	
AVOW CH5	
AVOW CH6	

STEP 13 Set the oscillator frequency to 1568 Hz using enough output to operate the counter.

STEP 14 Set the oscillator output level to -23 dBm. (This will produce -29 dBm into a 600 ohm load). Disconnect the input to AVOW CH1 IN, reset the CH1 indicator by pushing the CH1 selector switch.

STEP 15 Monitor the Channel 1 pushbutton selector switch while connecting the input to AVOW CH1 IN, the CH1 selector shall flash within 2 seconds of connection.

STEP 16 Repeat step 15 for Channels 2 through 6.

STEP 17 Set the oscillator level to -9 dBm (this will produce -15 dBm into a 600 ohm load) at a frequency of 1632 Hz.

STEP 18 Repeat steps 15 and using Channels 1 through 6.

Measurement Signaling Sensitivity			Requirement
Input at 1568 Hz -29 dBm	CH1	CH4	CH flashes within 2 seconds of application of signal.
	CH2	CH5	
	CH3	CH6	
Input at 1643 Hz -15 dBm	CH1	CH4	
	CH2	CH5	
	CH3	CH6	

STEP 19 Verify that the requirements of 3.7.4.2, 3.7.4.3, 3.7.4.4, 3.7.4.5, 3.7.4.6 and 3.7.4.7 are met.

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4.7.4.1 Input and output impedance. (See 3.7.4.1)

STEP 1 Connect the equipment as shown in Figure 10.

STEP 2 Set the control panel AVOW toggle switches 1 through 6 to terminate.

STEP 3 With the oscillator connected in, turn to the signals listed below at a frequency of 1 kHz with an output level of 1.0 volt as measured on the V1 voltimeter. Measure and record the V2 voltage.

STEP 4 Repeat step 3 at 300 Hz and 3400 Hz

Signal	Conn	UUT Point of Measurement Pin No. High	Low	Ground
CH 1 AVOW IN	J6	B	C	A
CH 2 AVOW IN	J6	D	E	A
CH 3 AVOW IN	J6	F	G	A
CH 4 AVOW IN	J6	H	J	A
CH 5 AVOW IN	J6	K	L	A
CH 6 AVOW IN	J6	M	N	A
CH 1 AVOW OUT	J6	P	R	A
CH 2 AVOW OUT	J6	S	T	A
CH 3 AVOW OUT	J6	U	V	A
CH 4 AVOW OUT	J6	W	X	A
CH 5 AVOW OUT	J6	Y	Z	A
CH 6 AVOW OUT	J6	a	b	A

STEP 5 Calculate the input and output impedance using the formula:

$$Z = 600 \frac{V_2}{V_1} \text{ ohms}$$

STEP 6 Verify that the requirements of 3.7.4.1 are met.

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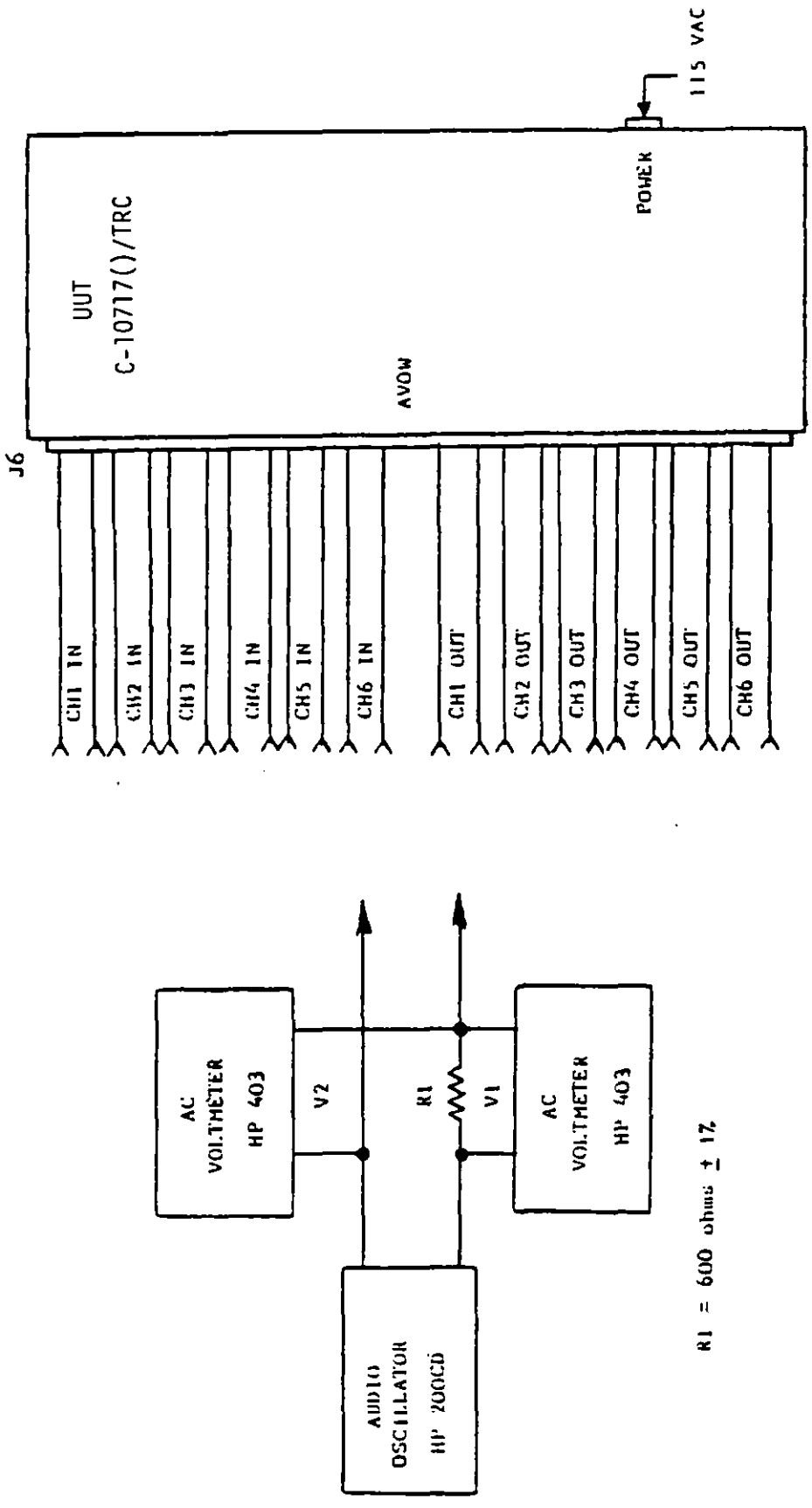


FIGURE 10. AVOW input and output impedance.

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4.7.5 AVOW headset interface. (See 3.7.5)

4.7.5.1 Output. (See 3.7.5.1)

STEP 1 Connect the equipment as shown in Figure 11.

STEP 2 Ensure that the following connections are made.

Item	UUT Connector and Pin Numbers
Oscillator In	J6 B, C CH1 OW In
RMS Voltmeter	J16 A, B Headset RCV

STEP 3 Set the oscillator output voltage to -12.2 dBm (this will produce -18.2 dBm into 600 ohms) at a frequency of 1 kHz.

STEP 4 Push the analog Channel 1 operator control switch on the control panel.

STEP 5 Measure and record the output at J16 (headset out).

Measurement	Requirement
Headset Out (Headset Rx)	0.614 to 0.867 volts into 150 ohms

STEP 6 Verify that the headset output is within the required limits.

4.7.5.2 Input impedance. (See 3.7.5.2)

STEP 1 Connect the equipment as shown in Figure 12.

STEP 2 Ensure that the follwoing connections are made.

Item	UUT Connections
Oscillator	Headset XMT Pins C, D

STEP 3 Adjust the oscillator at a frequency of 1000 Hz to produce a V1 voltage level of 1.0 volt, then measure and record the V2 voltage. Push ANALOG "B" push-button switch.

STEP 4 Calculate the impedance using the formula:

$$Z = 600 \frac{V_2}{V_1} \text{ ohms}$$

STEP 5 Verify that the input impedance is within the required limits.

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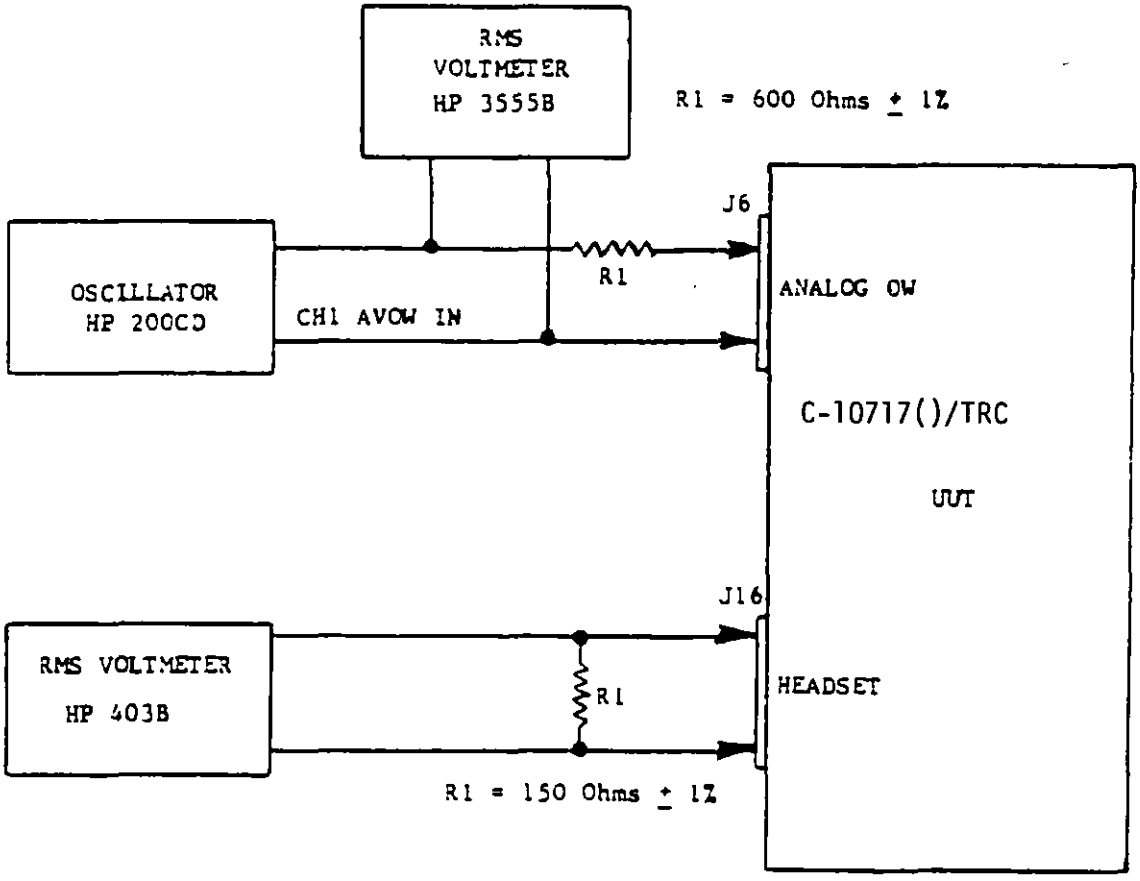


FIGURE 11. AVOW headset output test.

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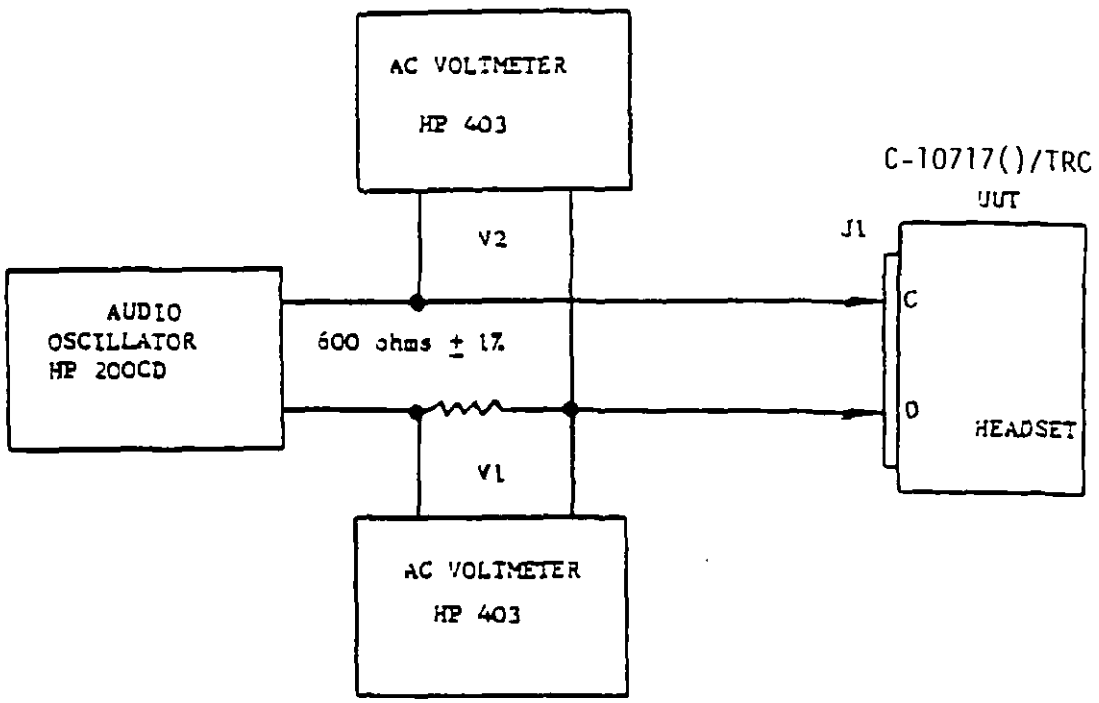


FIGURE 12. AVOW headset input impedance.

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4.7.5.3 Input level. (See 3.7.5.3)

STEP 1 Connect the equipment as shown in Figure 13.

STEP 2 Ensure that the following connections are made.

Item	UUT Connector and Pin Nos.
Oscillator	J16 P, R Headset Xmt
RMS Voltmeter	J6 P, R CH1 AVOW OUT

STEP 3 Set the oscillator output voltage to 150 mV rms, at 1 kHz.

STEP 4 Measure and record the output at AVOW CH1.

Measurement	Requirement
AVOW CH1 OUT	+20.3 to +22.3 dBm

STEP 5 Verify that the AVOW CH1 output is within the required limits.

4.7.5.4 Frequency response. (See 3.7.5.4)

STEP 1 With the oscillator set to 150 mV rms at the frequencies listed below, measure and record the output at AVOW CH1.

Measurement Frequency Response	Requirement
300 Hz	±1.5 dB
400 Hz	±1.5 dB
1000 Hz (Ref)	-
3000 Hz	±1.5 dB
3400 Hz	±1.5 dB

STEP 2 Verify that the frequency response is within ±15 dB maximum from the reference value at 1000 Hz.

4.7.6 Alarm interface. (See 3.7.6)

4.7.6.1 Relay contact alarms. (See 3.7.6.1)

STEP 1 Connect the equipment as shown in Figure 14.

STEP 2 Apply and remove one at a time each of the alarms given in Table XXII.

STEP 3 Verify that as each alarm is applied (contact closure), the corresponding error message and status is observed on the UUT and the serial data analyzer.

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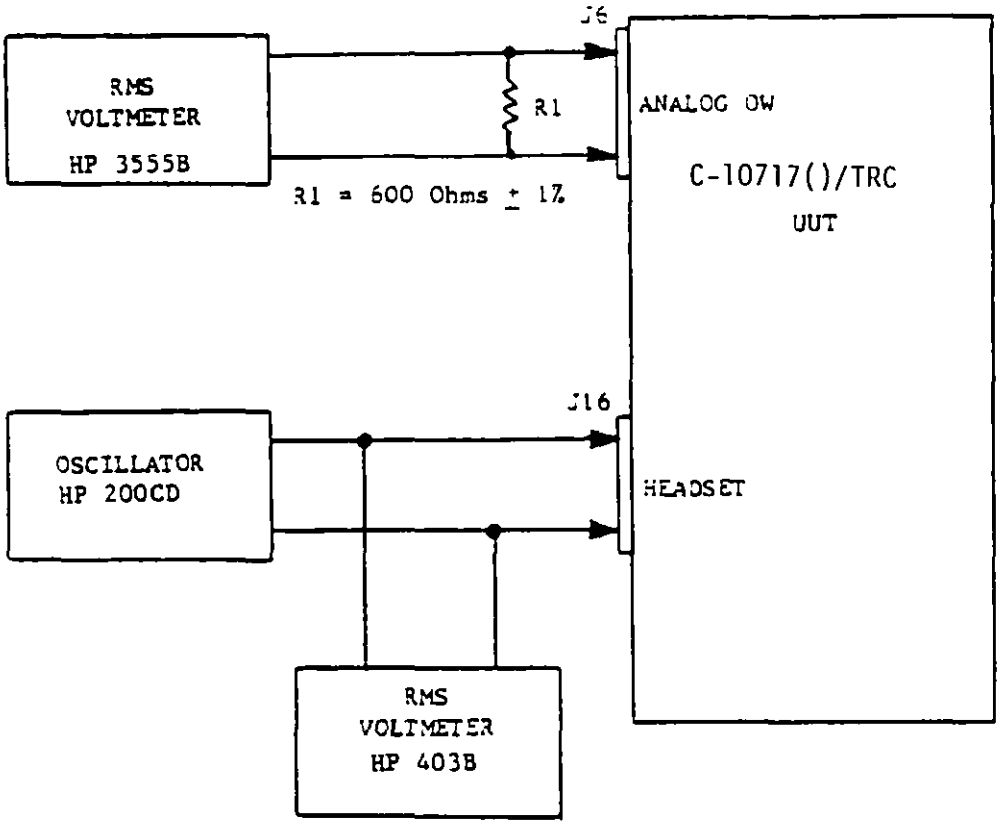


FIGURE 13. Input level/frequency.

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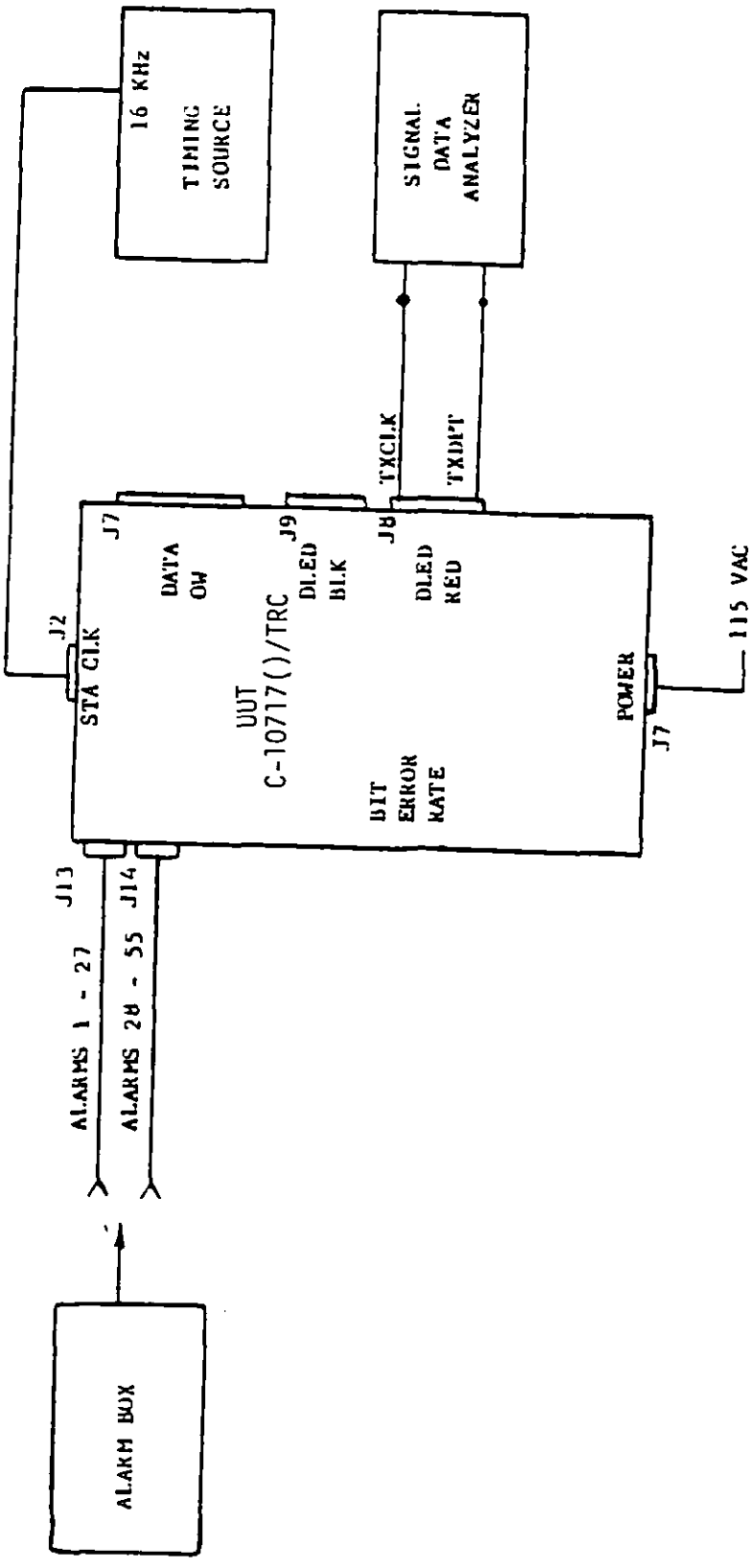


FIGURE 14. Alarm interface.

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TABLE XXII. Relay contact alarms.

Measurement Alarm No.	Requirement		
	Character 7-12 Error Message	Status Red	Character 5 Goes to E
1	RCVSA1		
2	RCVSA2		
3	RCVSA3		
4	RCVRS1		
5	RCVRS2		
6	RCVRS3		
7	MGMSA1		
8	MGMSA2		
9	MGMFS1		
10	MGMFS2		
11	GMSA1		
12	GMSA2		
13	GMSA3		
14	GMSA4		
15	GMLS1		
16	GMLS2		
17	GMLS3		
18	GMLS4		
19	HCDSA1		
20	HCDSA2		
21	HCDLS1		
22	HCDLS2		
23	LCDSA1		
24	LCDSA2		
25	LCDSA3		
26	LCDLS1		
27	LCDLS2		
28	LCDLS3		
29	TGM1		
30	TGM2		

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TABLE XXII. Relay contact alarms. (Cont'd)

Measurement Alarm No.	Requirement		
	Character 7-12 Error Message	Status Red	Character 5 Goes to E
31	RMC1		
32	RMC2		
33	RMC3		
34	RMC4		
35	CNVSA1		
36	CNVSA2		
37	RCDSA1		
38	RCDSA2		
39	RCDSA3		
40	RCDSA4		
41	RCDLS1		
42	RCDLS2		
43	RCDLS3		
44	RCDLS4		
45	XMTR1		
46	XMTR2		
47	XMTR3		
48*	OCU		

* To initiate Alarm 48, push fault lamp on power supply.

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4.7.6.2 Logic driver alarms. (See 3.7.6.2)

STEP 1 The equipment setup is the same as in 4.7.6.1.

STEP 2 Apply +2 volts to Alarm 49. The control panel Summary Status shall indicated red.

STEP 3 Apply +6 volts to Alarm 49. The control panel Summary Status shall remain red.

STEP 4 Apply +0.8 volt to Alarm 49. The control panel Summary Status shall indicate green.

STEP 5 Apply -6 volts to Alarm 49. The control panel Summary Status shall remain green.

STEP 6 Repeat steps 2 through 5 for Alarms 50 through 55. Use the connectors and pin numbers indicated in Table XXIII.

TABLE XXIII. Logic driver alarms.

Alarm No.	Conn	UUT Pin Connections	
		Pin No.	Ground
49	J14	W	DD
50		Y	DD
51		AA	DD
52		CC	DD
53		EE	DD
54		GG	DD
55		BB	DD

STEP 7 Using the test setup shown in Figure 15, connect it in turn to each signal input Alarms 45 through 55 and adjust the audio oscillator to a frequency of 10 Hz to produce an output voltage at V1 of 1.0 volt. Then measure and record the V2 voltage.

STEP 8 Calculate all the input impedances using the formulas below:

$$R = R\frac{V2}{V1} \text{ ohms} \qquad R = \frac{V2}{V1} \times 6000 \text{ ohms}$$

STEP 9 Verify that the input impedance is no less than 6 kilohms.

STEP 10 Verify that the requirements of 3.7.6.2 are met.

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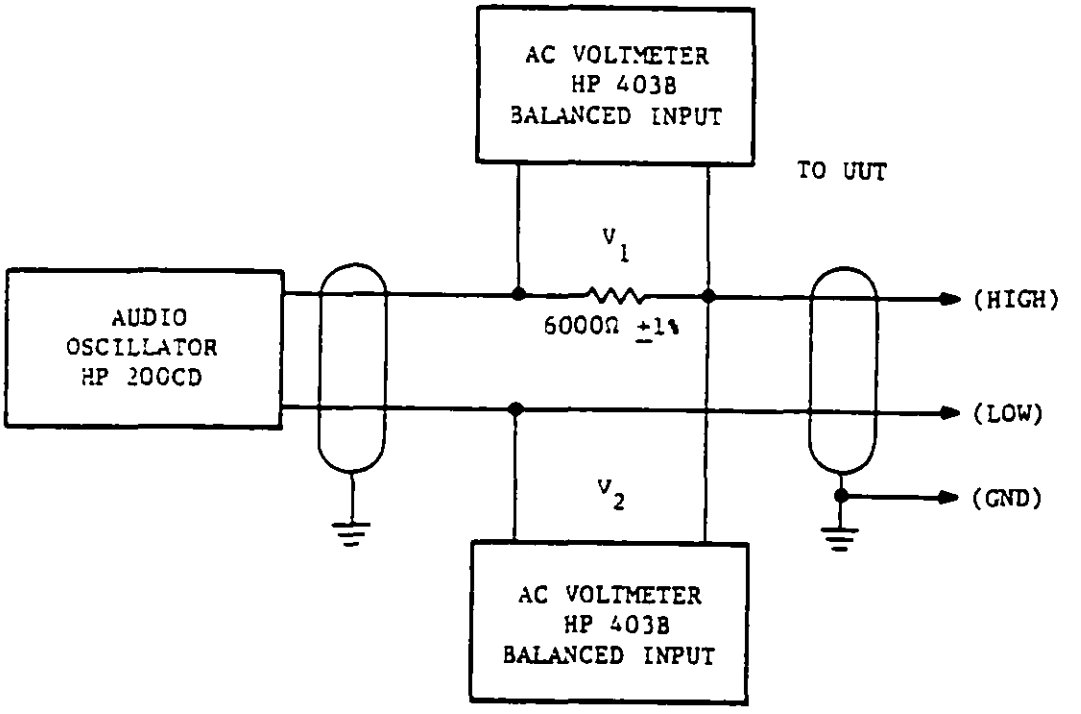


FIGURE 15. Logic alarm impedance test.

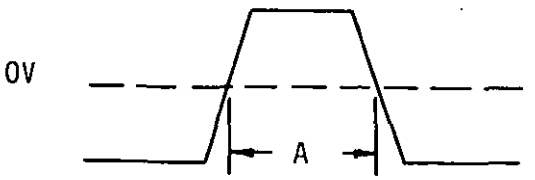
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4.7.7 Bit error rate interface. (See 3.7.7) The signal characteristics of the strobe data lines, including the requirements of 3.7.7.2, slew time and 3.7.7.3, strobe voltages are tests as part of the balanced equipment-side characteristics tests of 4.7.1.

4.7.7.1 Strobe pulse width. (See 3.7.7.1)

STEP 1 The equipment test setup is the same as in 4.7.1.1.

STEP 2 Measure each strobe pulse width (A) at zero output as indicated below.



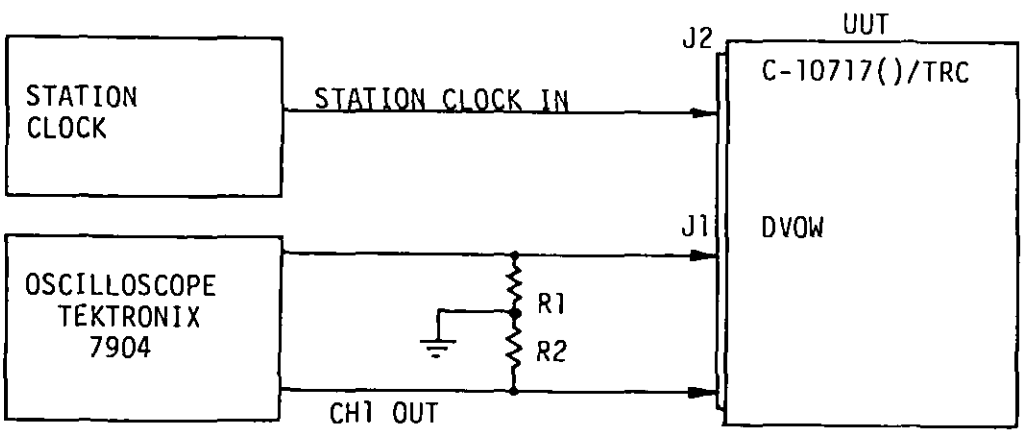
STEP 3 Verify that the strobe pulse widths are within the requirements specified.

Measurement	Requirement
Strobe Out 1	25 to 70 μs
Strobe Out 2	
Strobe Out 3	

4.7.8 Digital voice orderwire (DVOW) performance. (See 3.7.8)

4.7.8.1 Signaling code generation. (See 3.7.8.1)

STEP 1 Connect the equipment as shown in Figure 16.



R1 & R2 = 50 ohms ±1%

FIGURE 16. DVOW signaling code generation.

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STEP 2 Ensure that the following connections are made.

Signal	Conn	UUT Pin Numbers
STATION CLOCK	J2	STATION CLOCK IN w x Ground A
OSCILLOSCOPE	J1	CH1 DATA OUT q r Ground A

STEP 3 Set the UUT and control panel switches as shown in Tables XXIV and XXV.

STEP 4 Set the RING CODE switch on the control panel to position 1.

STEP 5 Push the Channel 1 digital pushbutton, then push the RING button.

STEP 6 Verify that the ring code (1) is as specified.

Measurement Ring Code			
Dial Position	Party	Code	(✓) Indicates Compliance
1	1	10000001	
2	2	10000010	
3	3	10000100	
4	4	10001000	
5	5	10000111	
6	6	10001110	
7	7	10011100	
8	8	10111000	
9	9	10110001	
10	10	10101001	
11	11	10110100	
12	12	10110010	
13	13	10011111	
14	14	10101111	
15	15	10110111	
16	16	10111011	

STEP 7 Repeat steps 4 through 6 for each remaining ring code.

STEP 8 Using the oscilloscope, measure the time between the first and last bits of Code 16, verify that the ring code duration is within the required limits.

Signal Duration	Requirement
Ring Code 16	2 - 3 seconds

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TABLE XXIV. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	ON
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Code A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring S1	Codes 1 thru 16 *	1
	Codes S2		1
	S3		1

*See Front Cover for code switch positions.

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TABLE XXV. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1	X							
	2	X							
	3	X							
	4	X							
	5	X							
	6	X							
	7	X							
	8	X							
	9	X							
	10	X							
	11	X							
	MG	X							
	P	X							

Switch	Position			
	S	1	2	3
Timing Buffer	X			

Switch	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
Timing Station											X

Switch	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Digital Ring Code							As Required									

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4.7.8.2 Signaling detection. (See 3.7.8.2)

STEP 1 Connect the equipment as shown in Figure 17.

STEP 2 Ensure that the following connections are made.

Signal	Conn	UUT Connections
STATION CLOCK IN	J2	w x A Ground
DVOW CH 1 Tim In	J1	W X A Ground
DVOW CH 1 Data Out	J1	q r A Ground
to		
DVOW CH 1 Data In	J1	B C A Ground
DVOW CH 2 Data In	J1	D E A Ground
DVOW CH 3 Data In	J1	F G A Ground
DVOW CH 4 Data In	J1	H J A Ground
DVOW CH 5 Data In	J1	K L A Ground
DVOW CH 6 Data In	J1	M N A Ground
DVOW CH 7 Data In	J1	P R A Ground
DVOW CH 8 Data In	J1	S T A Ground
DVOW CH 9 Data In	J1	U V A Ground
DVOW CH 10 Data In	J2	W X A Ground
DVOW CH 11 Data In	J2	Y Z A Ground
DVOW CH MG Data In	J2	a b A Ground
DVOW CH P Data In	J2	c d A Ground

STEP 3 Set the UUT switch positions and control panel switches as shown in Tables XXVI and XXV.

STEP 4 Set all receive ring codes (S1, S2 and S3 on CCA A8) to code position 1.

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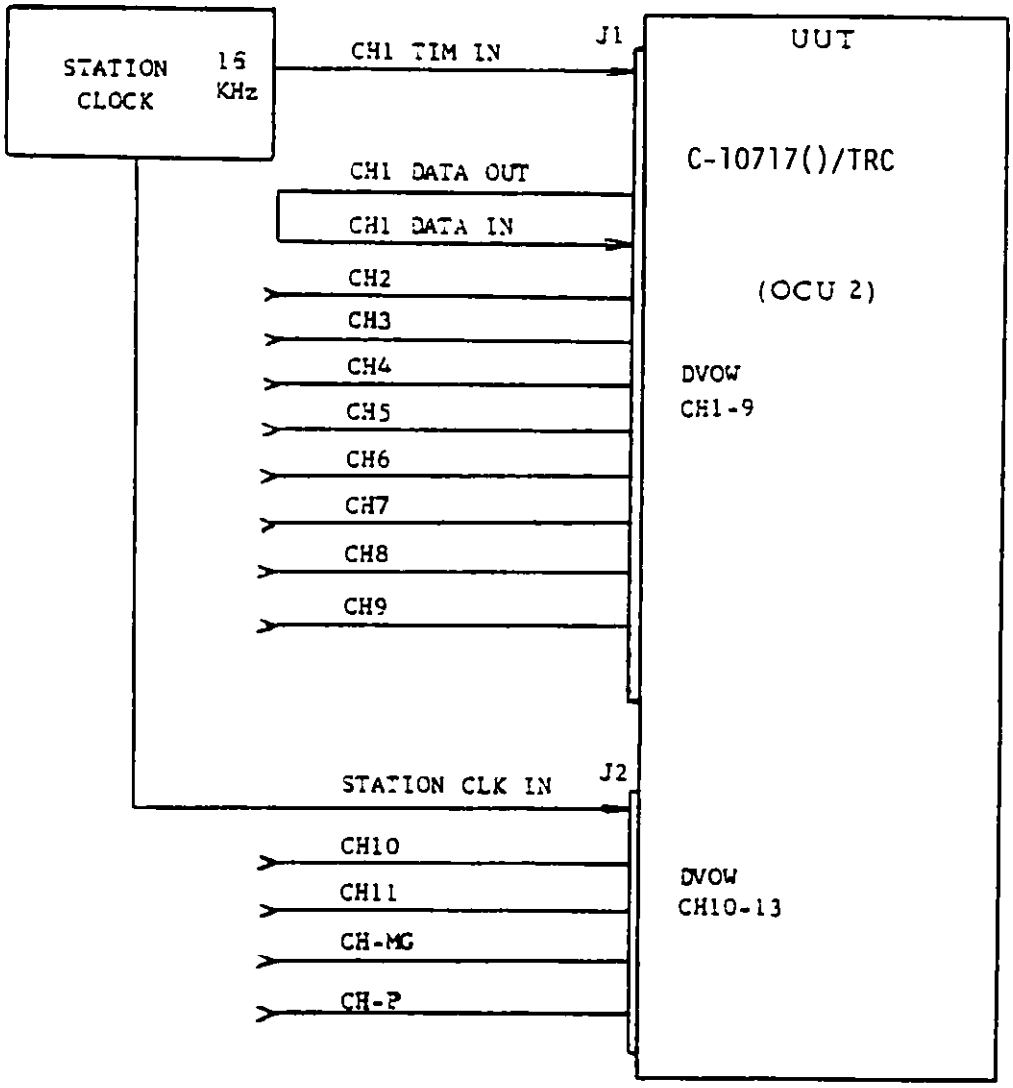


FIGURE 17. DVOW signaling code detection.

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TABLE XXVI. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	As Required
	S2		As Required
	S3		As Required

*See Front Cover for code switch positions.

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STEP 5 Connect DVOW CH1 OUTPUT to DVOW CH1 INPUT and 16 KCLK to CH1 TIM IN. Push CH1 button. Push the RING button on the control panel. Verify that only Channel 1 flashes. Then push the DVOW CH1 pushbutton twice.

STEP 6 Repeat step 5 substituting CH2 through P for CH1 (one at a time). Verify that only the selected channel flashes.

STEP 7 Reconnect the input to Channel 1 and set the digital ring code to positions 2 through 16 one at a time. Ring at each setting and verify that no channel flashes.

STEP 8 Set the receive ring codes (S1, S2 on S3 on CCA A5) to Code 2. Then ring with Codes 1 through 16. Verify that Channel 6 flashes on Code 2 only.

STEP 9 Repeat this procedure, steps 5 through 8, using codes 2 through 16.

4.7.8.2.1 Detection with errors. (See 3.7.8.2.1)

STEP 1 Connect the equipment as shown in Figure 18.

STEP 2 Ensure that the following connections are made.

Signal	Conn	UUT Connections	
ERROR INSERTER	J1	CH1 TIM IN	W, X
		CH1 DATA IN	B, C
		CH1 DATA OUT	q, r
STATION CLOCK	J2	STA CLK IN	w, x

STEP 3 Set the UUT and control panel switch positions as shown in Tables XXVII and XXI.

STEP 4 Push DIGITAL DVOW 1 selector pushbutton on the control panel.

STEP 5 Set the Error Inserter to give a 10^{-3} error rate with bursts of 20 percent errors at a duty cycle of 5 percent.

STEP 6 Set the transmit digital ring code to 1 and ring 20 times. Channel 1 shall respond at least 18 times.

Measurement Detection With Errors	Requirement
CH1 Response	CH1 responds 18 out of 20 times.

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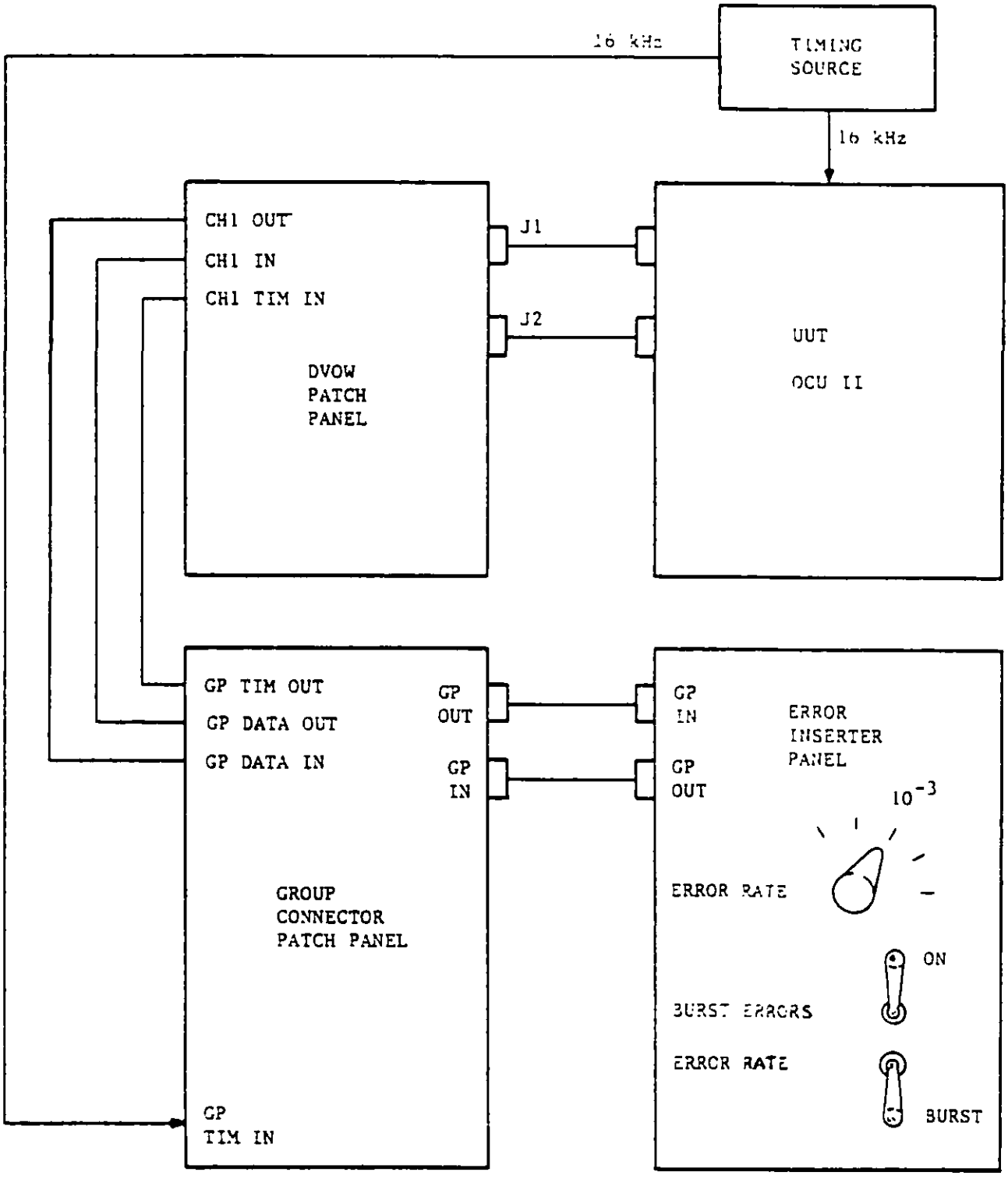


FIGURE 18. DVOW signaling detection with errors.

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TABLE XXVII. OCU II CCA switch position matrix.

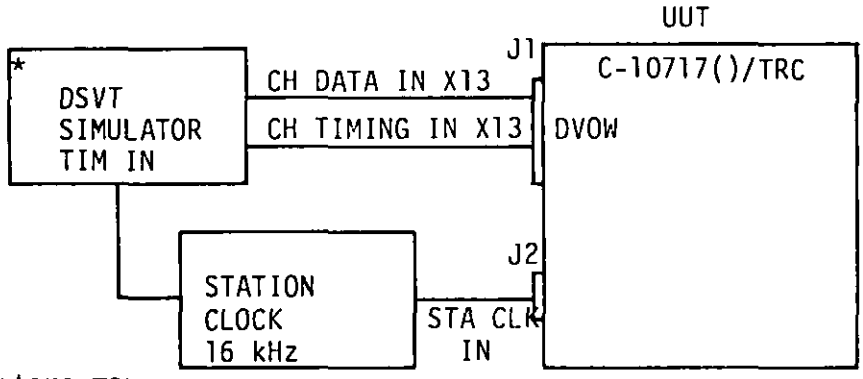
CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s / 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4* Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8* Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4* Codes 0 thru 9	0
	Channel Select S2	5 thru 8* Codes 1 thru 13	1
	BER Threshold S3	5 thru 8* Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16*	1
	S2		1
	S3		1

*See Front Cover for code switch positions.

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4.7.8.2.2 False ring detection. (See 3.7.8.2.2)

STEP 1 Connect the equipment as shown in Figure 19.



* 2 DSVT Simulators may be required.

FIGURE 19. False ring detection.

Item	Conn	UUT Connections		
DSVT Simulator	J1	CH1 DATA IN	B, C	A Ground
		CH2 DATA IN	D, E	
		CH3 DATA IN	F, G	
		CH4 DATA IN	H, J	
		CH5 DATA IN	K, L	
		CH6 DATA IN	M, N	
		CH7 DATA IN	P, R	
		CH8 DATA IN	S, T	
	J1	CH9 DATA IN	U, V	
	J2	CH10 DATA IN	W, X	
		CH11 DATA IN	Y, Z	
		CH MG DATA IN	a, b	
	J2	CH P DATA IN	c, d	
	J1	CH1 TIM IN	W, X	
		CH2 TIM IN	Y, Z	
		CH3 TIM IN	a, b	
	J1	CH4 TIM IN	c, d	A Ground

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Item	Conn	UUT Connections		
DSVT Simulator	J1	CH5 TIM IN	e,f	A Ground
		CH6 TIM IN	g,h	
		CH7 TIM IN	i,j	
		CH8 TIM IN	k,m	
	J1	CH9 TIM IN	n,p	
	J2	CH10 TIM IN	e,f	
		CH11 TIM IN	g,h	
		CH MG TIM IN	i,v	
	J2	CH P TIM IN	k,m	A Ground
Station Clock	J2	STA CLK IN	w,x	A Ground

STEP 3 The UUT switches and control panel switches are the same as in Tables XXVIII and XXI. Verify each channel works by connecting operator to channel and ringing it.

STEP 4 Run the unit for 24 hours. There shall be no ring detections.

Measurement	Requirement
24 hour ring detection	No ring detections for 24 hours.

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TABLE XXVIII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	14
	S2		15
	S3		16

*See Front Cover for code switch positions.

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4.7.8.3 Operator access and bridging (DVOW). (See 3.7.8.3)

STEP 1 Connect the equipment as shown in Figure 20.

STEP 2 Ensure that the following connections are made.

Signal	Connector	UUT Connections Pins		
		Low	High	Ground
DSVT DVOW CH1 DATA IN	J1	B	C	A
DVOW CH1 DATA OUT		q	r	A
DVOW CH2 DATA IN		D	E	A
DVOW CH2 DATA OUT		s	I	A
DVOW CH3 DATA IN		F	G	A
DVOW CH3 DATA OUT		u	v	A
DVOW CH4 DATA IN		H	J	A
DVOW CH4 DATA OUT		w	x	A
DVOW CH5 DATA IN		K	L	A
DVOW CH5 DATA OUT		y	z	A
DVOW CH6 DATA IN		M	N	A
DVOW CH6 DATA OUT		AA	BB	A
DVOW CH7 DATA IN		P	R	A
DVOW CH7 DATA OUT		CC	DD	A
DVOW CH8 DATA IN		S	T	A
DVOW CH8 DATA OUT		EE	FF	A
DVOW CH9 DATA IN		U	V	A
DVOW CH9 DATA OUT	J1	GG	HH	A
DVOW CH10 DATA IN	J2	W	X	A
DVOW CH10 DATA OUT		n	p	A
DVOW CH11 DATA IN		Y	Z	A
DVOW CH11 DATA OUT		q	r	A
DVOW MG DATA IN		a	b	A
DVOW MG DATA OUT		s	t	A
DVOW P DATA IN		c	d	A
DVOW P DATA OUT	J2	u	v	A
STATION CLOCK	J2	w	x	A
VINSON DATA IN	J3	-	B	A
VINSON DATA OUT	J3	-	C	A

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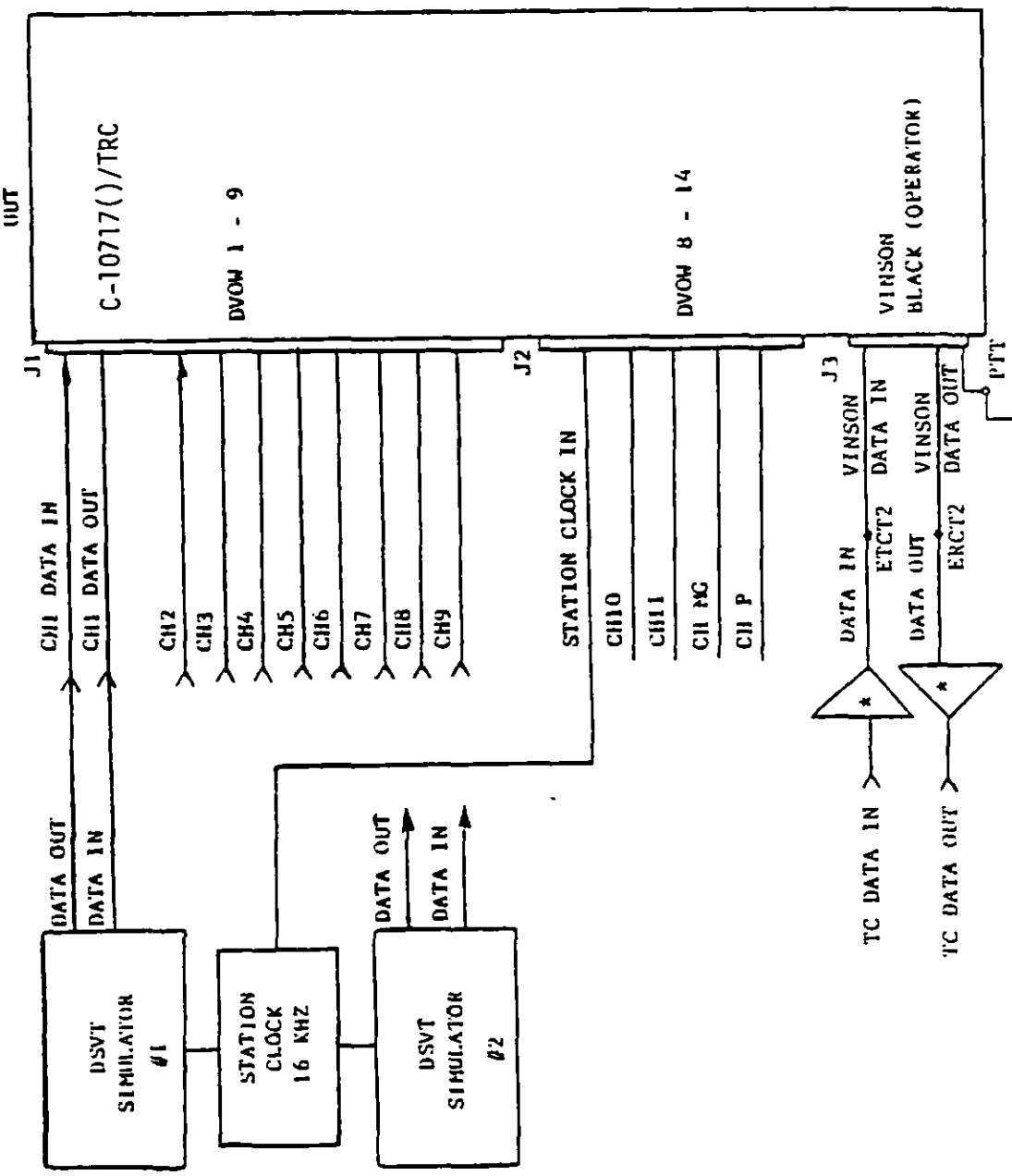


FIGURE 20. Operator access and bridging (DVOW).

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STEP 3 Set the unit switches and control panel switches as specified in Tables XXVII and XV.

STEP 4 Push the digital OW B pushbutton on the control panel.

STEP 5 Connect the DSVT Simulators to each of the combinations of channels marked T shown below.

NOTE: Use only 1 DSVT Simulator at the combinations marked O.

DSVT No. 2

DSVT No. 1	1	2	3	4	5	6	7	8	9	10	11	MC	P	O
1	O	T	T	T	T	T	T	T	T	T	T	T	T	T
2	-	O	T	T	T	T	T	T	T	T	T	T	T	T
3	-	-	O	T	T	T	T	T	T	T	T	T	T	T
4	-	-	-	O	T	T	T	T	T	T	T	T	T	T
5	-	-	-	-	O	T	T	T	T	T	T	T	T	T
6	-	-	-	-	-	O	T	T	T	T	T	T	T	T
7	-	-	-	-	-	-	O	T	T	T	T	T	T	T
8	-	-	-	-	-	-	-	O	T	T	T	T	T	T
9	-	-	-	-	-	-	-	-	O	T	T	T	T	T
10	-	-	-	-	-	-	-	-	-	O	T	T	T	T
11	-	-	-	-	-	-	-	-	-	-	O	T	T	T
MG	-	-	-	-	-	-	-	-	-	-	-	O	T	T
P	-	-	-	-	-	-	-	-	-	-	-	-	O	T
OP	-	-	-	-	-	-	-	-	-	-	-	-	-	O

STEP 6 Connect the DSVT Simulator #1 Data Input to Channel 1 Data Output and DSVT Simulator #2 Data Input to Channel 2 Data Output. Connect the DSVT #1 Simulator Data Output to Channel 1 Data Input and verify error free transmission to DSVT Simulator #2 for at least 10 seconds.

STEP 7 Leaving these connections in place, connect the Data Output from DSVT Simulator #2 to Channel 2 Data Input. Verify that error free transmission is maintained at DSVT Simulator #2.

STEP 8 Disconnect the Data Output from DSVT Simulator #1 and verify error free transmission to DSVT Simulator #1 for at least 10 seconds.

STEP 9 Repeat steps 6 through 8 for all combinations in the table which are marked T.

STEP 10 Disconnect DSVT Simulator #2. Connect DSVT Simulator #1 Data Input and Output to Channel 1 Data Output and Input.

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STEP 11 Verify that no activity is received by DSVT Simulator #1.

STEP 12 Repeat steps 10 and 11 for all channels marked 0 in the table.

STEP 13 Push the DVOW 1 button. Connect DSVT 1 to CH1 and DSVT 2 to operator Verify error free transmission in both directions. Connect DSVT 1 Data Out and TIM OUT and Data Out to CH2. Verify no activity from Channel 2 to operator.

STEP 14 Repeat step 13 using channels 2 through P to the operator. Verify error free transmission for each selected channel and no activity to one other channel.

4.7.8.4 Bridge seizure and release. (See 3.7.8.4)

STEP 1 Connect the equipment as shown in Figure 21. Set the UUT and control panel switches as shown in Tables XXVII and XXIX.

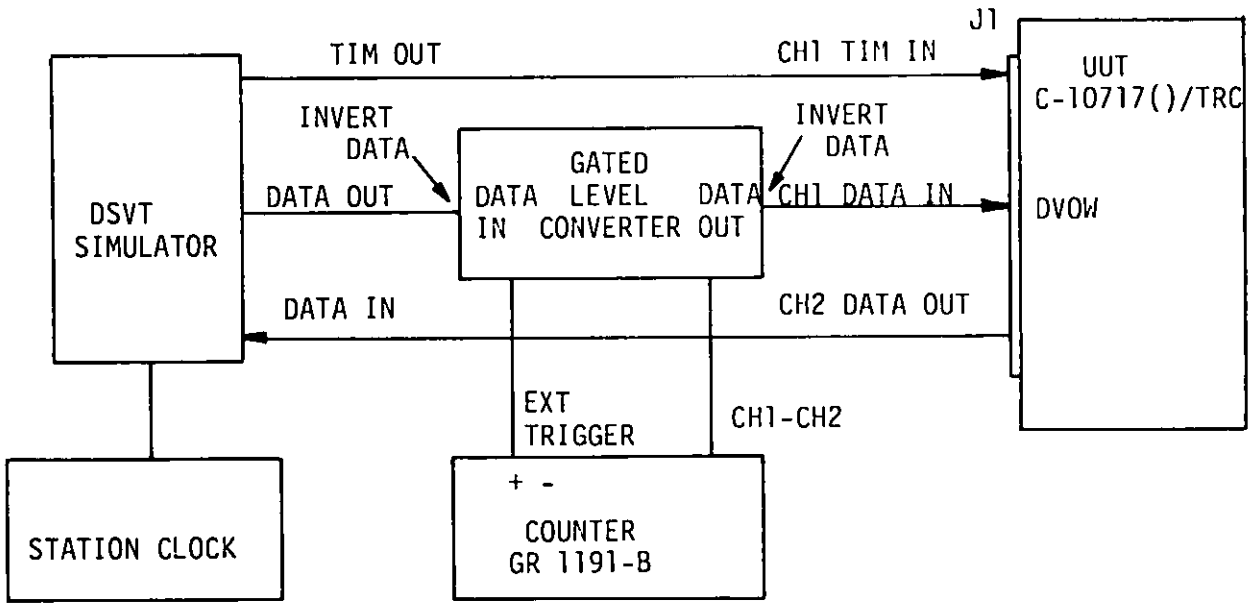


FIGURE 21. Bridge seizure and release.

STEP 2 Trigger the counter (+) upon application of data. Then measure the time from the start of the counter to the appearance of data (bridge seizure).

STEP 3 Repeat step 2 for a total of 20 times.

STEP 4 Invert the data out on the level converter and trigger the counter (-). Then measure the time from the start of the trace to the appearance of all ones (bridge release).

STEP 5 Repeat step 4 for a total of 20 times.

STEP 6 Verify that the specified requirements of 3.7.8.4 are met.

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TABLE XXIX. OCU I control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1		X						
	2		X						
	3	X							
	4	X							
	5	X							
	6	X							
	7	X							
	8	X							
	9	X							
	10	X							
	11	X							
	MG	X							
	P	X							

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
											X

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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4.7.8.5 VINSON buffer. (See 3.7.8.5)

STEP 1 Connect the equipment as shown in Figure 22.

STEP 2 Ensure that the following connections are made.

Signal	UUT Connections		
STATION CLOCK	J2	STATION Clock In	w, x, Ground A
DSVT SIMULATOR	J1	CH1 Data Out	q, r, Ground A
FREQUENCY SYNTHESIZER	J2	Buffer Clock 1	B, C, Ground A
FREQUENCY SYNTHESIZER	J2	Buffer Clock 2	D, E, Ground A
FREQUENCY SYNTHESIZER	J2	Buffer Clock 3	F, G, Ground A
PTT	J3	PTT In	D Ground M, N
LEVEL CONVERTER	J3	Data In	B Ground M, N

STEP 3 Set the UUT and control panel switches as shown in Tables XXVII and XXI.

STEP 4 Connect the frequency synthesizer to Buffer Timing 1 at a frequency of 16000.48 Hz (+30 ppm).

STEP 5 Set BUFFER TIMING switch to 1.

STEP 6 Push DVOW 1 selector on the control panel and the PTT switch.

STEP 7 Measure and record the time between BCI losses.

STEP 8 Repeat steps 4 through 7 with the synthesizer set to 15999.52 Hz (-30 ppm).

STEP 9 Release the PTT switch. Verify that Channel 1 output is all ones.

STEP 10 Repeat steps 4 through 9 substituting Buffer Timing 1 with Buffer Timing 2 and 3.

STEP 11 Verify that the specified requireements of 3.7.8.5 are met.

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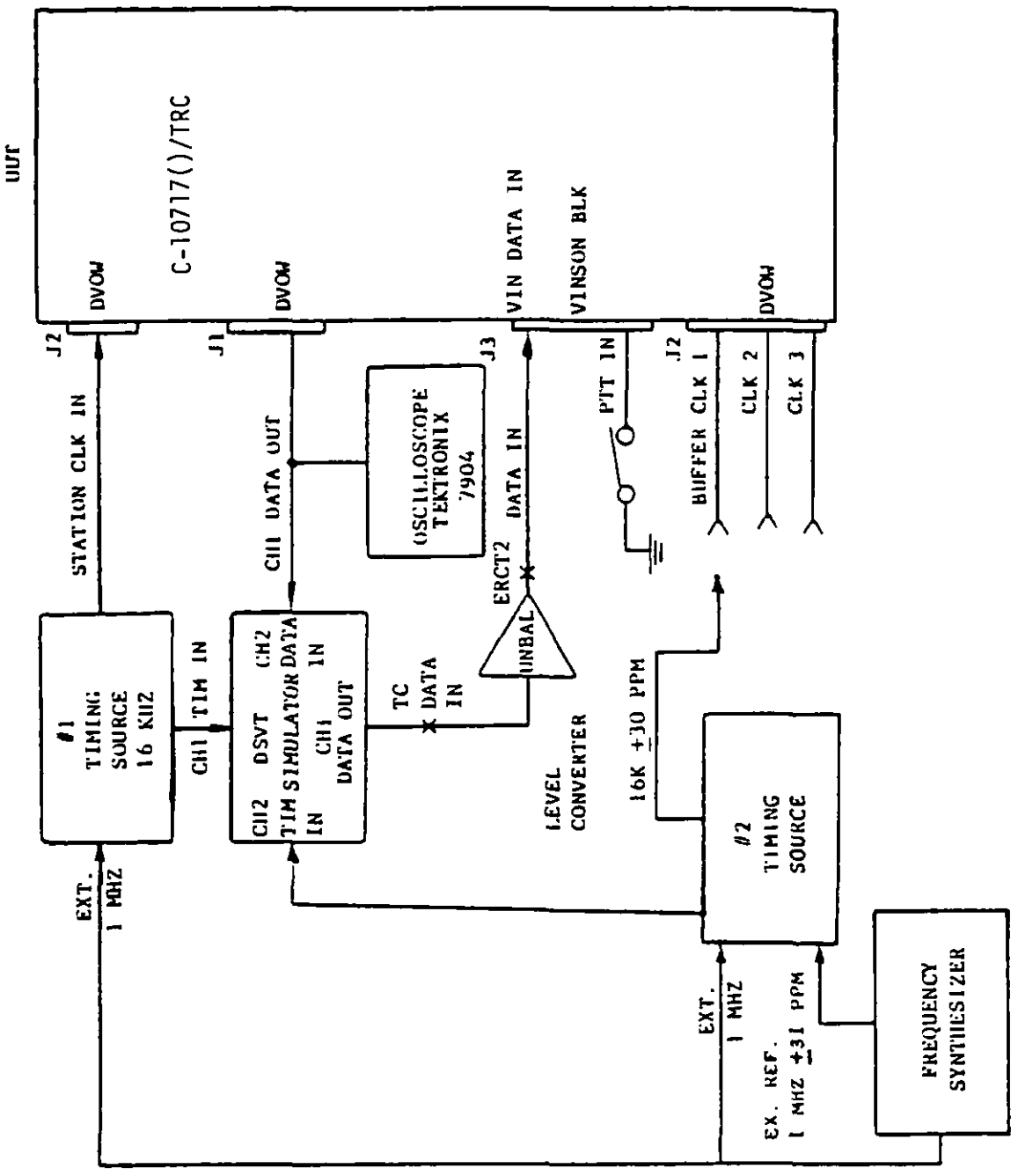


FIGURE 22. VINSON buffer operation and timing.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T03:59Z
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Measurement BCI Losses	Requirement
Buffer Timing 1 at 16000.52 Hz (+30 ppm)	2.2 Minutes Minimum
at 15999.48 Hz (-30 ppm)	
Buffer Timing 2 at 16000.52 Hz (+30 ppm)	2.2 Minutes Minimum
at 15999.48 Hz (-30 ppm)	
Buffer Timing 3 at 16000.52 Hz (+30 ppm)	2.2 Minutes Minimum
at 15999.48 Hz (-30 ppm)	

4.7.8.6 DVOW multiplexing. (See 3.7.8.6)

STEP 1 Connect the equipment as shown in Figure 23.

STEP 2 Set the UUT and control panel switches as shown in Tables XXX and XXI.

STEP 3 Set the Group Simulator and OCU at both the 256 (16 on GP SIM CH RATE) and 512 kb/s (32 on GP SIM CHANNEL RATE) rates to the positions listed below at each position. Verify that the specified requirements are met.

Group Simulator Position	Requirement
No Frame	FR is sent by UUT.
FR	FRA is sent by UUT.
FRA	MSF is sent by UUT.
MSF	MSF is sent by UUT.

STEP 4 Disconnect the Group Simulator. Put on Group loop plug at end of 100 feet of group cable. Set DVOW to 256 kb/s.

STEP 5 Connect the DSVT DATA IN and OUT to DVOW Channel 1 IN and OUT. Observe the oscilloscope. Sync the scope to the timing signal. Locate the (OH) framing bit zero (see below). Verify that activity appears on the selected channel bit. Sync scope to 2 kHz DCM clock.

MUX BIT	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
DVOW SELECTED CHANNEL	OH	D	D	D	1	2	3	4	5	6	7	8	9	10	11	MG

SELECTED CHANNEL BITS

OH = OVERHEAD (FRAMING BIT); D = DUMMY BITS, SET TO ONES; P = PRIVATE CHANNEL

STEP 6 Reconnect the Group Simulator at 256 kb/s (RING DET switch to GP, PVT to MG switch on ON, PVT CHAN switch to MUX) LOOPBACK DATA IN to DATA OUT on Channels 1 through 11 and MG. Look at GROUP DATA OUT on oscilloscope and verify activity on all channels.

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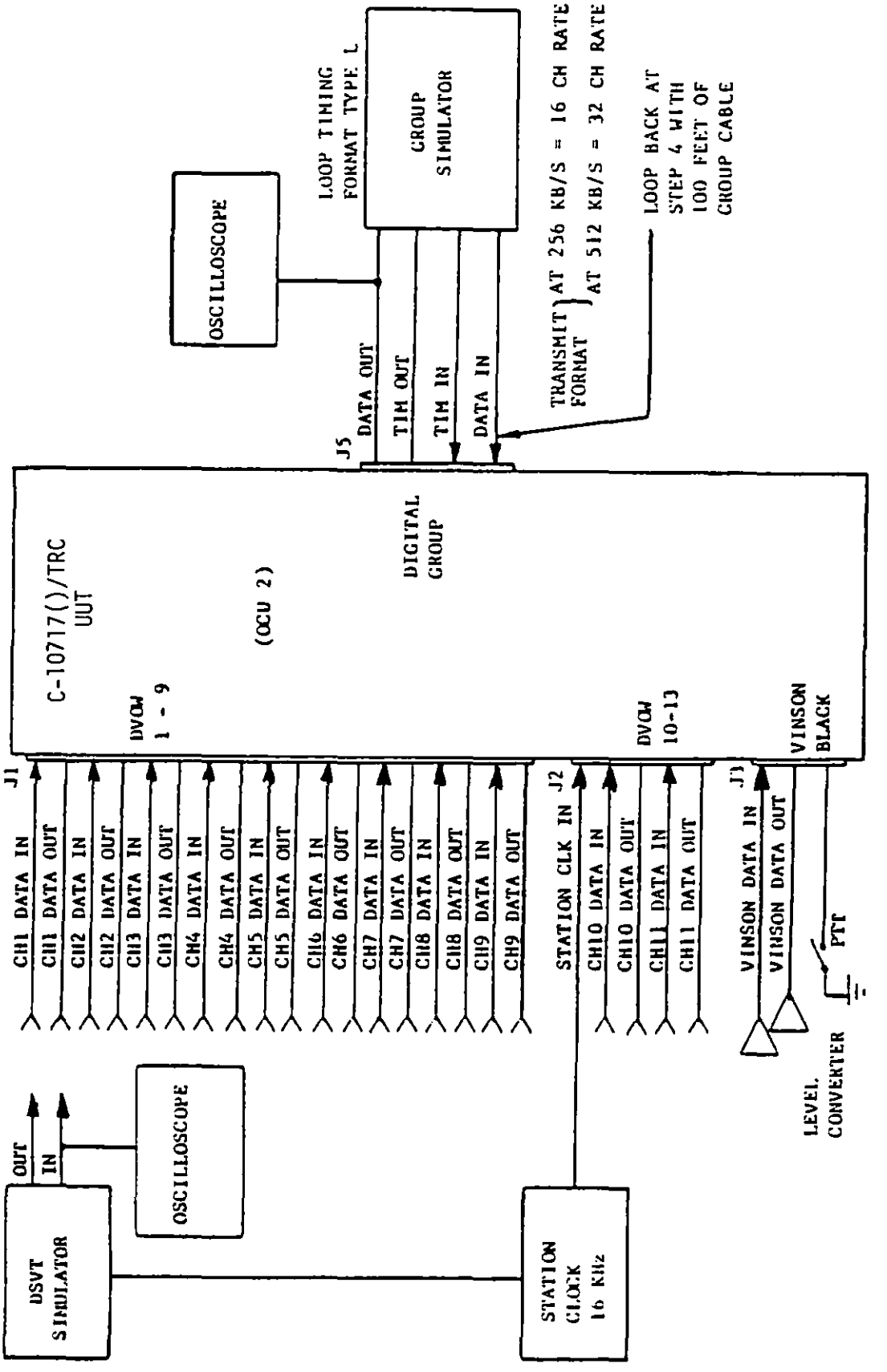


FIGURE 23. DVOW MUX frame format.

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TALBE XXX. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	OFF
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	AS USED
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	ON
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	MUX
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring S1	Codes 1 thru 16 *	1
	Codes S2		1
	S3		1

*See Front Cover for code switch positions.

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STEP 7 Set the MUX ON/OFF switch on CCA A7 to the OFF position. Observe the oscilloscope. Verify that all DVOW channels are ones.

STEP 8 Repeat steps 6 and 7 at 512 kb/s.

STEP 9 Set the MUX ON/OFF switch on CCA A7 to the ON position.

STEP 10 Disconnect Group Simulator and connect 100 feet of group cable with loopback plug. Set OW MUX to 256 kb/s.

STEP 11 Measure and record the CH1 DVOW BER over a 10 second period.

STEP 12 Repeat steps 4 through 8 for each remaining Channel 2 through P. P to MG then MG to P.

NOTE: When measureing Channel "P", go into MG port (P to MG switch is ON) PVT CHAN switch is ON.

STEP 13 Measure MG to MG with PVT CHAN switch in the MUX position and PVT to MG switch in the ON position.

STEP 14 Measure MG to P and P to MG with the PVT CHAN switch in the EXT position.

STEP 15 Measure P to P with the PVT CHAN switch in the EXT position and the P pushbutton switch ON.

STEP 16 Connect the equipment as shown in Figure 24. The UUT and control panel switch positions are the same as shown in Tables XXX and XXI.

STEP 17 Set the frequency synthesizer to 1,000,001 Hz (timing source 16,000.016 Hz).

STEP 18 Measure the times between BCI loss as indicated by errors on the DSVT Simulator number 2.

STEP 19 Set the frequency synthesizer to 999,999 Hz (timing source 15,999.984 Hz) and repeat step 3. Record the results.

STEP 20 Repeat steps 2 through 4 for DVOW channels 2 through P.

Measurement MUX BCI	Requirement
DVOW CH1	1 minute minimum
DVOW CH2	
DVOW CH3	
DVOW CH4	
DVOW CH5	
DVOW CH6	
DVOW CH7	
DVOW CH8	
DVOW CH9	
DVOW CH10	
DVOW CH11	
DVOW CH MG	
DVOW CH P	

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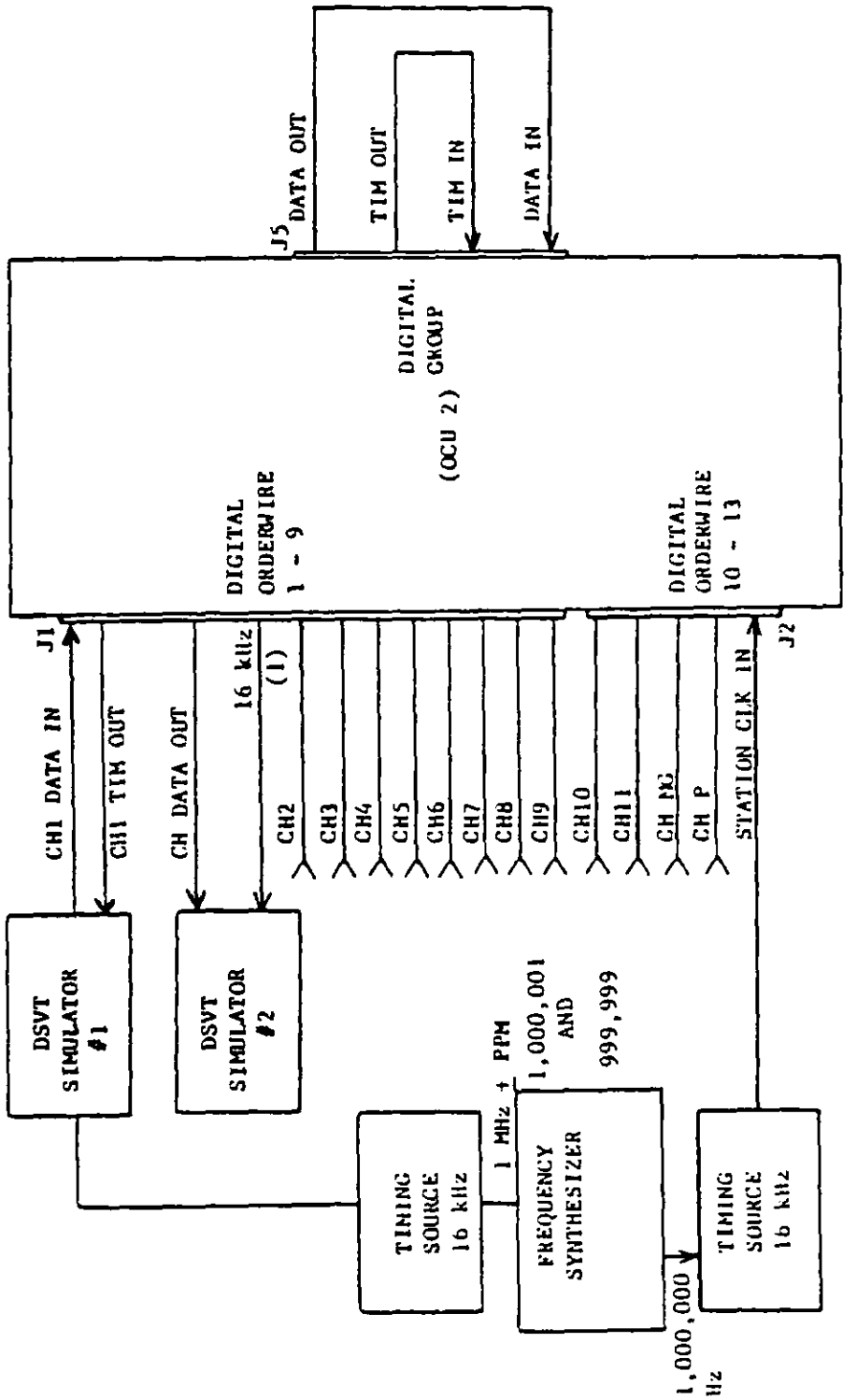


FIGURE 24. DVOW channel BCI test.

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4.7.8.7 DVOW Demux framing. (See 3.7.8.7)

STEP 1 Connect the equipment as shown in Figure 25.

STEP 2 Ensure that the following connections are made.

Signals	Conn	UUT Connections Pins High	Low	Ground
Station Clk - Station Clk In	J2	w	x	A
Group Data Out	J5	E	F	P
Group Tim Out	J5	G	H	R
Group Data In	J5	A	B	M
Group Tim In (OCU II A or B)	J5	C	D	N

STEP 3 Set the UUT and control panel switches as shown in Tables XXXI and XXXII.

STEP 4 Set up Error Inserter Test Set for 10^{-3} error rate.

STEP 5 For the 512 kb/s group rate perform steps 6 and 7 to determine the frame-in time of the UUT.

STEP 6 Squelch the group data at the Error Inserter Test Set to cause loss of frame. Release squelch to restore data.

STEP 7 Measure the time from resumption of data to frame-in. Use the squelch release signal at the Error Inserter Test Set to start time-interval measurement, and the negative goind edge of TP8 on the Group Framing PCB to stop the time interval counter. Take 50 measurements and record the results.

STEP 8 For the 256 kb/s rate perform steps 6 and 7 to determine the frame-in time of the UUT.

STEP 9 Verify that at each rate, high and low, no more than 5 of 50 measurements exceed 100 ms.

Measurement Frame-In Test 0.1% BER	Requirement
Frame In - High Rate	No more than 5 of 50 measure-
Frame In - Low Rate	ments exceed 100 ms.

STEP 10 The test setup is the same as in Figure 25. The UUT and control panel switches are set per Tables XXI and XXXII.

STEP 11 Set the DVOW Group RATE SELECT switch to 512 kb/s. Set pulse generator for 2.5 µs pw, 4V. Measure the time between reset of the Mux Frame counter and the reacquisition of frame. Take 20 measurements and record the results below.

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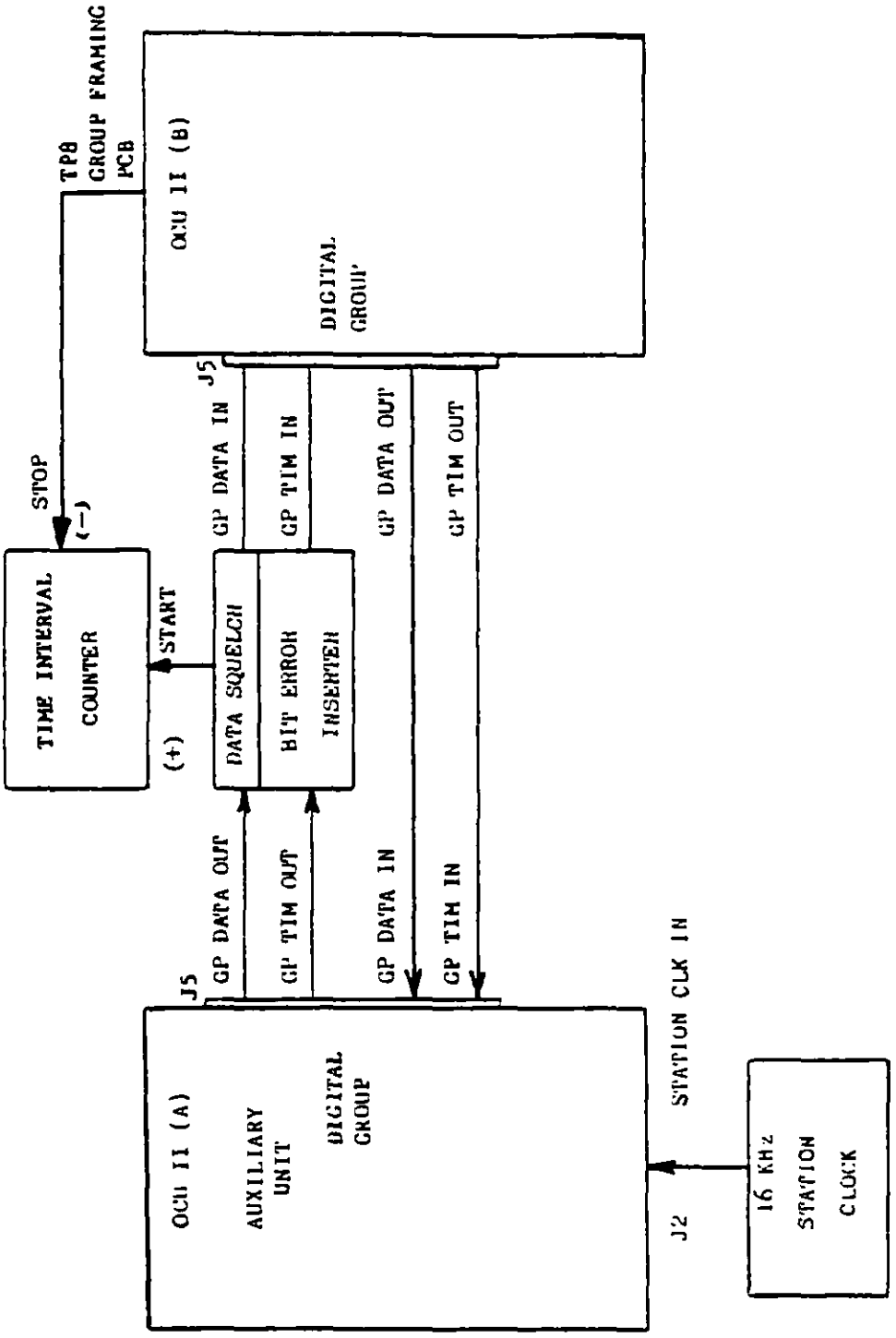


FIGURE 25. DVOW DEMUX framing.

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TABLE XXXI. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	OFF
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	AS REQUIRED
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring S1	Codes 1 thru 16 *	1
	Codes S2		1
	S3		1

*See Front Cover for code switch psotions.

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TABLE XXXII. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1	X							
	2	X							
	3	X							
	4	X							
	5	X							
	6	X							
	7	X							
	8	X							
	9	X							
	10	X							
	11	X							
	MG	X							
	P	X							

Switch	Position			
	S	1	2	3
Timing Buffer	X			

Switch	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
Timing Station								X			

Switch	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Digital Ring Code	X															

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STEP 12 Repeat Step 2 with the DVOW Group RATE SELECT switch at 256 kb/s. Set pulse generator for 6 µs pw, 4V.

STEP 13 Verify that at each rate, high and low, no more than 2 of 20 measurements exceed 200 ms.

Measurement Frame Drop-Out Reacquisition	Requirement
High Rate 512 kb/s	No more than 2 of 20 measurements exceed 200 ms.
Low Rate 256 kb/s	

STEP 14 Connect the equipment as shown in Figure 26. Set UUT and control panel switch positions the same as shown in Table XXXI and XXXII.

STEP 15 Ensure that the following connections are made.

Signals	Connector	UUT Connections Pins Low	High	Ground
STA Clock - Station Clock In	J2	$\overline{W}$	$\overline{X}$	A
DSVT SIM - DVOW CH1 Data In	J1	$\overline{B}$	$\overline{C}$	A
Data Out 1 DVOW CH2 Data In	J1	D	E	A
Data Out 1 DVOW CH3 Data In	J1	F	G	A
Data Out 1 DVOW CH4 Data In	J1	H	J	A
Data Out 1 DVOW CH5 Data In	J1	K	L	A
Data Out 1 DVOW CH6 Data In	J1	M	N	A
Data Out 2 DVOW CH7 Data In	J1	P	R	A
Data Out 2 DVOW CH8 Data In	J1	S	T	A
Data Out 2 DVOW CH9 Data In	J1	U	V	A
Data Out 2 DVOW CH10 Data In	J2	W	X	A
Data Out 2 DVOW CH11 Data In	J2	Y	Z	A
OCU II A or B				
Group Data Out	J5	E	F	P
Group Tim Out	J5	G	H	R
Group Data In	J5	A	B	M
Group Tim In	J5	C	D	N

STEP 16 Set the Error Inserter to 10^{-3} mode.

STEP 17 Set the PRG Test Set Group Rate to 512 kb/s.

STEP 18 Monitor TP8 (Positive going edge will indicate Out-of-Frame) with a counter, for 24 hours, to detect loss of frame. The unit shall not lose frame during the 24 hour test period.

Measurement	Requirement
24 hour frame maintenance test	No loss of frame

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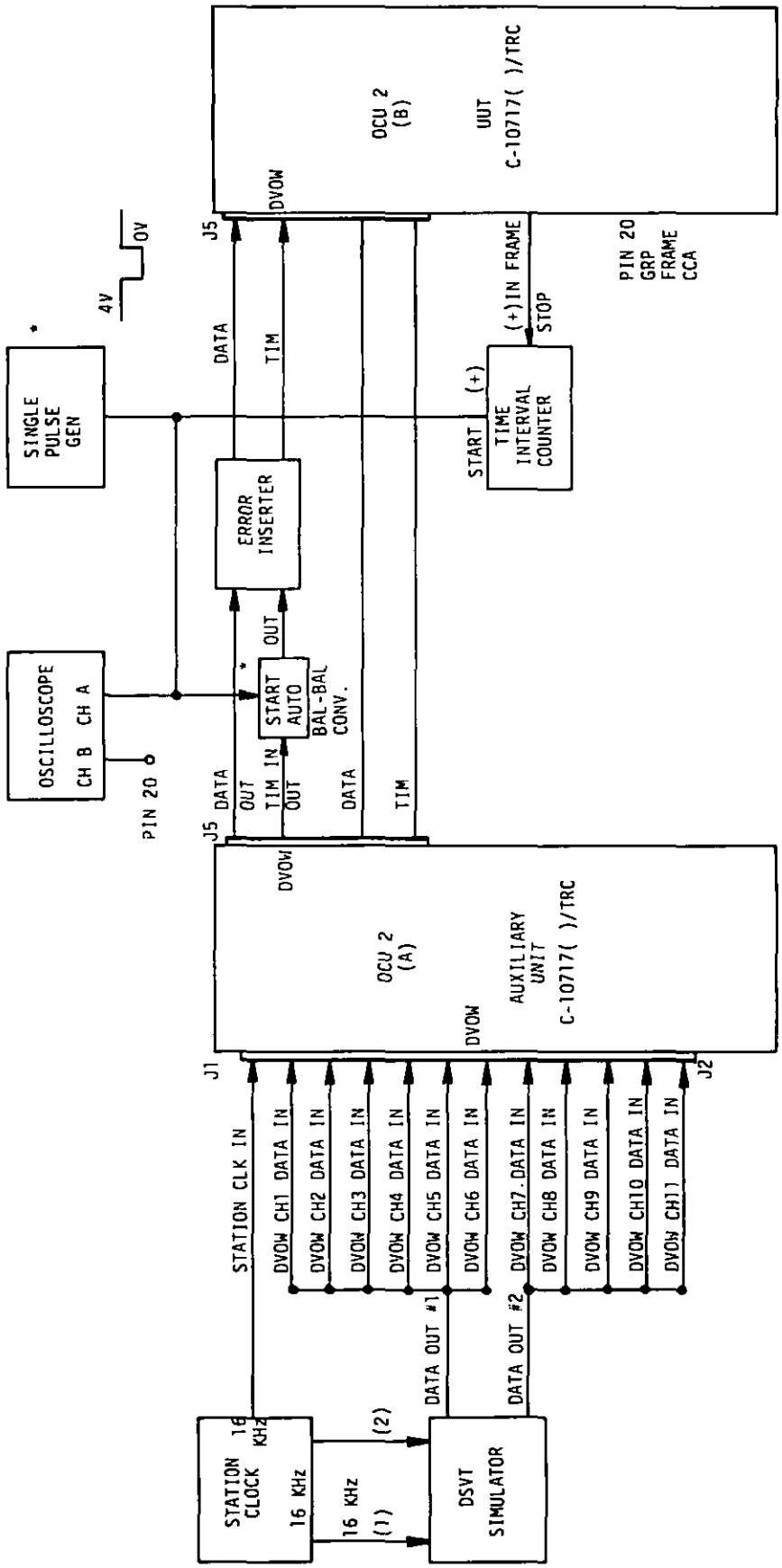


FIGURE 26. 24 hour frame maintenance.

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4.7.9 Timing distribution. (See 3.7.9)

STEP 1 Connect the equipment as shown in Figure 27.

STEP 2 Ensure that the following connections are made.

Signals	Connector	UUT Connections Pins Low	High	Ground
SYNTHESIZER				
CH1 TIM IN	J1	W	X	A
CH2 TIM IN	J1	Y	X	A
CH3 TIM IN	J1	a	b	A
CH4 TIM IN	J1	c	d	A
CH5 IN	J1	e	f	A
MG TIM IN	J2	i	j	A
STATION CLOCK IN	J2	w	x	A
COUNTER				
16 kHz CLK 1 OUT	J2	H	J	A
16 kHz CLK 2 OUT	J2	K	L	A
16 kHz CLK 3 OUT	J2	M	N	A
32 kHz CLK 1 OUT	J2	P	R	A
32 kHz CLK 2 OUT	J2	S	T	A
32 kHz CLK 3 OUT	J2	U	V	A

STEP 3 Set the UUT switches and control panel switches as shown in Tables XXXIII and XXXIV.

STEP 4 Set the control panel STATION TIMING thumbwheel switch to position 1.

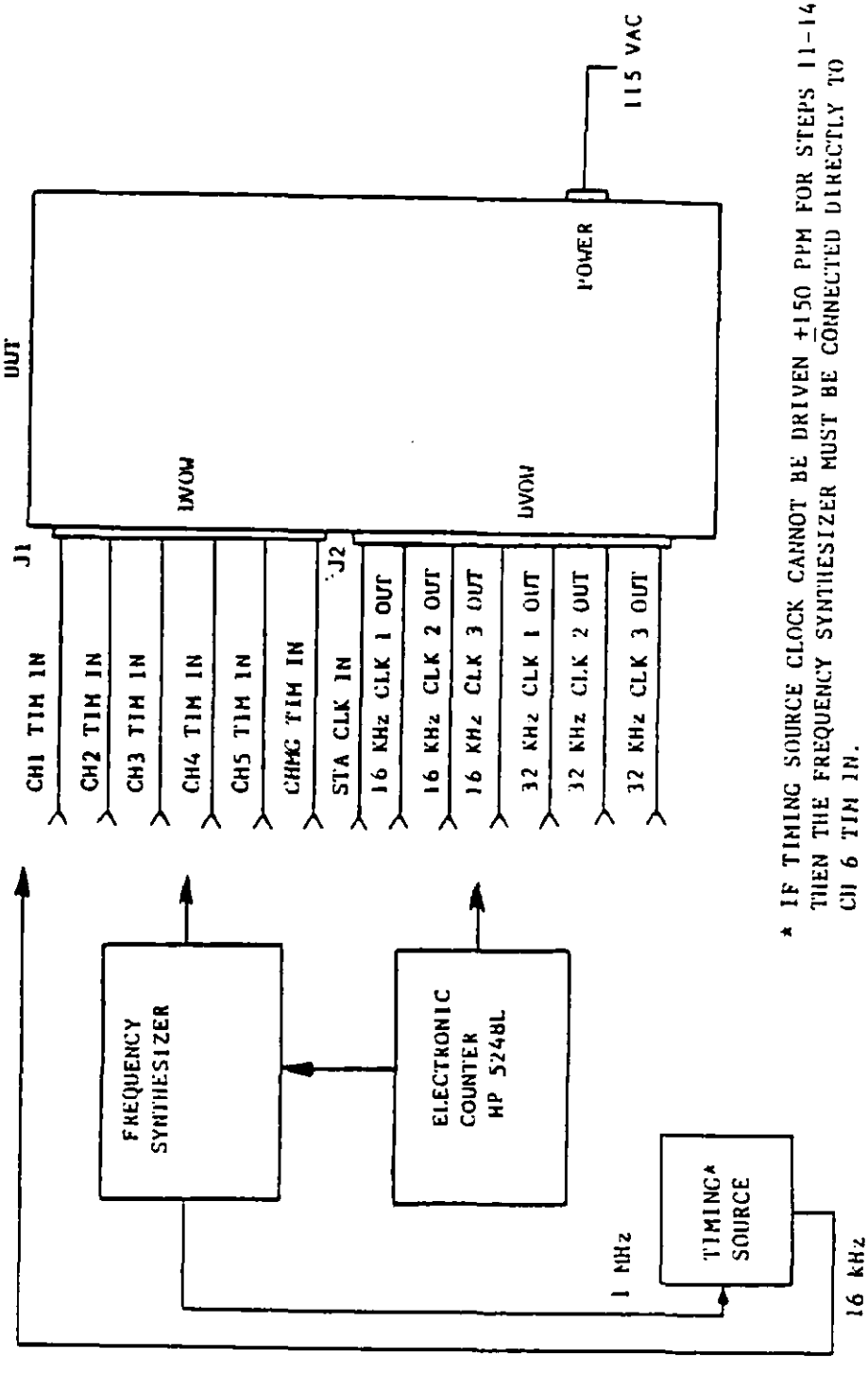
STEP 5 Connect the frequency synthesizer to CH1 Timing In at a frequency of 16000 Hz. Measure and record the frequency Out of J2 16K CLK OUT 1 1 over a 100 second period. Verify that the output frequency is equal to the input frequency within one count.

STEP 6 Repeat step 5 using input Channels 2 through 5, FG, and Station Clock. Set the STATION TIMING thumbwheel switch to the corresponding input; for MG input set thumbwheel to 5, for STA CLK input set thumbwheel to S1.

STEP 7 With the input connect to CH MG, measure the outputs of the 16K CLK OUT 2 and 16K CLK OUT 3.

Measurement 16K CLK Distribution	Requirement
CH1 IN - 16K CLK 1 OUT	Output frequency is equal to input frequency ±1 count.
CH2 IN - 16K CLK 1 OUT	
CH3 IN - 16K CLK 1 OUT	
CH4 IN - 16K CLK 1 OUT	
CH5 IN - 16K CLK 1 OUT	
CH MG IN - 16K CLK 1 OUT	
STA CLK - 16K CLK 1 OUT	
MG IN - 16K CLK 2 OUT	
MG IN - 16K CLK 3 OUT	

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* IF TIMING SOURCE CLOCK CANNOT BE DRIVEN ±150 PPM FOR STEPS 11-14, THEN THE FREQUENCY SYNTHESIZER MUST BE CONNECTED DIRECTLY TO CH 6 TIM IN.

FIGURE 27. Timing distribution.

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TABLE XXXIII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	OFF
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

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TABLE XXXIV. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1	X							
	2	X							
	3	X							
	4	X							
	5	X							
	6	X							
	7	X							
	8	X							
	9	X							
	10	X							
	11	X							
	MG	X							
	P	X							

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
					As Required						

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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STEP 8 With the 16.000 kHz input connected to Channel MG and TIMING switch on STA 6, CLK output 1, 2, and 3 over a 10 second period.

Measurement 32 kHz CLK Distribution	Requirement	Measured Value
MG IN 32 kHz CLK 1 OUT	Output frequency shall be twice the input frequency ±1 count. Reference frequency 16.0000 kHz.	
MG IN 32 kHz CLK 2 OUT		
MG IN 32 kHz CLK 3 OUT		

STEP 9 Connect synthesizer to STA CLK IN.

STEP 10 Set synthesizer to 32000 Hz. Set STA TIM switch to S2.

STEP 11 Measure 32 kHz CLK 1 OUT.

STEP 12 Verify that the ouptut frequency is equal to the input frequency ±1 count.

Measurement 32 kHz CLK	Requirement	Measured Value
32000 Hz STA CLK IN	Output frequency is equal to input ±1 count.	

STEP 13 With the input connected to MG set STA TIM to 6 and adjust the frequency synthesizer to a frequency of 16,002.4 Hz (this is +150 ppm). Then disconnect and reconnect the input.

STEP 14 Measure and record the 16K CLK OUT 1 signal over a 100 second period.

STEP 15 Repeat steps 13 and 14 with the synthesizer set to 15997.6 Hz. (This is -150 ppm).

Measurement	Requirement
MG at 16002.4 Hz	Output frequency is equal to input frequency ±1 count.
MG at 15997.6 Hz	

STEP 16 Verify that the specified requirements in 3.7.9 is met.

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4.7.10 AVOW performance. (See 3.7.10)

4.7.10.1 Signaling generation. (See 3.7.10.1)

STEP 1 Connect the equipment as shown below in Figure 28.

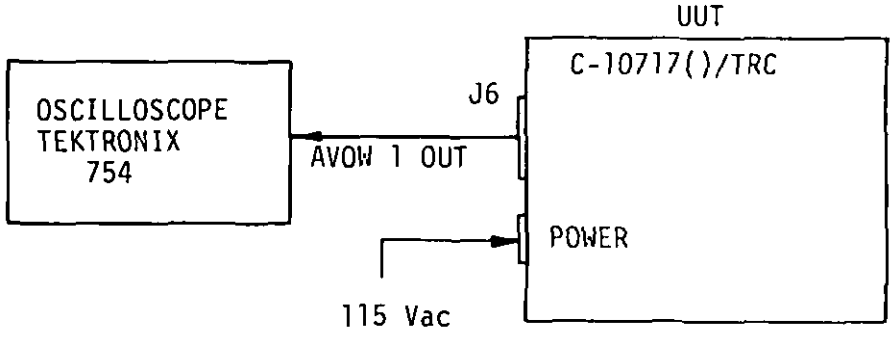


FIGURE 28. Signaling generation.

STEP 2 Ensure that the following connections are made.

Signal	Conn	UUT Connections
Oscilloscope	J6	AVOW 1 OUT P,R Ground A

STEP 3 Push the AVOW 1 pushbutton on the control panel, and the RING button.

STEP 4 Measure and record the duration of the ring signal.

Measurement	Requirement
Ring Duration	Between 2 and 3 seconds

STEP 5 Verify that the requirement of 3.7.10.1 is met.

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4.7.10.2 Signaling detection. (See 3.7.10.2)

STEP 1 Connect the equipment as shown in Figure 29.

STEP 2 Set all channel switches to terminate.

STEP 3 Set the oscillator to 1632 Hz at a level of -15dBm. (noise off)

STEP 4 Turn off the oscillator gate and set the noise generator to -30 dBm (true rms reading).

STEP 5 Key the oscillator on for 2 seconds a minimum of 20 times and measure and record the number of detections.

STEP 6 Repeat steps 3 through 5 with the oscillator set to -29 dBm and a noise level of -44 dBm.

STEP 7 Repeat steps 3 through 6 with an oscillator frequency of 1568 Hz.

1632 Hz -15 dBm Noise -30 dBm					
Test	Detected	Not Detected	Test	Detected	Not Detected
1			11		
2			12		
3			13		
4			14		
5			15		
6			16		
7			17		
8			18		
9			19		
10			20		

STEP 8 Verify that the following requirements of 3.7.10.2 are met.

Measurement	Requirement
1632 Hz -15 dBm Noise -30 dBm	Detection: 18 out of 20 times
1632 Hz -20 dBm Noise -44 dBm	
1568 Hz -15 dBm Noise -30 dBm	
1568 Hz -29 dBm Noise -44 dBm	

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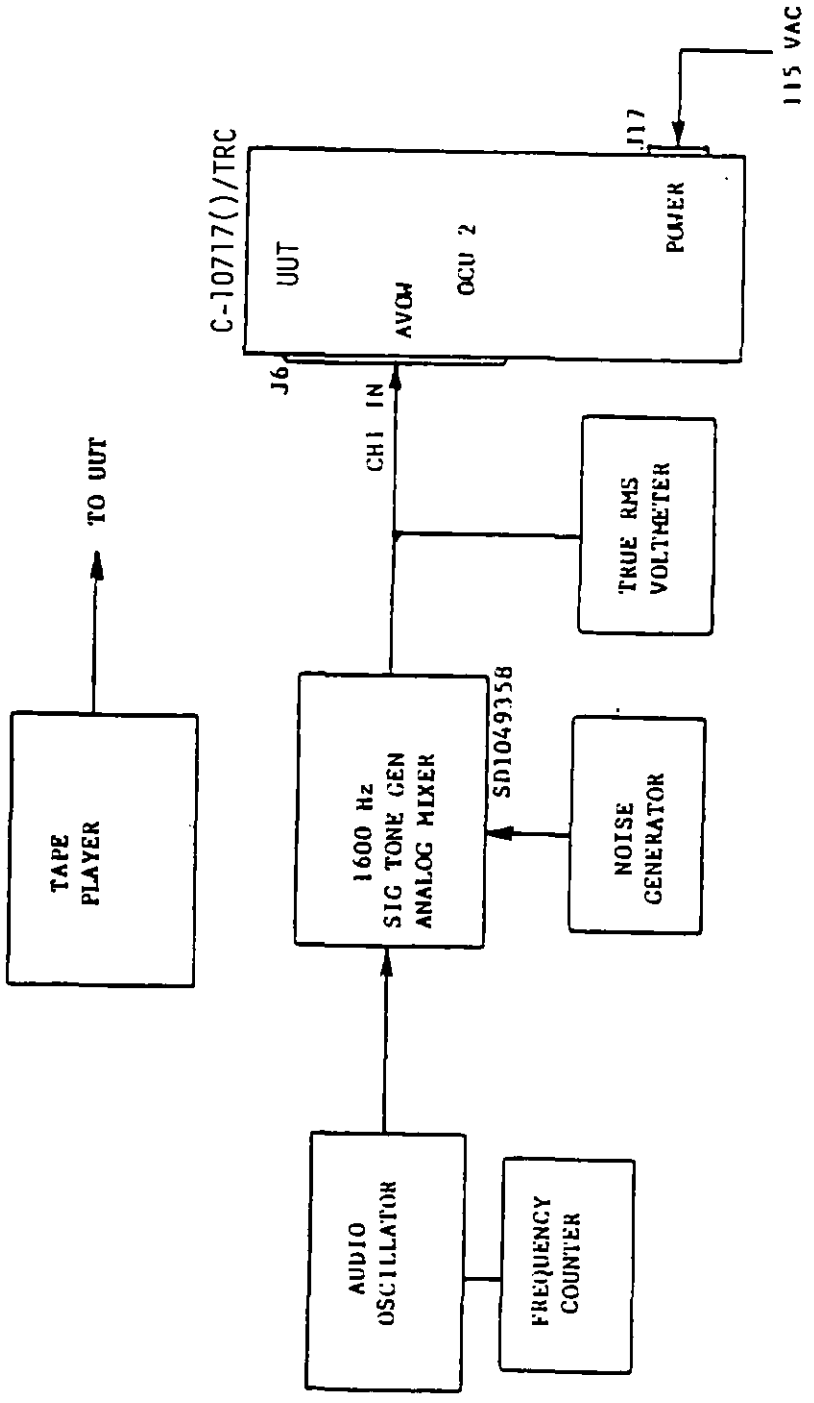


FIGURE 29. Signaling detection.

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4.7.10.3 False signaling detection. (See 3.7.10.3)

STEP 1 Setup same as 4.7.10.2.

STEP 2 Turn off the oscillator and set the noise level to -18.2 dBm. Monitor for detections a minimum of 1 hour.

STEP 3 Apply the tape player with the level set to normal conversation level for 1 hour and monitor for detection.

Measurement	Requirement
Noise -18.2 dBm Tape Noise	No more than one detection during the time specified.

STEP 4 Verify that the requirements of 3.7.10.3 are met.

4.7.10.4 Operator access and bridging (AVOW). (See 3.7.10.4)

STEP 1 Connect the equipment as shown in Figure 30.

STEP 2 Set all the AVOW toggle switches on the control panel to bridge and push the AVOW B button.

STEP 3 Connect the oscillators and voltmeters in turn to the channels shown in Table XXXV.

STEP 4 The oscillator shall be set as follows:

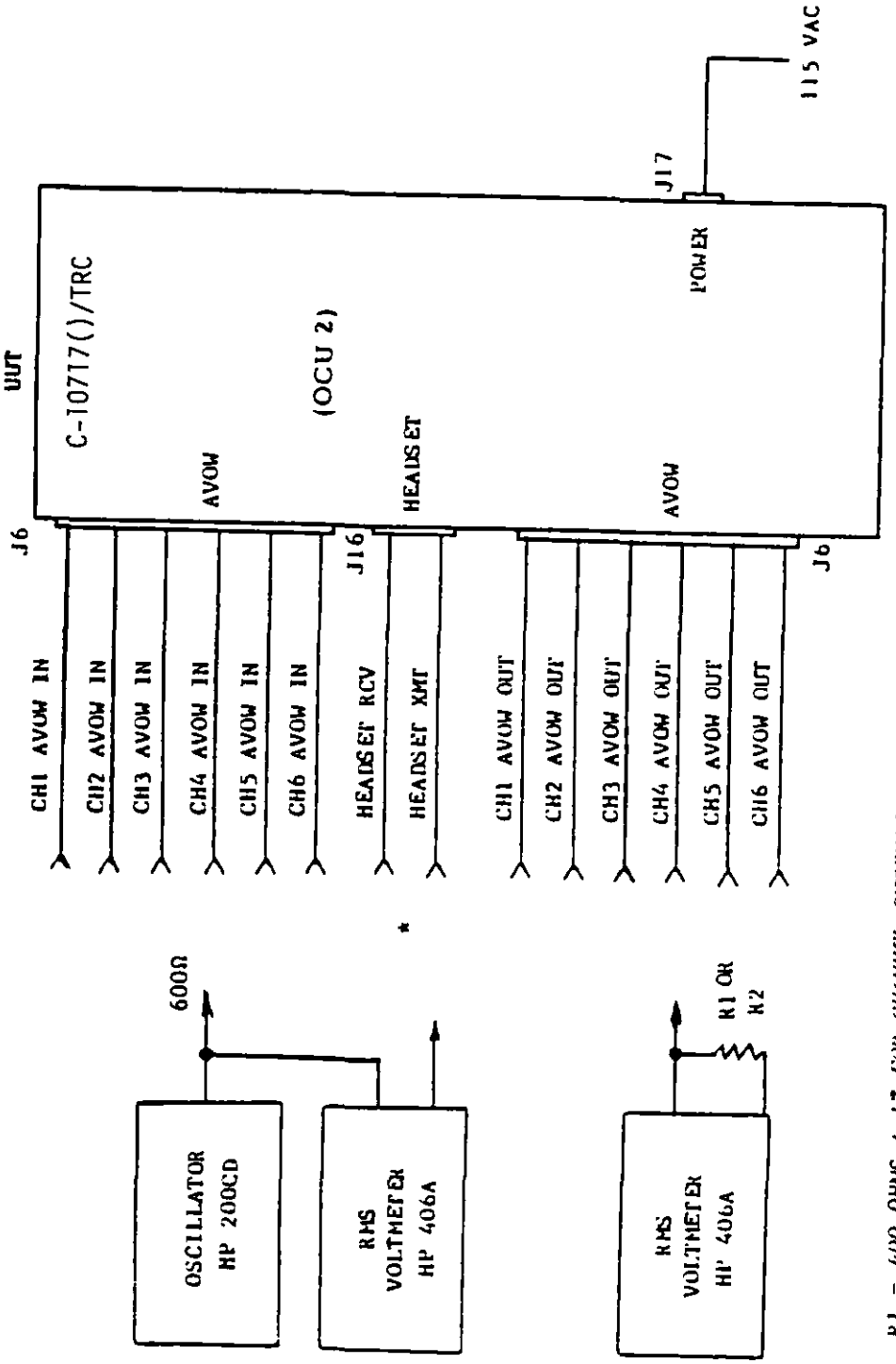
Set oscillator to 1 kHz.

-18.2 dBm for channel inputs and 150 mV rms for operator inputs.

STEP 5 Use the following UUT connections.

Signal	Conn	UUT Pin Numbers
AVOW CH1 In	J6	B, C Ground A
AVOW 2 In	J6	D, E Ground A
AVOW 3 In	J6	F, G Ground A
AVOW 4 In	J6	H, J Ground A
AVOW 5 In	J6	K, L Ground A
AVOW 6 In	J6	M, N Ground A
AVOW 1 Out	J6	P, R Ground A
AVOW 2 Out	J6	S, T Ground A
AVOW 3 Out	J6	U, V Ground A
AVOW 4 Out	J6	W, X Ground A
AVOW 5 Out	J6	Y, Z Ground A
AVOW CH6 Out	J6	a, b Ground A
Headset Xmt	J16	C, D
Headset Rcv	J16	A, B

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R1 = 600 OHMS ± 1% FOR CHANNEL OUTPUTS ONLY
R2 = 150 OHMS ± 1% FOR HEADSET OUTPUTS ONLY

* USE HEADSET CABLE & REMOVE 600Ω RESISTOR FROM OSCILLATOR WHEN PERFORMING OPERATIONAL MEASUREMENTS.

FIGURE 30. Operator access and bridging (AVOW).

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TABLE XXXV. Operator access and bridging channel connections.

Measurement		Requirement CH1	Measured Value CH1	Requirement CH2	Measured Value CH2	Requirement CH3	Measured Value
Input	CH1	+0.3dBm max		+20.3 - 22.3 dBm		+20.3 - 22.3 dBm	
	CH2	+20.3 - 22.3 dBm		+0.3dBm max			
-18.2dBm	CH3			+20.3 - 22.3 dBm		+0.3dBm max	
	CH4					+20.3 - 22.3 dBm	
	CH5						
	CH6						
150 mV	OP	+4 to +7dBm		+4 to +7dBm		+4 to +7dBm	

Measurement		Requirement CH4	Measured Value CH4	Requirement CH5	Measured Value CH5	Requirement CH6	Measured Value CH6
Input	CH1	+20.3 - 22.3 dBm		+20.3 - 22.3 dBm		+20.3 - 22.3 dBm	
	CH2						
	CH3						
-18.2dBm	CH4	+0.3dBm max					
	CH5	+20.3 - 22.3 dBm		+0.3dBm max			
	CH6			+20.3 - 22.3 dBm		+0.3dBm max	
150 mV	OP	+4 to +7 dBm		+4 to +7 dBm		+4 to +7 dBm	

Measurement		Requirement Operator	Measured Value Operator
Input	CH1	+4 to +7 dBm	
	CH2	.614 to .867 Volts.	
-18.2dBm	CH3		
	CH4		
	CH5		
	CH6		
150 mV	OP	NA	NA

NOTE: All readings with loads connected to outputs.

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STEP 6 At each combination measure and record the output voltages. Use HP 3555B for outputs and HP 403 for operator interfaces.

STEP 7 Set AVOW 1 CH1 toggle switch to terminate.

STEP 8 Push the AVOW 1 button on the control panel (removes operator from the bridge).

STEP 9 Connect the oscillator and voltmeter to the 36 combinations as shown in Table XXXVI.

STEP 10 Set the oscillator output to -18.2 dBm and verify that each output is less than -39.7 dBm.

STEP 11 If an output exceeds the -39.7 dBm limit, reduce the oscillator level to zero, and measure the residual noise then correct the reading by rms subtraction.

TABLE XXXVI. Operator access and bridging channel combinations.

		Selected Channel (Oscillator) (✓) Indicates Compliance					
Selected		1	2	3	4	5	6
Channel	1						
Voltmeter	2						
	3						
	4						
	5						
	6						

(✓) Indicates output is less than -39.7 dBm.

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4.7.11 Data orderwire (DOW) performance. (See 3.7.11)

4.7.11.1 Format. (See 3.7.11.1)

STEP 1 Connect the equipment as shown in Figure 31.

STEP 2 *Ensure that* the following connections are made.

Signal	Conn	UUT Pin Numbers
SD Analyzer	J8	TXCLK G
SD Analyzer	J8	TXDPT F
BER Generator	J15	Strobe 1 R, S Ground A
BER Generator	J15	Strobe 2 K, L Ground A
BER Generator	J15	Strobe 3 M, N Ground A
BER Generator	J15	BER BIT 1 B, C Ground A
BER Generator	J15	BER BIT 2 D, E Ground A
BER Generator	J15	BER BIT 3 F, G Ground A
BER Generator	J15	BER BIT 4 H, J Ground A
Alarm Box	J13	Alarm 1 hi B
Alarm Box	J13	Alarm 1 lo C
Alarm Box	J13	Alarm 2 hi D
Alarm Box	J13	Alarm 2 lo E
Alarm Box	J13	Alarm 3 hi F
Alarm Box	J13	Alarm 3 lo G
Alarm Box	J13	Alarm 4 hi H
Alarm Box	J13	Alarm 4 lo J
Alarm Box	J13	Alarm 5 hi K
Alarm Box	J13	Alarm 5 lo L
Alarm Box	J13	Alarm 6 hi M
Alarm Box	J13	Alarm 6 lo N

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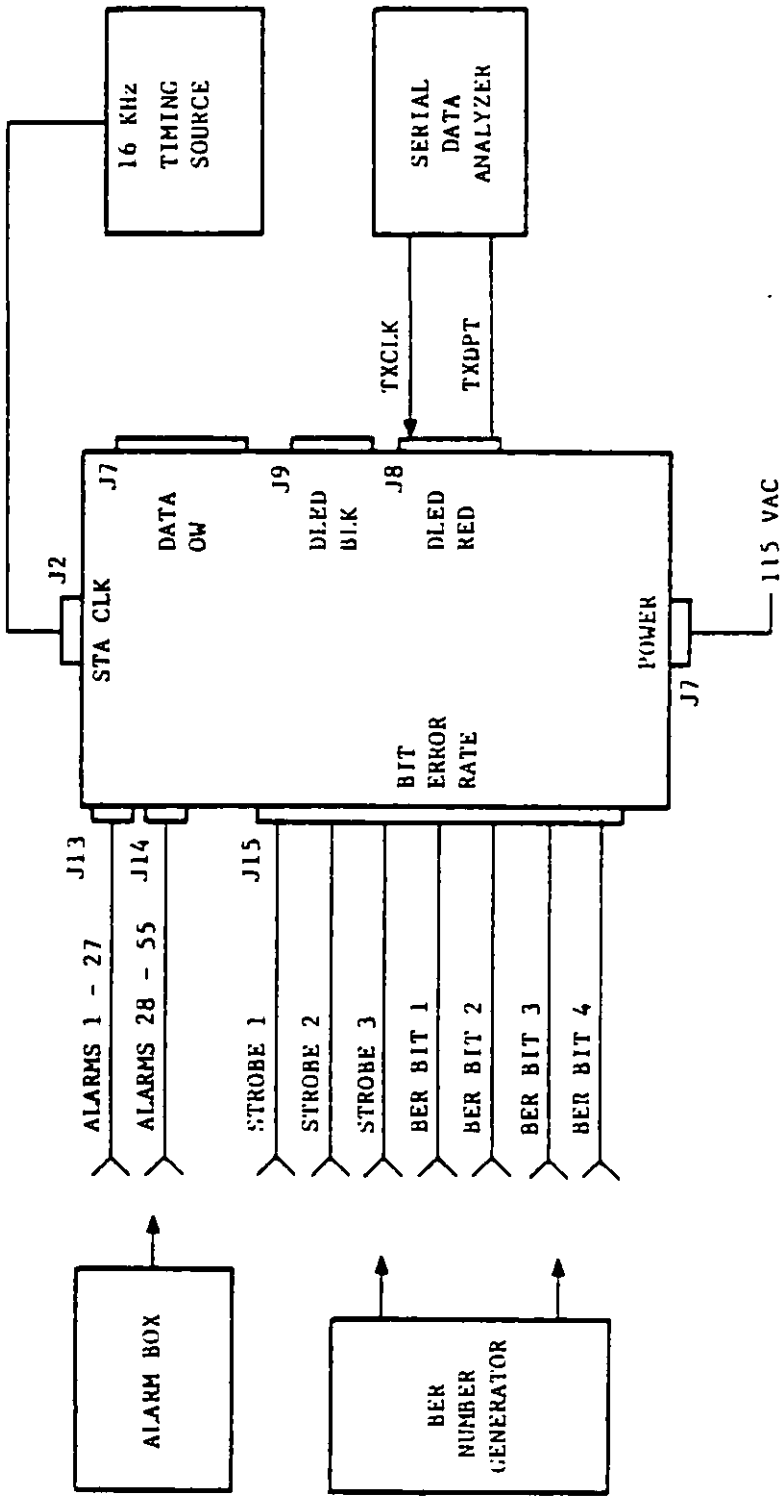


FIGURE 31. DOW performance.

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Signal	Conn	UUT Pin Numbers	
Alarm Box	J13	Alarm 7 hi	P
Alarm Box	J13	Alarm 7 lo	R
Alarm Box	J13	Alarm 8 hi	S
Alarm Box	J13	Alarm 8 lo	T
Alarm Box	J13	Alarm 9 hi	U
Alarm Box	J13	Alarm 9 lo	V
Alarm Box	J13	Alarm 10 hi	W
Alarm Box	J13	Alarm 10 lo	X
Alarm Box	J13	Alarm 11 hi	Y
Alarm Box	J13	Alarm 11 lo	Z
Alarm Box	J13	Alarm 12 hi	a
Alarm Box	J13	Alarm 12 lo	b
Alarm Box	J13	Alarm 13 hi	c
Alarm Box	J13	Alarm 13 lo	d
Alarm Box	J13	Alarm 14 hi	e
Alarm Box	J13	Alarm 14 lo	f
Alarm Box	J13	Alarm 15 hi	g
Alarm Box	J13	Alarm 15 lo	h
Alarm Box	J13	Alarm 16 hi	i
Alarm Box	J13	Alarm 16 lo	j
Alarm Box	J13	Alarm 17 hi	k
Alarm Box	J13	Alarm 17 lo	m
Alarm Box	J13	Alarm 18 hi	n
Alarm Box	J13	Alarm 18 lo	p
Alarm Box	J13	Alarm 19 hi	q
Alarm Box	J13	Alarm 19 lo	r
Alarm Box	J13	Alarm 20 hi	s
Alarm Box	J13	Alarm 20 lo	t
Alarm Box	J13	Alarm 21 hi	u
Alarm Box	J13	Alarm 21 lo	v
Alarm Box	J13	Alarm 22 hi	w
Alarm Box	J13	Alarm 22 lo	x
Alarm Box	J13	Alarm 23 hi	y
Alarm Box	J13	Alarm 23 lo	z

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Signal	Conn	UUT Pin Numbers	
Alarm Box	J13	Alarm 24 hi	AA
Alarm Box	J13	Alarm 24 lo	BB
Alarm Box	J13	Alarm 25 hi	CC
Alarm Box	J13	Alarm 25 lo	DD
Alarm Box	J13	Alarm 26 hi	EE
Alarm Box	J13	Alarm 26 lo	FF
Alarm Box	J13	Alarm 27 hi	GG
Alarm Box	J13	Alarm 27 lo	HH
Alarm Box	J13	Ground	A
Alarm Box	J14	Alarm 28 hi	B
Alarm Box	J14	Alarm 28 lo	C
Alarm Box	J14	Alarm 29 hi	D
Alarm Box	J14	Alarm 29 lo	E
Alarm Box	J14	Alarm 30 hi	F
Alarm Box	J14	Alarm 30 lo	G
Alarm Box	J14	Alarm 31 hi	H
Alarm Box	J14	Alarm 31 lo	J
Alarm Box	J14	Alarm 32 hi	K
Alarm Box	J14	Alarm 32 lo	L
Alarm Box	J14	Alarm 33 hi	M
Alarm Box	J14	Alarm 33 lo	N
Alarm Box	J14	Alarm 34 hi	P
Alarm Box	J14	Alarm 34 lo	R
Alarm Box	J14	Alarm 35 hi	S
Alarm Box	J14	Alarm 35 lo	T
Alarm Box	J14	Alarm 36 hi	U
Alarm Box	J14	Alarm 36 lo	V
Alarm Box	J14	Alarm 37 hi	W
Alarm Box	J14	Alarm 37 lo	X
Alarm Box	J14	Alarm 38 hi	Y
Alarm Box	J14	Alarm 38 lo	Z

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Signal	Conn	UUT Pin Numbers
Alarm Box	J14	Alarm 39 hi a
Alarm Box	J14	Alarm 39 lo b
Alarm Box	J14	Alarm 40 hi c
Alarm Box	J14	Alarm 40 lo d
Alarm Box	J14	Alarm 41 hi e
Alarm Box	J14	Alarm 41 lo f
Alarm Box	J14	Alarm 42 hi g
Alarm Box	J14	Alarm 42 lo h
Alarm Box	J14	Alarm 43 hi i
Alarm Box	J14	Alarm 43 lo j
Alarm Box	J14	Alarm 44 hi k
Alarm Box	J14	Alarm 44 lo m
Alarm Box	J14	Alarm 45 hi n
Alarm Box	J14	Alarm 45 lo p
Alarm Box	J14	Alarm 46 hi q
Alarm Box	J14	Alarm 46 lo r
Alarm Box	J14	Alarm 47 hi s
Alarm Box	J14	Alarm 47 lo t
Alarm Box	J14	Alarm 48 hi u
Alarm Box	J14	Alarm 48 lo v
Alarm Box	J14	Alarm 49 w
Alarm Box	J14	Alarm 50 y
Alarm Box	J14	Alarm 51 AA
Alarm Box	J14	Alarm 52 CC
Alarm Box	J14	Alarm 53 EE
Alarm Box	J14	Alarm 54 GG
Alarm Box	J14	Alarm 55 BB
Alarm Box	J14	Sig Gnd DD,HH,z,FF,x
Alarm Box	J14	Shield Gnd A

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STEP 3 Set the control panel and CCA switches as shown in Tables XXXVII and XXI.

STEP 4 Verify that the status indicator on the control panel is green, and that the combined format is as shown in Figure 32. Using the Serial Data Analyzer verify that the idle character state is one's.

STEP 5 Verify that the single status block for Channel 1 is as specified below, and that all other channels are 1.

Measurement Character No.	Requirement Symbol	Measurement Character No.	Requirement Symbol
1	STX	11	↑
2	A	12	↑
3	0	13	↑
4	0	14	↑
5	P	15	↑
6	9	16	↑
7	↑	17	↑
8	↑	18	↑
9	↑	19	ETX
10	↑	20	ETB

STEP 6 Repeat step 5 substituting in turn Channels 2 through 13.

NOTE: It will be necessary as each channel is selected to set the S2 (CCA15) DIP switch (positions 5,6,7 and 8 to the corresponding code for selected channel.

STEP 7 Verify that the following requirements were met.

Measurement	Requirement
CH1 CH2 CH3 CH4 CH5 CH6 CH7 CH8 CH9 CH10 CH11 CH12 CH13	Status block for each channel is as specified. All other channels are 1.

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TABLE XXXVII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	OFF
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	RCV
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	AS REQUIRED
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	AS REQUIRED
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	AS REQUIRED
	Channel Select S2	5 thru 8 * Codes 1 thru 13	AS REQUIRED
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	AS REQUIRED
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

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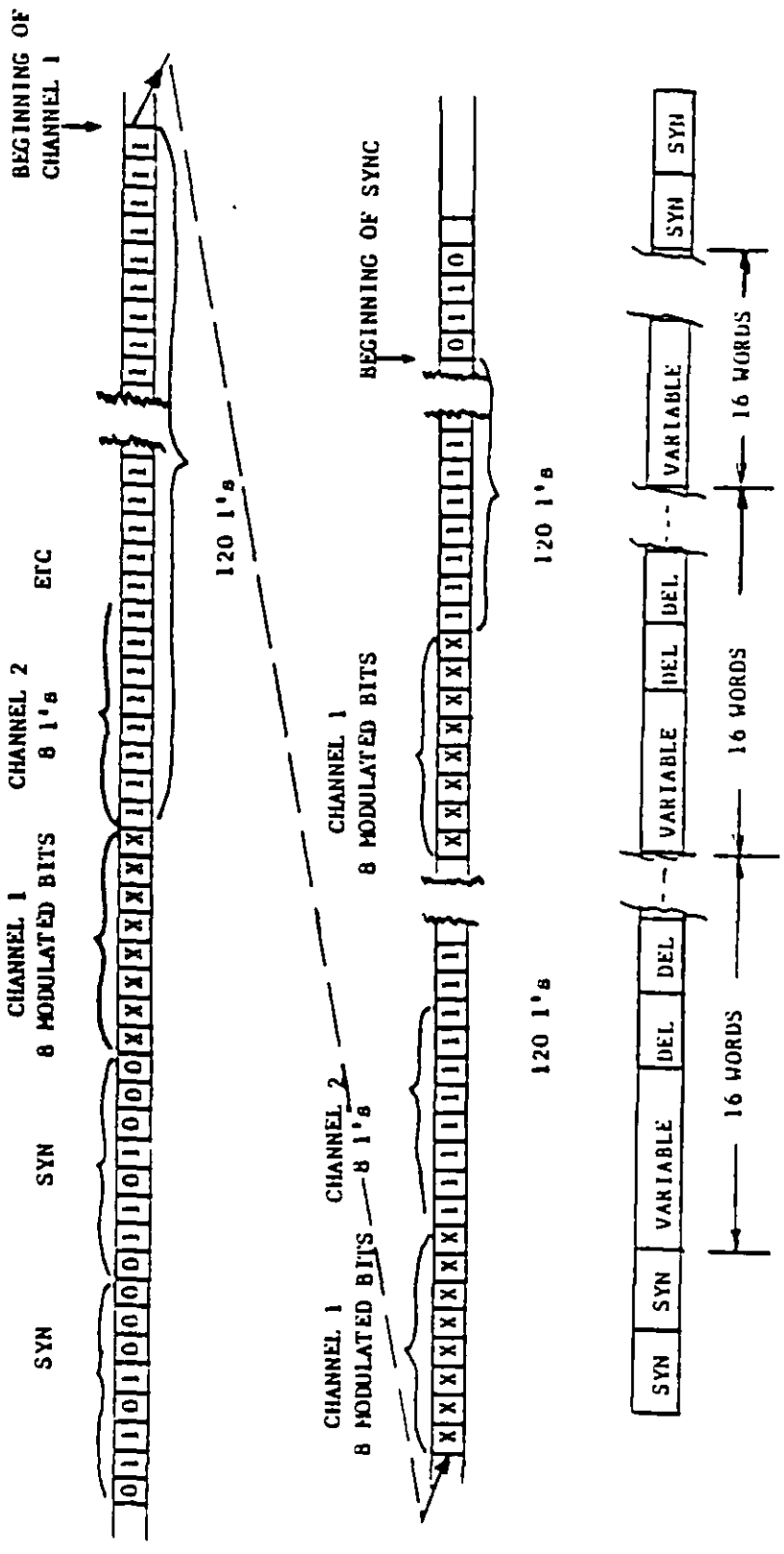


FIGURE 32. Telemetry combiner format.

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STEP 8 Reset the channel select settings for Channel 1.

STEP 9 Verify that channel character 2 is A.

STEP 10 Select in turn each of the station ID alpha positions; Set S1 (CCA A15) positions 1, 2, 3 and 4 to the corresponding positions indicated on the front cover. As each position is selected verify that the CH 1 character 2 is as specified below.

Measurement Station ID	Requirement Character 2	(✓) Indicates Compliance	Measurement Station ID	Requirement Character 2	(✓) Indicates Compliance
A	A		L	L	
B	B		M	M	
C	C		P	P	
D	D		R	R	
E	E		S	S	
F	F		T	T	
G	G		X	X	
H	H		Z	Z	

STEP 11 Select in turn each of the station numeric 1 positions. Set S1 (CCA A5) positions 5, 6, 7 and 8 to the corresponding positions indicated on the front cover. As each position is selected, verify that the CH 1 character 2 is as specified below.

Measurement Station Numeric 1	Requirement Character 3	(✓) Indicates Compliance
0	0	
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	
7	7	
8	8	
9	9	

STEP 12 Select in turn each of the station numeric 2 positions. Set S2 (CCA A15) positions 1, 2, 3, and 4 to the corresponding positions indicated on the front cover. As each position is selected, verify that the CH 1 character 4 is as specified below.

Measurement Station Numeric 2	Requirement Character 4	(✓) Indicates Compliance
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	
7	7	
8	8	
9	9	

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STEP 13 Apply and remove one at a time each of the alarms given in Table XXXVIII.

STEP 14 Verify that the specified requirements for each alarm are met.

STEP 15 Apply any three alarms simultaneously from Table XXXVIII.

STEP 16 Verify that the two block status message for Channel 1 is in accordance with Table XXXIX and that the status error messages selected appear in the appropriate locations.

STEP 17 Apply any five alarms simultaneously from Table XXXVIII.

STEP 18 Verify that the three block status message for channel is in accordance with Table XL and that the status error messages selected appear in the appropriate locations.

STEP 19 Apply any seven alarms simultaneously from Table XXXVIII.

STEP 20 Verify that the four block status message for channel is in accordance with Table XLI and that the status error messages selected appear in the appropriate locations.

STEP 21 Apply any nine alarms simultaneously from Table XXXVIII.

STEP 22 Verify that the single block status message for Channel 1 is as specified in Table XLII.

STEP 23 Remove all alarms, verify that the status indicator is green.

4.7.11.2 Telemetry decombiner framing. (See 3.7.11.2)

STEP 1 Connect the equipment as shown in Figure 33.

STEP 2 Ensure that the following connections are made.

Signal	UUT Connections
Station Clock	TC Tim In Xmt J7 Y,Z
Error Inserter	TC Data Out J7 U,V
Error Inserter	TD Tim In Rcv 1 J7 K,L
Error Inserter	TD Data In J7 a,b
Jumper	ETXC2 J9 L to TX CLK J8 F
Jumper	ERXC1 J9 E to RX CLK J8 C
Counter	SYNC CMD TX J9 D
Counter	TP6 on TCG CCA
Counter	ETCT1 J9 F
Connections	ERCT1 J9 F thru setup to RX DPT J8 B

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TABLE XXXVIII. Alarm switch position matrix.

Measurement Alarm No.	Requirement			Verification
	Character 7-12 Error Message	Status Red	Character 5 Goes to E	(✓) Indicates Compliance
1	RCVSA1			
2	RCVSA2			
3	RCVSA3			
4	RCVRS1			
5	RCVRS2			
6	RCVRS3			
7	MGMSA1			
8	MGMSA2			
9	MGMFS1			
10	MGMFS2			
11	GMSA1			
12	GMSA2			
13	GMSA3			
14	GMSA4			
15	GMLS1			
16	GMLS2			
17	GMLS3			
18	GMLS4			
19	HCDSA1			
20	HCDSA2			
21	HCDLS1			
22	HCDLS2			
23	LCDSA1			
24	LCDSA2			
25	LCDSA3			
26	LCDLS1			
27	LCDLS2			
28	LCDLS3			
29	TGM1			
30	TGM2			

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TABLE XXXVIII. Alarm switch position matrix. (Cont'd)

Measurement Alarm No.	Requirements			Verification
	Character 7-12 Error Message	Status Red	Character 5 Goes to E	(✓) Indicates Compliance
31	RMC 1			
32	RMC 2			
33	RMC 3			
34	RMC 4			
35	CNVSA1			
36	CNVSA2			
37	RCDSA1			
38	RCDSA2			
39	RCDSA3			
40	RCDSA4			
41	RCDLS1			
42	RCDLS2			
43	RCDLS3			
44	RCDLS4			
45	XMTR1			
46	XMTR2			
47	XMTR3			
48**	OCU			
49*	DLED			
50*	DLED			
51*	TED1			
52*	TED2			
53*	TED3			
54*	TED4			
55*	VINSON			

* Logic Level

** To initiate Alarm 48, push fault lamp on power supply.

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TABLE XXXIX. Two block status message.

Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance	Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance
1	STX		21	FF	
2	A		22	A	
3	0		23	0	
4	0		24	0	
5	E		25	E	
6	9		26	9	
7			27		
8	First*		28	Third*	
9	Alarm		29	Alarm	
10	Word		30	Word	
11			31		
12			32		
13			33	↑	
14	Second*		34	↑	
15	Alarm		35	↑	
16	Word		36	↑	
17			37	↑	
18			38	↑	
19	↑		39	ETX	
20	ETB		40	ETB	

*Record the error message.

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TABLE XL. Three block status message.

Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance	Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance	Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance
1	STX		21	FF		41	FF	
2	A		22	A		42	A	
3	0		23	0		43	0	
4	0		24	0		44	0	
5	E		25	E		45	E	
6	9		26	9		46	9	
7	First*		27	Third*		47	Fifth*	
8	Alarm		28	Alarm		48	Alarm	
9	Word		29	Word		49	Word	
10			30			50		
11			31			51		
12			32			52		
13	Second*		33	Fourth*		53	:	
14	Alarm		34	Alarm		54	:	
15	Word		35	Word		55	·	
16			36			56	·	
17			37			57	·	
18			38			58	·	
19	:		39	·		59	ETX	
20	ETB		40	ETB		60	ETB	

*Record error message

Verified _____

_____

_____

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TABLE XLI. Four block status message.

Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance	Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance
1	STX		21	FF	
2	A		22	A	
3	0		23	0	
4	0		24	0	
5	E		25	E	
6	9		26	9	
7			27		
8	First*		28	Third*	
9	Alarm		29	Alarm	
10	Word		30	Word	
11			31		
12			32		
13			33		
14	Second*		34	Fourth*	
15	Alarm		35	Alarm	
16	Word		36	Word	
17			37		
18			38		
19	†		39	†	
20	ETB		40	ETB	

*Record the error message.

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TABLE XLI. Four block status message. (Cont'd)

Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance	Measurement CH1 Character	Requirement Symbol	(✓) Indicates Compliance
41	FF		61	FF	
42	A		62	A	
43	0		63	0	
44	0		64	0	
45	E		65	E	
46	9		66	9	
47			67		
48			68		
49	Fifth* Alarm Word		69	Seventh* Alarm Word	
50			70		
51			71		
52			72		
53			73	↑	
54			74	↑	
55	Sixth* Alarm Word		75	↑	
56			76	↑	
57			77	↑	
58			78	↑	
59	↑		79	ETX	
60	ETB		80	ETB	

*Record the error message.

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TABLE XLII. Single block status message.

Measurement CH1 Character	Requirement Character	(✓) Indicates Compliance
1	STX	
2	A	
3	0	
4	0	
5	E	
6	9	
7	M	
8	A	
9	J	
10	0	
11	R	
12	↑	
13	M	
14	A	
15	J	
16	0	
17	R	
18	↑	
19	ETX	
20	ETB	

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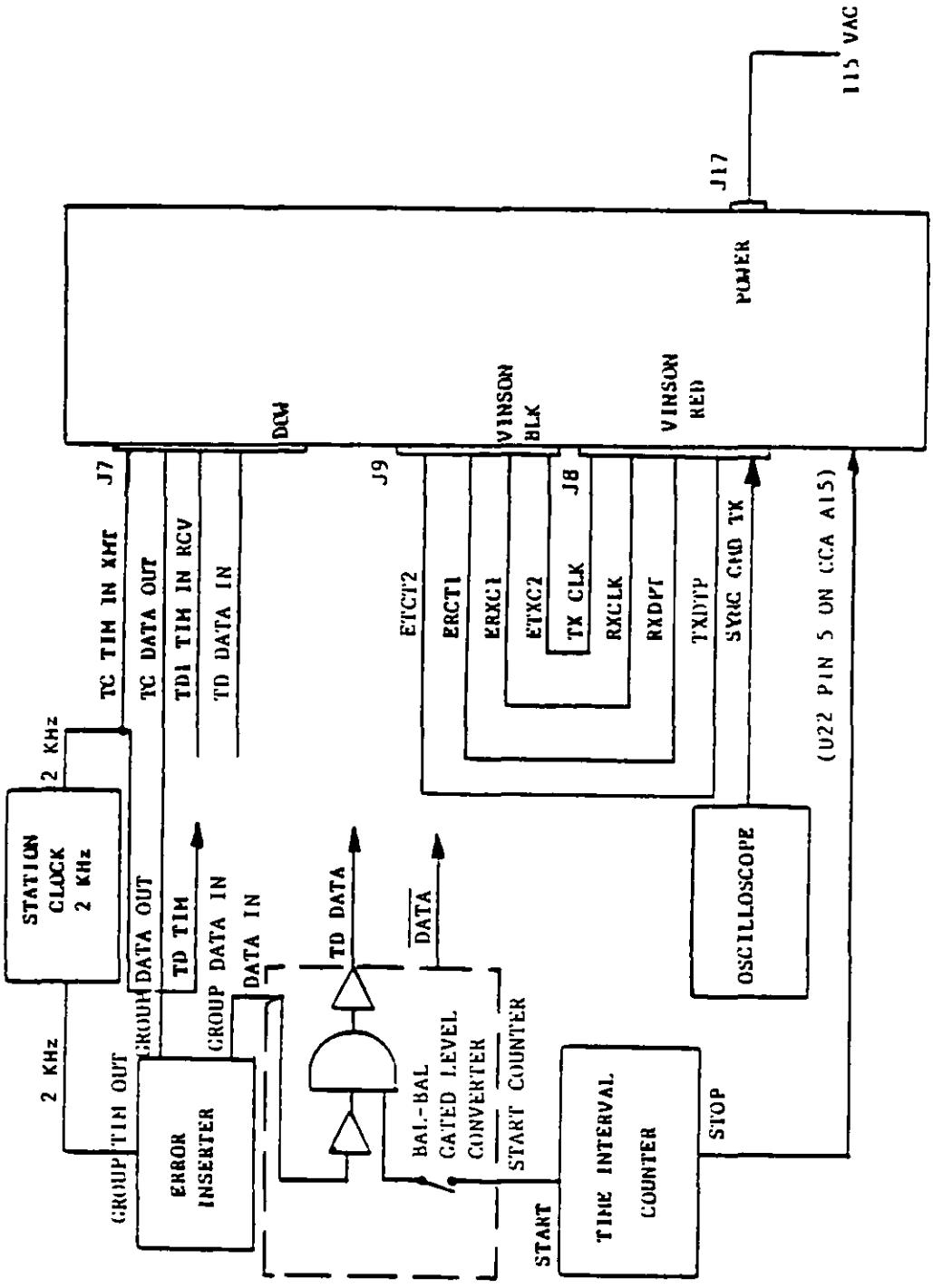


FIGURE 33. Telemetry decombiner framing.

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STEP 3 Set the UUT and control panel switches as shown in Tables XLIII and XXI.

STEP 4 Connect the stop channel of the interval counter to U22 Pin 5 on the TCG CCA (OOF signal) (falling edge).

STEP 5 Connect the start channel of the interval counter to trigger upon application of data (rising edge).

STEP 6 Set the BER on the error inserter to 10^{-3}, measure and record the lock in time on the counter 20 times.

Measurement Lock In Time Test	Measured Value	Measurement Lock In Time Test	Measured Value
1		11	
2		12	
3		13	
4		14	
5		15	
6		16	
7		17	
8		18	
9		19	
10		20	

STEP 7 Verify that the following requirement is met.

Measurement	Requirement
Lock In Time	No more than 2 of 20 measurements shall exceed 0.4 seconds

STEP 8 Connect the stop channel of the counter to the R17 OOF on CCA A15.

STEP 9 Set the start channel to trigger upon cessation of the data.

STEP 10 Connect TDI DATA IN to the DATA OUT of the BAL-BAL Gated Level Converter.

STEP 11 Measure and record the drop-out time on the counter 20 times.

STEP 12 Set the BER to 10^{-3} and gate the data on.

STEP 13 Connect the oscilloscope to LED OOF, monitor the LED for 24 hours. There shall be no more than 1 indication during the 24 hour period.

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TABLE XLIII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	OFF
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	RCV
	TLMY Decomb 1	ON/OFF	ON
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

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4.7.11.3 Resync command pulse rate. (See 3.7.11.3)

STEP 1 The equipment test setup is the same as in 4.7.11.1.

STEP 2 Gate the DATA OFF; using the oscilloscope, measure the time between resync commands. Verify that the time is between 1.6 and 5 seconds.

Measurement	Requirement
Maintenance	No more than 1 indication in 24 hours
Resync Command Pulse	Between 1.6 and 5.0 seconds

STEP 3 Verify that the Resync Command Pulse meets the requirements of 3.7.11.3.

4.7.11.4 DOW BCI. (See 3.7.11.4)

STEP 1 Connect the equipment as shown in Figure 34.

STEP 2 Ensure that the following connections are made.

Signal	UUT Point of Connection
Logic analyzer	16 Bit Buss CCA A15
2 kHz station clock	J7 TC Tim In Xmt Y, Z
Frequency synthesizer	J8 RX CLK C
Jumper	J8 TX CLK G to J9 ETXC2 L

STEP 3 Set the UUT and control panel switches as shown in Tables XLIII and XXI.

STEP 4 With the frequency set to 2002 Hz and 1998 Hz, measure and record the time between reset indications (this is 1000 times the specified offset). Locate the 16 bit word for FIFO Reset on the Logic Analyzer. Then measure the time between consecutive FIFO reset indications.

NOTE: FIFO RESET=015A
0000/0001/0101/1010
} PROGRAM ADDRESS

Measurement Offset	Requirement
At 2000.2 Hz	900 ms minimum
At 1999.8 Hz	

STEP 5 Verify that the specified requirement of 3.7.11.4 is met.

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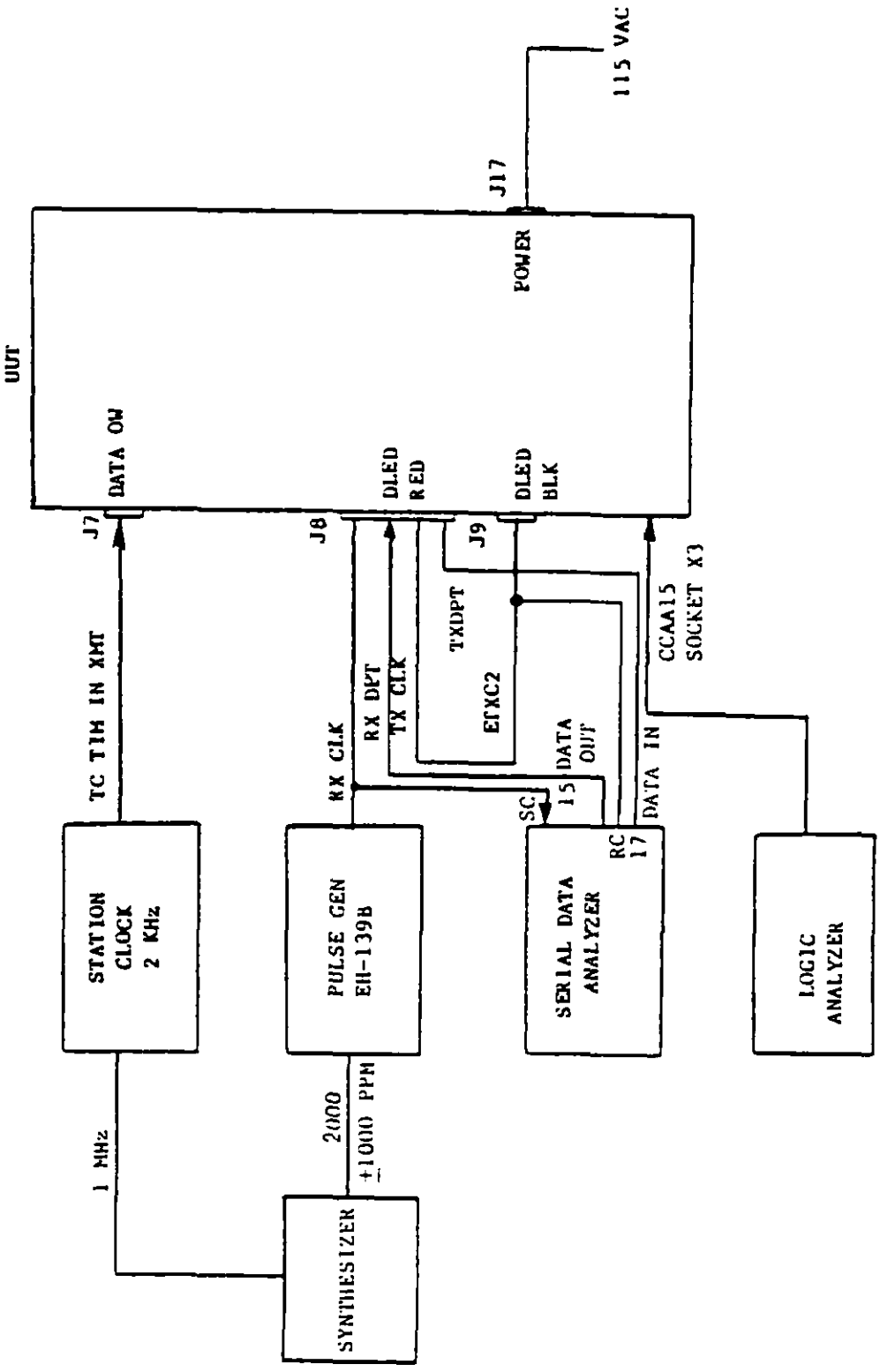


FIGURE 34. DOW BCI.

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4.7.11.5 DCM multiplexing. (See 3.7.11.5)

STEP 1 Connect the equipment as shown in Figure 35.

STEP 2 Set the UUT switch positions as shown in Tables XXXIII and XXI.

STEP 3 With the oscilloscope, observe the DCM 1 frame format.

DCM 1 and 2 Frame Format

	FRAME BIT	CH1	CH2	CH3	CH4	CH5	CH6	CH7
	0	1	1	1	1	1	1	1
	0	1	1	1	1	1	1	1
	0	1	1	1	1	1	1	1
DATA BIT -	1	1	1	1	1	1	1	1

STEP 4 Connect the DSVT simulator to DCM Channels 1 through 7 one at a time.

STEP 5 Verify that as each channel is selected, the one data bit on the selected channel changes to random data and that the data bits move down the frame in sequence with the selected channel.

STEP 6 Repeat this procedure for DCM 2 Channels 8 through 14.

STEP 7 Connect the equipment as shown in Figure 36.

STEP 8 Adjust the frequency synthesizer to 2000.020 Hz. Measure the time between losses of BCI as observed by errors on DSVT Simulator #2.

STEP 9 Repeat step 3 with the frequency synthesizer set to 1999.980 Hz. Record the results.

DCM Channel No.	Requirement Time Between Losses	DCM BCI @2000.020 Hz	DCM BCI @1999.980 Hz
1	18 sec. min.	50	25
2		50	25
3		50	25
4		50	25
5		50	25
6		50	25
7		50	25
8		50	25
9		50	25
10		50	25
11		50	25
12		50	25
13		50	25
14		50	25

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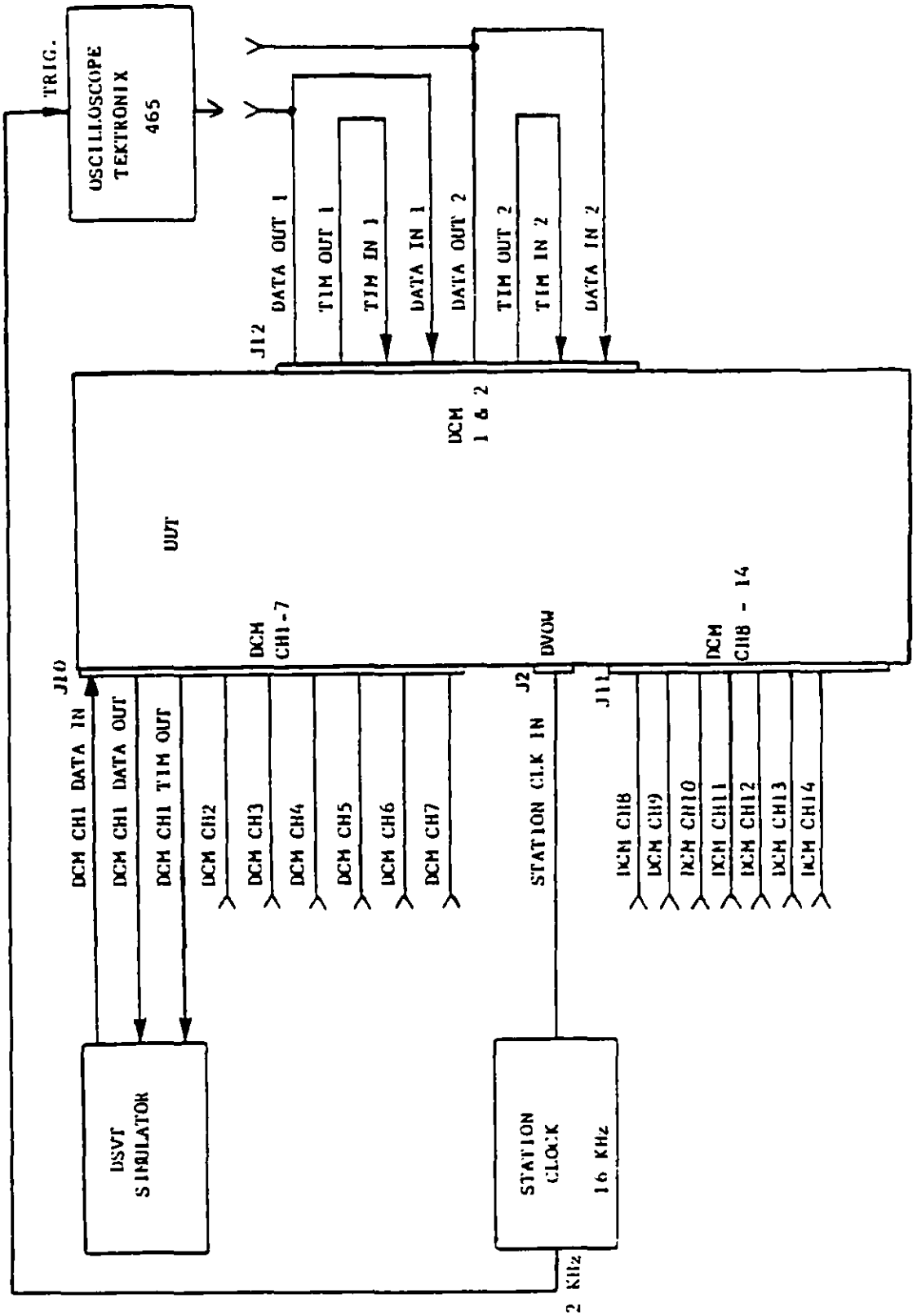


FIGURE 35. DCM multiplexing.

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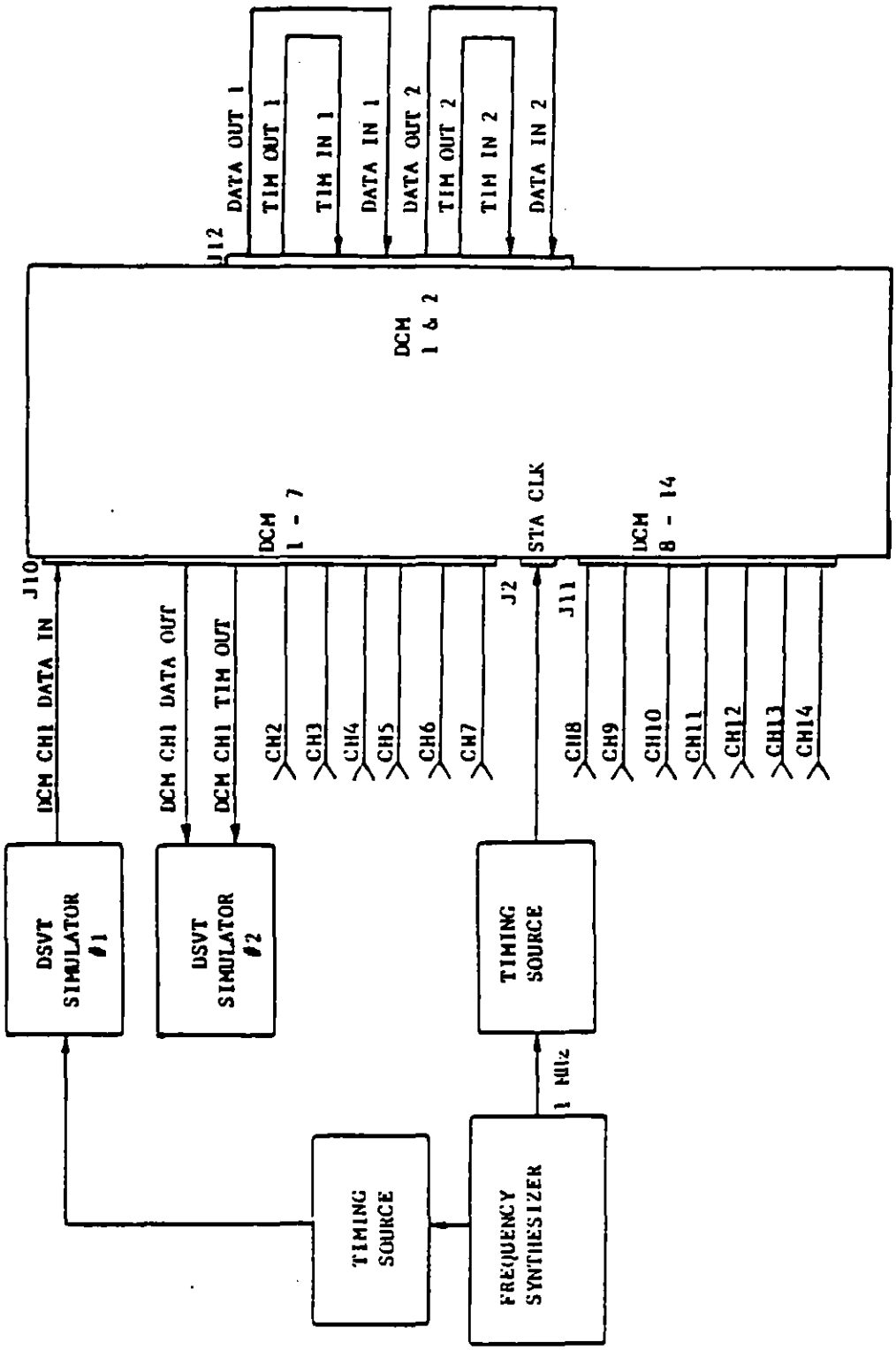


FIGURE 36. DCM channel BCI test.

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4.7.11.6 DCM framing test. (See 3.7.11.6)

STEP 1 Connect the equipment as shown in Figure 37.

STEP 2 Set the UUT switches as shown in Tables XXXIII and XXI.

STEP 3 Set up the error inserter for 10^{-3} error rate.

STEP 4 Squelch the group data at Error Inserter Test Set to cause loss of frame. Release squelch to restore data.

STEP 5 Measure the time from resumption of data to frame-in. Use the squelch release signal at the Error Inserter Test Set to start the interval measurement, and the signal at U21-1 (1=IN FRAME, 0= OUT-OF-FRAME) to stop the time interval measurement. Take 20 measurements and record the results.

No.	Time	No.	Time	No.	Time	No.	Time
1		6		11		16	
2		7		12		17	
3		8		13		18	
4		9		14		19	
5		10		15		20	

STEP 6 Verify that no more than 2 of 20 measurements exceed 100 ms.

Measurement	Requirement
DCM Frame-In Test	No more than 2 of 20 measurements exceed 100 ms.

STEP 7 Set the error inserter for 10^{-3} error rate.

STEP 8 Monitor the frame signal to detect loss of frame for 240 hours. The unit shall not lose frame more than once.

Measurement	Requirement
Frame Maintenance	No more than one time in 240 hours.

STEP 9 Connect the equipment as shown in Figure 38.

STEP 10 Set the Error Inserter Test Set for a 10^{-3} error rate.

STEP 11 Connect the time interval counter to start at Data Squelch and stop at In-Frame condition. (U21-1, 1= IN FRAME, 0=OUT-OF-FRAME).

STEP 12 With the DCM in frame, reset the counter and measure the time between reset and reacquisition of frame. Take 20 measurements and record the data.

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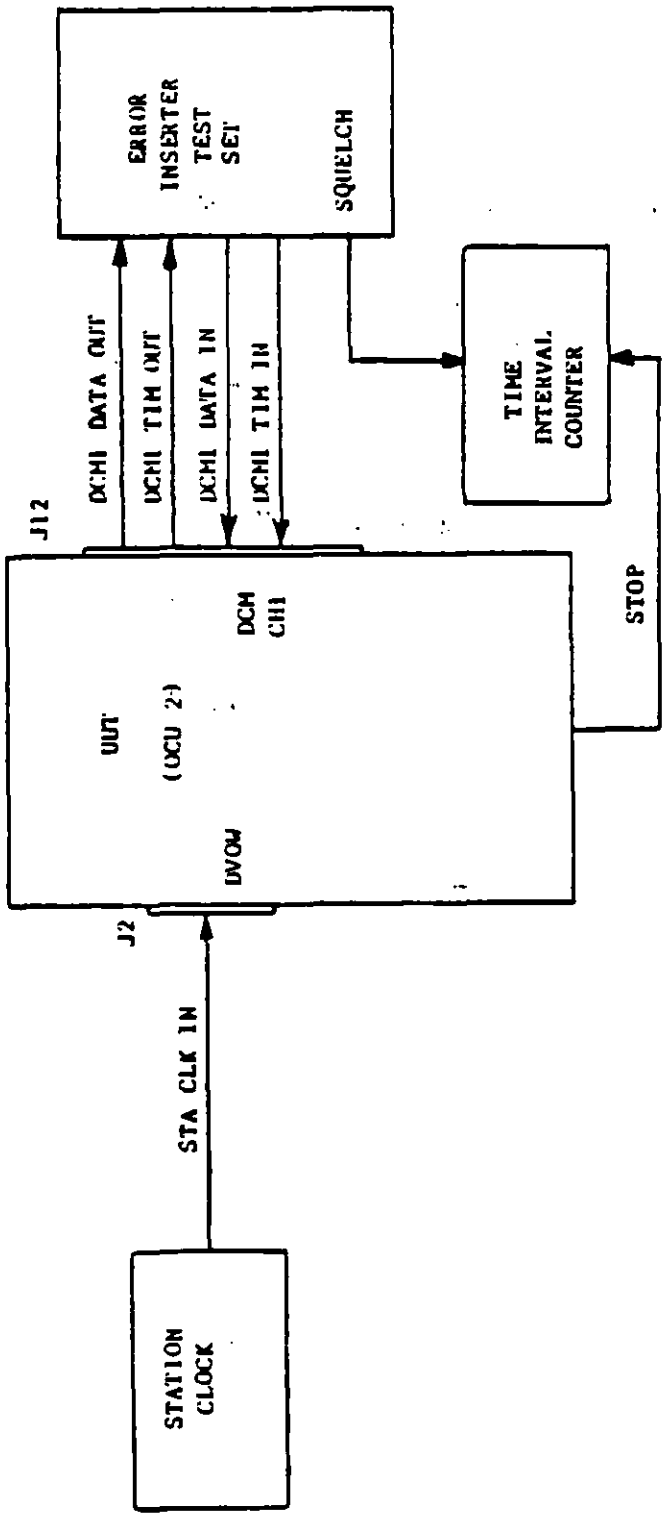


FIGURE 37. DCM framing test.

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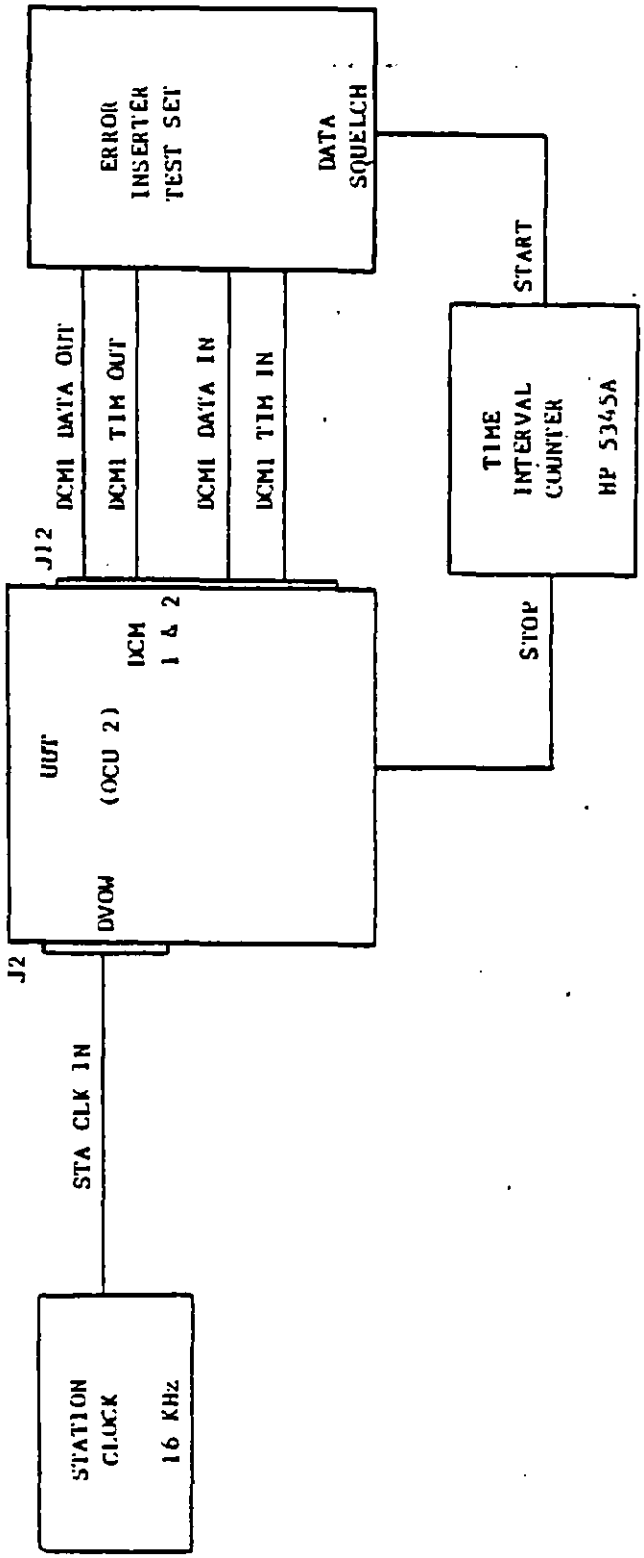


FIGURE 38. DCM frame loss and reacquisition test.

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No.	Time	No.	Time	No.	Time	No.	Time
1		6		11		16	
2		7		12		17	
3		8		13		18	
4		9		14		19	
5		10		15		20	

STEP 13 Verify that no more than 2 of 20 measurements exceed 200 ms.

Measurement	Requirement
Loss of frame and Reacquisition	No more than 2 of 20 measurements exceed 200 ms.

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4.7.12 Power variation test. (See 3.7.12)

STEP 1 Connect the equipment as shown in Figure 39.

STEP 2 Ensure that the following connections are made.

Signal	Connector	UUT Connections	
DSVT Simulator	J1	DVOW Data In	B,C
DSVT Simulator	J1	DVOW Data Out	P,R
Station Clock	J2	STA CLK In	c,d
Station Clock	J7	TC Tim In Xmt	Y,Z
Station Clock	J7	TD Tim In	K,L
Jumper	J7	TD Data Out	U,V
		to	
		TD Data In	a,b
RMS Voltmeter	J6	AVOW CH 2 Out	S,T
RMS Voltmeter	J6	AVOW CH 3 Out	U,V
RMS Voltmeter	J6	AVOW CH 4 Out	W,X
RMS Voltmeter	J6	AVOW CH 5 Out	Y,Z
RMS Voltmeter	J6	AVOW CH 6 Out	A,c
Oscillator	J6	AVOW CH 1 In	B,C
Jumper	J2	Test Data Out	W,X
	J2	Test Data In	Y,Z
Jumper	J9	ERXC1	E
		to	
	J8	RX CLK	C
Jumper	J9	ERTC1	D
		to	
	J8	RX DPT	B
Jumper	J9	ETCT2	K
		to	
		TX DPT	F
Jumper	J9	ETXC2	L
		to	
	J8	TX CLK	G
Power Source	J7	High	A
		Low	C
		Ground	B

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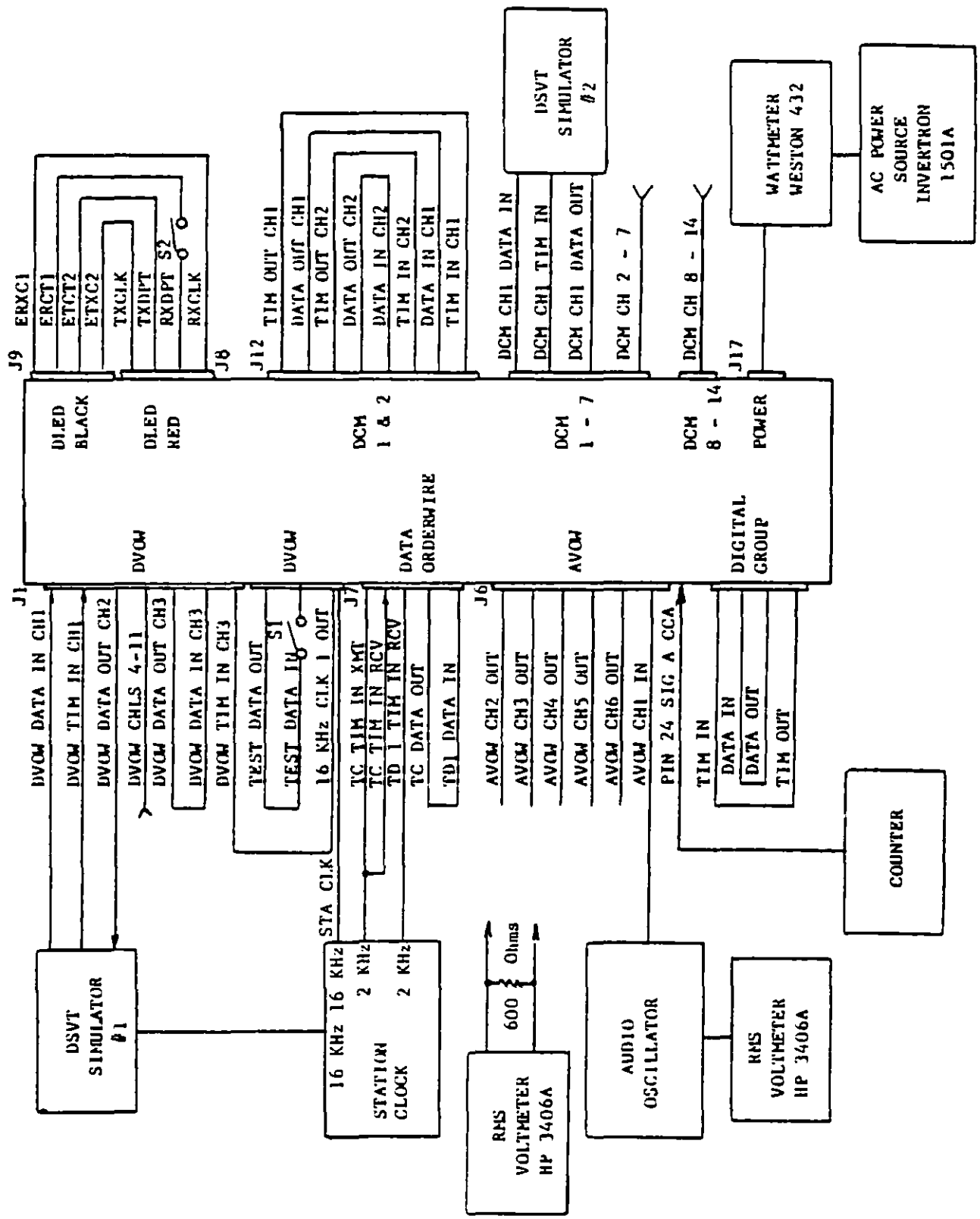


FIGURE 39. Power variation.

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STEP 3 Set the UUT and control panel switches as shown in Tables XLIV and XLV. Adjust the input power to 115 Vac at 60 Hz.

STEP 4 With the oscillator connected to AVOW CH1 IN, adjust the oscillator output to -18.2 dBm at 1600 Hz. Verify and record the AVOW CH1 pushbutton flashes and that there are no alarms or BITE indications.

NOTE: Alarms may come on during remainder of test.

STEP 5 Set the oscillator frequency to 1000 Hz, and push the AVOW CH1 pushbutton twice. Verify and record that the AVOW 1 pushbutton light goes out.

STEP 6 With the oscillator connected to AVOW CH1, measure and record the AVOW 2 through 6 output level.

STEP 7 Connect the frequency counter to AVOW CH1 OUT, push CH1 button and push RING button, measure and record the output frequency.

STEP 8 Push the DVOW CH3 pushbutton, set the ring code to position 1 and push the RING button. Verify that DVOW Channel 3 button flashes.

STEP 9 Remove the Channel 3 data and timing inputs.

STEP 10 With the DSVT Simulator DATA IN and TIM IN connected to DVOW Channel 1, measure and record the DVOW Error Rate at Channel 2 over a 30 second period.

STEP 11 Connect the DSVT DATA and TIMING to Channel 2. Measure and record the DVOW Error Rate at Channel 1 over a 30 second period.

STEP 12 Repeat steps 10 and 11 for Channels 3 and 4, Channels 5 and 6, Channels 7 and 8, Channels 9 and 10, Channels 11 and 12.

STEP 13 Open the connection between TEST DATA IN and TEST DATA OUT (S1). Verify that the ERROR LED is lit. Close (S1) and verify that the ERROR LED goes out.

STEP 14 Open the connection between RXDPT and ERCT1 (S2). Verify that the LOI is indicated. Close (S2) and verify that the LOI and OOF are off (no indication).

STEP 15 Connect DSVT Simulator #2 to DCM Channel 1. Measure and record the DOW error rate over a 30-second period.

STEP 16 Repeat step 15 for DCM Channels 2 through 14.

STEP 17 Measure and record the input power (line voltage, frequency and current).

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TABLE XLIV. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	256
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	OFF
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

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TABLE XLV. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1		X
	2		X
	3		X
	4		X
	5		X
	6		X

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1		X						
	2		X						
	3		X						
	4		X						
	5		X						
	6		X						
	7		X						
	8		X						
	9		X						
	10		X						
	11		X						
	MG		X						
	P		X						

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
		X									

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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STEP 18 Repeat steps 3 through 16 for each of the combinations of voltage and frequency specified in Table XLVI.

STEP 19 Apply 156 Vac at 60 Hz for 6 seconds maximum. Verify that there is no power supply out of tolerance indication and that the pilot light remains ON.

STEP 20 Apply 115 Vac at 60 Hz and verify that there is no power supply out of tolerance indication and that the pilot lamp is ON.

TABLE XLVI. Power variation.

Measurement	Requirement	Voltage and Frequency				
		115 V 60 Hz	103.5 V 47.5 Hz	103.5V 420 Hz	126.5 V 47.5 Hz	126.5 V 420 Hz
Normal Operation	No BITE indications					
AVOW CH 1	AVOW Light Flashes					
	AVOW Light Goes Out					
AVOW OUTPUT LEVEL CH2	20.3 to 22.3 dBm					
CH3						
CH4						
CH5						
CH6						
Frequency	1600 Hz ± 32 Hz					
DVOW Ring	DVOW 3 Button Flashes					
BER DVOW CH1 to CH2	Zero Errors in 30 sec					
CH2 to CH1						
CH3 to CH4						
CH4 to CH3						
CH5 to CH6						
CH6 to CH5						
CH7 to CH8						
CH8 to CH7						
CH9 to CH10						
CH10 to CH9						
CH11 to CH2						
CH2 to CH11						

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TABLE XLVI. Power variation. (Cont'd)

Measurement	Requirement	Voltage and Frequency				
		115 V 60 Hz	103.5 V 47.5 Hz	103.5 V 420 Hz	126.5 V 47.5 Hz	126.5 V 420 Hz
TEST DATA IN/OUT OPEN (S1)	Error LED illuminates					
CLOSED	Error LED goes out					
RXDPT/ERCT1 (S2) OPEN	LOI illuminates					
CLOSED	LOI and OOF not illuminated					
DCM BER CH1	Zero errors in 30 seconds					
CH2						
CH3						
CH4						
CH5						
CH6						
CH7						
CH8						
CH9						
CH10						
CH11						
CH12						
CH13						
CH14						
Line Voltage	N/A					
Line Frequency	N/A					
Input Current	N/A					
Input Power	80 W maximum					
Voltage Transient 156 Vac for 6 sec maximum	No power supply out tolerance or faults before and after application.	(✓) indicates compliance				

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4.7.13 BITE and alarms. (See 3.7.13)

STEP 1 Connect the equipment as shown in Figure 40.

STEP 2 Ensure that the following connections are made.

Signal/Unit	Connector	UUT Connections
GM GP1 TLM Tim Out J15 C,D	J7	TC Tim In Rcv W,X
GM GP1 TLM Data In J15 A,B	J7	TC Data Out U,V
GM GP2 TLM Data Out J15 J,K	J7	TD Data In a,b
GM GP2 TLM Tim Out J15 L,M	J7	TD Tim In Rcv K,L
Station Clock	J2	STA CLK In c,d
Jumper thru Switch	J2	Test Data Out W,X
	to J2	to Test Data In Y,Z
Long term with short term switch	J2	16 kHz H
	to J2	to 16 kHz J
Jumper	J9	ERXC1 D
	to J8	to RX CLK C
Jumper	J9	ERCT1 F
	to J8	to RX DPT B
Jumper	J9	ETCT2 K
	to J8	to TX DPT F
Jumper	J9	ETXC2 L
	to J8	to TX CLK G

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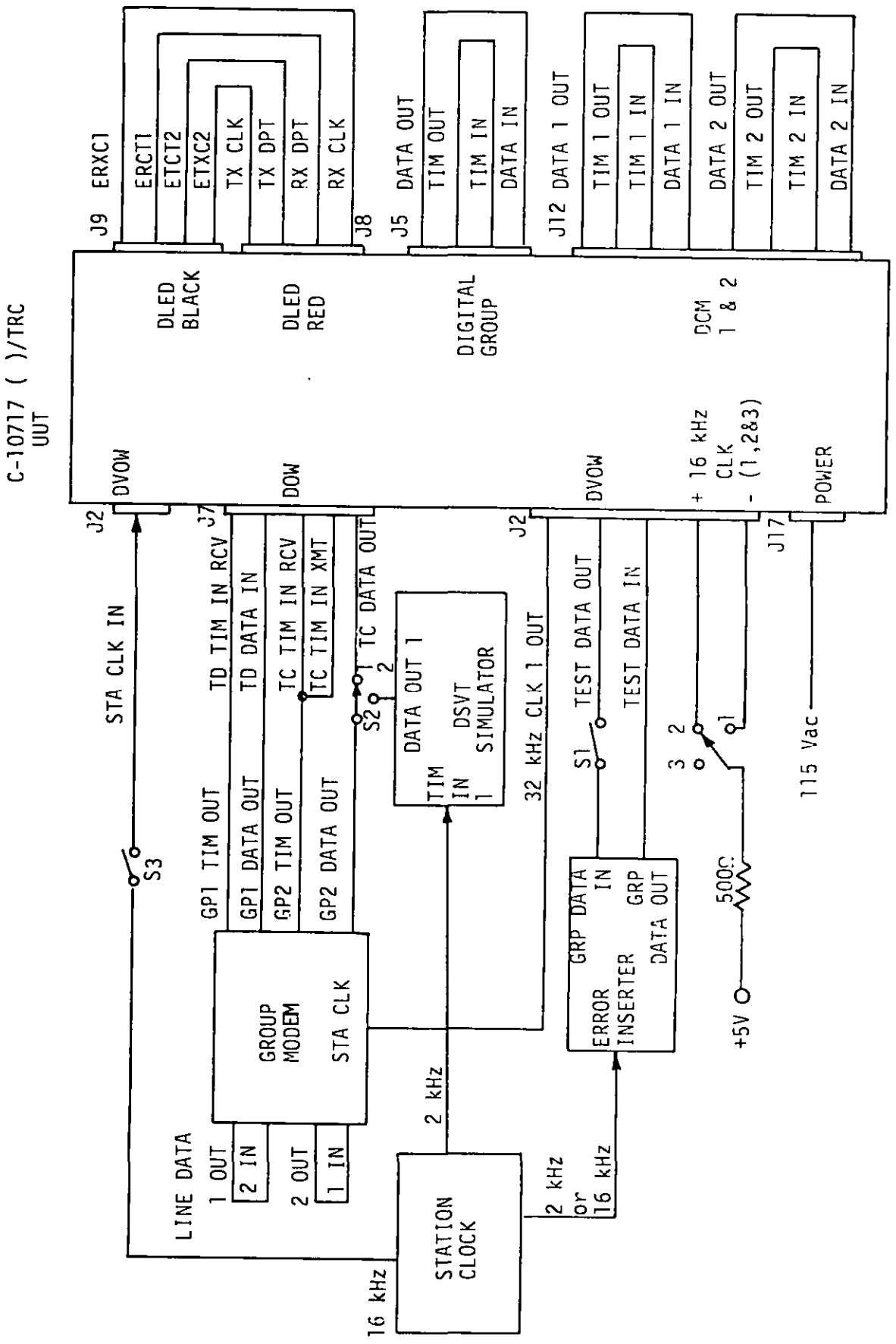


FIGURE 40. BITE and alarms.

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STEP 3 Set the UUT switches and control panel switches as shown in Tables XLVII and XXI.

STEP 4 Verify that the CCA A1 RATE SELECT switch and timing source are set to 16 kb/s and PSEUDO RANDOM GENERATOR (PRG) switch is OFF; and the error insert is set to minimum error rate.

STEP 5 Open S1, verify that the ERROR LED on the BITE CCA is OFF.

STEP 6 Turn on the PRG switch, verify that the ERROR LED on the BITE CCA is ON.

STEP 7 Close S1, verify that the ERROR LED goes OFF.

STEP 8 Repeat steps 4 through 7 with the RATE SELECT switch and timing source set to 2 kb/s.

STEP 9 Record the results.

STEP 10 Set the error inserter to 10^{-4}. Verify that LED flashes (ERROR LED).

Operation	Requirement	Verified	
		16 kb/s	2 kb/s
STEP 5 S1 Open	LED ON BITE OFF		
STEP 6 PRG ON	LED ON BITE ON		
S1 Closed	LED ON BITE OFF		
STEP 10	LED Flashes		

STEP 11 Repeat step 10 at 16 kb/s.

STEP 12 Return error inserter to minimum error rate.

No Fault Operation

STEP 13 Verify that the UUT switches and control panel switches are set as specified in Tables IX and XIV, S1 closed, S2 to TC DATA OUT, S3 closed, and S4 position 3.

STEP 14 Verify that the following requirements are met.

Measurement	Requirement
Summary Alarm and Out-of-Frame	Summary FAULT ON
	OOF LED on TCG is ON
	Status indicator remains green

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TABLE XLVII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOCAL
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	RCV
	TLMY Decomb TMG	XMT/RCV	RCV
	TLMY Decomb 1	ON/OFF	ON
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2		1
	S3		1

* See Front Cover for code switch positions.

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Summary Alarm and Out-of-Frame

STEP 15 Set S2 to the DSVT Simulator. Verify the following requirements.

Measurement	Requirement
No Fault Operation	All LEDs are OFF
	Summary status is green
	Summary status red and amber are OFF

Summary Alarm and Loss of Input

STEP 16 Open S3, verify the following requirements

Measurement	Requirement
Summary Alarm and Loss Input	Summary FAULT ON
	LOI LED on BITE ON Error light is ON
	Status indicator remains green

Summary Alarm and OCU Alarm and Delay

STEP 17 Set S4 to position 2, close S3.

STEP 18 Repeat step 1 with S4 in position 1.

Measurement	Requirement
Summary and OCU II Alarms	Summary FAULT is ON
	LED ON timing distribution card is ON
	Green status indicator is OFF
	Red status indicator is ON
	Audible alarm sounds

STEP 19 Measure the time between application of S4 and appearance of red status indicator. Verify that the following requirement is met.

Measurement	Requirement
Alarm Delay	1 second min

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4.7.13.1 Alarm operation. (See 3.7.13.1)

STEP 1 Press the SUMMARY FAULT button on the power supply. Verify that the following requirements are met.

Measurement	Requirement
Alarm Test	Summary FAULT is ON
	Status lamp is red
	Green status lamp goes out
	Audible alarm sounds

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4.7.14 Operational test. (See 3.7.14)

a. UUT Preparation.

STEP 1 Remove the front cover to gain access to the printed circuit boards.

STEP 2 Set the UUT CCA switches and control panel switches as shown in Table XLVIII and XLIX.

b. Electrical test setup.

STEP 1 Connect the equipment as shown in Figure 41.

STEP 2 Ensure that the following connections are made.

Signal	Connector	UUT Connections		
		Pins		
		Low	High	Ground
DSVT Simulator:				
DVOW CH1 DATA IN	J1	B	C	A
DVOW CH1 DATA OUT		q	r	A
DVOW CH2 DATA IN		D	E	A
DVOW CH2 DATA OUT		s	t	A
DVOW CH3 DATA IN		F	G	A
DVOW CH3 DATA OUT		u	v	A
DVOW CH4 DATA IN		H	J	A
DVOW CH4 DATA OUT		w	x	A
DVOW CH5 DATA IN		K	L	A
DVOW CH5 DATA OUT		y	z	A
DVOW CH6 DATA IN		M	N	A
DVOW CH6 DATA OUT		AA	BB	A
DVOW CH7 DATA IN		P	R	A
DVOW CH7 DATA OUT		CC	DD	A
DVOW CH8 DATA IN		S	T	A
DVOW CH8 DATA OUT		EE	FF	A
DVOW CH9 DATA IN		U	V	A
DVOW CH9 DATA OUT	J1	GG	HH	A

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TABLE XLVIII. OCU II CCA switch position matrix.

CCA No.	Switch Name	Nomenclature	Position
A1	Rate Select	2 kb/s 16 kb/s	16 kb/s
	Pseudo Random Gen	ON/OFF	ON
A6	Frame Delay	ON/OFF	OFF
	TED	YES/NO	NO
A7	Bit Rate	512/256	512
	Mux	ON/OFF	ON
A10	PVT to MG	ON/OFF	OFF
A11	Timing	LOCAL/LOOP	LOCAL
A12	Timing	LOCAL/LOOP	LOOP
A13	Ring Detector	MUX/GP	GP
	PVT Channel	MUX/EXT	EXT
	TLMY Comb TMG	XMT/RCV	XMT
	TLMY Decomb TMG	XMT/RCV	XMT
	TLMY Decomb 1	ON/OFF	ON
A15	TLMY Decomb 2	ON/OFF	OFF
	TLMY Decomb 3	ON/OFF	OFF
	Station ID Alpha S1	1 thru 4 * Codes A thru Z	A
	Station ID Numeric (1) S1	5 thru 8 * Codes 0 thru 9	0
	Station ID Numeric (2) S2	1 thru 4 * Codes 0 thru 9	0
	Channel Select S2	5 thru 8 * Codes 1 thru 13	1
	Ber Threshold S3	5 thru 8 * Codes 1 thru 9	8
A8	VBRD Ring Codes S1	Codes 1 thru 16 *	1
	S2	Codes 1 thru 16 *	1
	S3		1

*See Front Cover for code switch positions.

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TABLE XLIX. OCU II control panel switch position matrix.

Switch		Position	
		Bridge	Terminate
Analog	1	X	
	2	X	
	3	X	
	4	X	
	5	X	
	6	X	

Switch		Position							
		T	B	1	2	3	4	5	6
Digital	1		X						
	2		X						
	3		X						
	4		X						
	5		X						
	6		X						
	7		X						
	8		X						
	9		X						
	10		X						
	11		X						
	MG		X						
	P		X						

Switch Timing Buffer	Position			
	S	1	2	3
	X			

Switch Timing Station	Position										
	S	1	2	3	4	5	6	M	S2	S2	S1
											X

Switch Digital Ring Code	Position															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	X															

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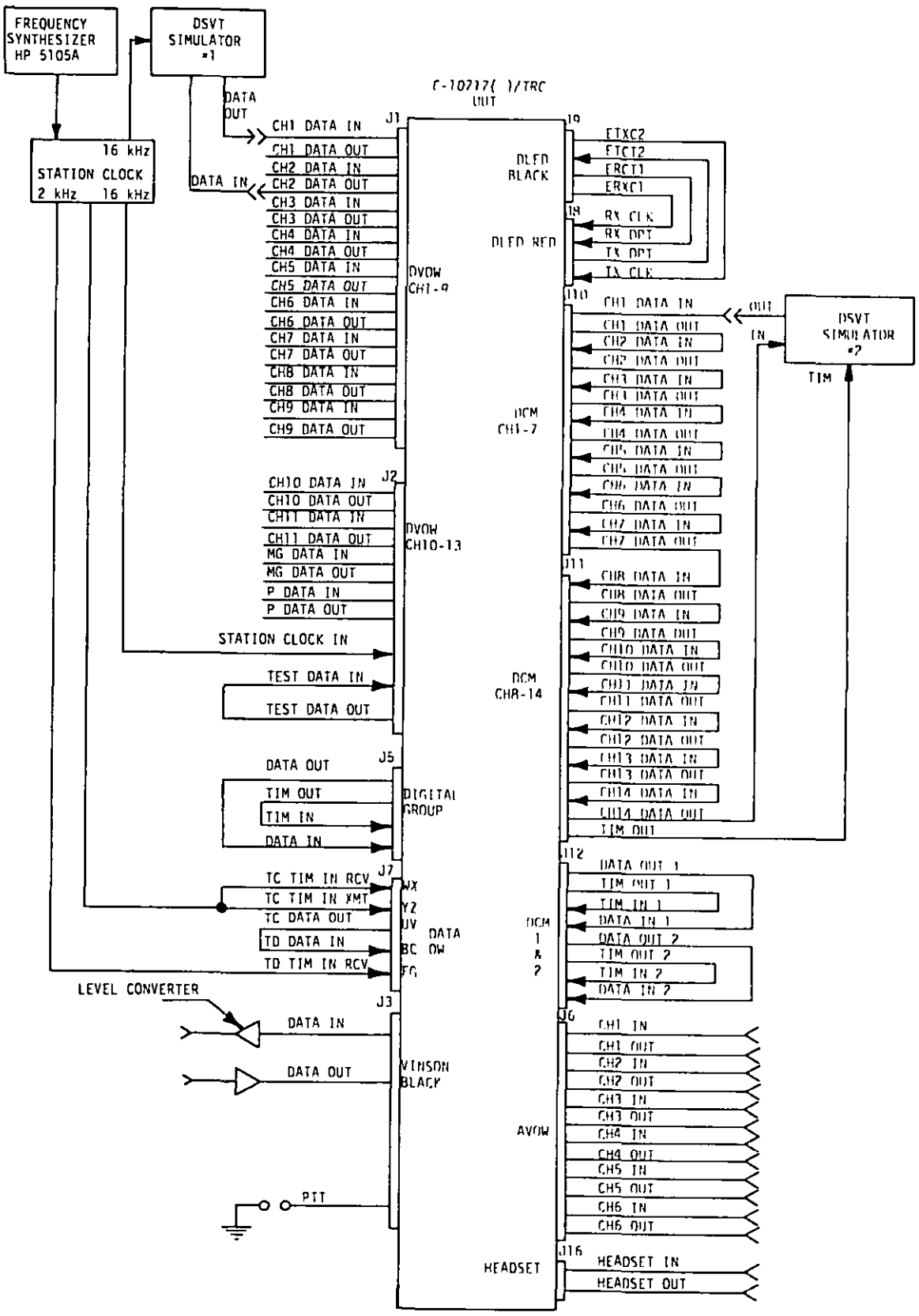


FIGURE 41. OCU II Unit operational electrical test.

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Signal	Connector	UUT Connections		
		Pins		
		Low	High	Ground
DVOW CH10 DATA IN	J2	W	X	A
DVOW CH10 DATA OUT		n	p	A
DVOW CH11 DATA IN		Y	Z	A
DVOW CH11 DATA OUT		g	r	A
DVOW CH MG DATA IN		a	b	A
DVOW CH MG DATA OUT		s	t	A
DVOW CH P DATA IN		c	d	A
DVOW CH P DATA OUT	J2	u	v	A
AVOW CH1 DATA IN	J6	B	C	A
AVOW CH1 DATA OUT		P	R	A
AVOW CH2 DATA IN		D	E	A
AVOW CH2 DATA OUT		S	T	A
AVOW CH3 DATA IN		F	G	A
AVOW CH3 DATA OUT		U	V	A
AVOW CH4 DATA IN		H	J	A
AVOW CH4 DATA OUT		W	X	A
AVOW CH5 DATA IN		K	K	A
AVOW CH5 DATA OUT		Y	Z	A
AVOW CH6 DATA IN		M	N	A
AVOW CH6 DATA OUT	J6	a	b	A
HEADSET DATA IN	J16	A	B	-
HEADSET DATA OUT		C	D	-

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Signal	Connector	UUT Connections		
		Pins		
		Low	High	Ground
Station Clock:				
2 kHz - TC TIM IN XMT	J7	Y	Z	A
2 kHz - TD TIM IN RCV	J7	W	X	A
16 kHz - STATION CLOCK	J2	$\underline{w}$	$\underline{x}$	A
16 kHz - DSVT SIMULATOR	-	-	-	-
Jumper:				
TC DATA OUT TO	J7	U	V	A
TD DATA IN	J7	B	C	A
ETXC2 TO	J9	L	-	-
TX CLOCK	J8	G	-	-
TXDPT TO	J8	K	-	-
ETCT2	J9	K	-	-
ERCT1 TO	J9	D	-	-
RXDPT	J8	B	-	-
ERXC1 TO	J9	E	-	-
RX CLK	J8	C	-	-
TEST DATA OUT TO	J2	$\underline{y}$	$\underline{z}$	A
TEST DATA IN	J2	$\overline{AA}$	$\overline{BB}$	A
Level Converter:				
VINSON DATA IN	J3	B	-	-
VINSON DATA OUT	J3	C	-	-
Time Interval Counter Through Switch to PTT	J3	D	-	-

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4.7.14.1 Alarm monitor check. (See 3.7.14.1) The BITE circuitry is used exclusively to continuously monitor the DOW circuits. Proper operation is monitored by verification that no BITE or fault lamps are illuminated. The green status light shall be ON.

STEP 1 Observe that there are no alarms or LED fault indications. Record the results.

STEP 2 Open S1 (TEST DATA IN/TEST DATA OUT). Verify that the LOI illuminates.

STEP 3 Close S1. Verify that LOI and OOF are extinguished.

STEP 4 Open S3 (TC DATA OUT/TD1 DATA IN). Verify that the LED on the PRG illuminates.

STEP 6 Close S2. Verify that the LED on the PRG is extinguished.

4.7.14.2 DVOW error rate test. (See 3.7.14.2)

STEP 1 Connect the DSVT Simulator #1 output to DVOW Channel 1 input and the DSVT Simulator #1 input to DVOW Channel 2 output.

STEP 2 Monitor the DVOW error detector counter for a period of 30 seconds. Verify that there are no errors generated. Record the results.

STEP 3 Connect the DSVT Simulator input consecutively to DVOW Channels 2 through 11, MG, P and the Operator channel input. Measure the error rate as in step 2 at each channel.

STEP 4 Connect the DSVT Simulator output to DVOW Channel 1 output. Measure the error rate as in step 2.

STEP 5 Repeat step 4 with the DSVT Simulator output connected to DVOW Channels 3 through 11, MG and P.

STEP 6 Connect the DSVT Simulator output to the operator input. Push the PTT switch. Measure the error rate as in step 2.

4.7.14.3 DVOW ring test. (See 3.7.14.3)

STEP 1 Connect the output of DVOW Channel 2 to the input of DVOW Channel 1.

STEP 2 Set the DVOW Channel 2 thumbwheel switch on the control panel to position T. Push the DVOW Channel 2 pushbutton.

STEP 3 Push the RING button on the control panel. Verify that the DVOW Channel 1 light flashes.

STEP 4 Push the DVOW Channel 1 button. Verify that the DVOW Channel 1 light is on steadily.

STEP 5 Push the DVOW Channel 1 button. Verify that the DVOW Channel 1 light goes out. Record the results.

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4.7.14.4 AVOW levels. (See 3.7.14.4)

STEP 1 Connect the oscillator output to Channel AVOW CH1 input and the RMS voltmeter to Channel 2 as shown in Figure 42.

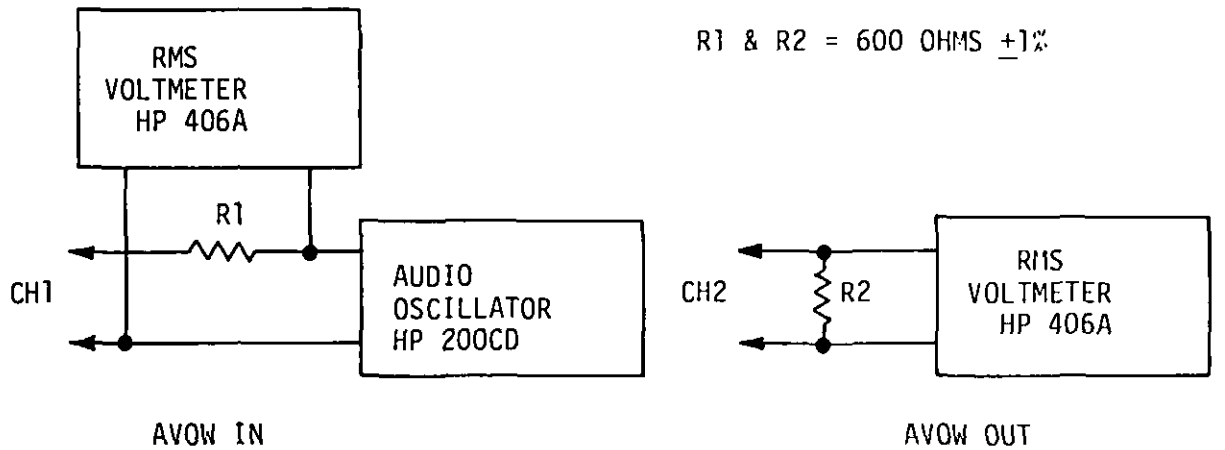


FIGURE 42. AVOW levels.

STEP 2 Adjust the audio oscillator at 1 kHz to produce an output level of -12.2 dBm. (This level will produce -18.2 dBm into 600 ohms).

STEP 3 Measure and record the output of Channels 2 through 6 into 600 ohms. Verify that the outputs are between +20.3 and +22.3 dBm.

STEP 4 Connect the oscillator to Channel 2 input and measure and record the output of Channel 1 into 600 ohms. Verify that the output is between +20.3 and +22.3 dBm.

STEP 5 Repeat step 4 with the oscillator connected to Channels 3 through 6.

STEP 6 With the oscillator connected to Channel input 2, set the oscillator output to -12.2 dBm.

STEP 7 Push the AVOW Channel 2 pushbutton on the control panel.

STEP 8 Measure and record the output of the Headset J2 on the control panel into 150 ohms. Verify that the output is between +4 and +7 dBm.

STEP 9 Connect the oscillator (without the 600 ohm load) to the headset input and set the oscillator output to 150 mV rms.

STEP 10 Measure and record the output of Channel 2 into 600 ohms. Verify that the output is between +20.3 and +22.3 dBm.

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4.7.14.5 AVOW ringing. (See 3.7.14.5)

STEP 1 Connect AVOW 2 output to AVOW 1 input through the 50 dB 600 ohm attenuator.

STEP 2 Set the AVOW Channel 2 toggle switch to terminate and press the AVOW Channel 2 pushbutton.

STEP 3 Push the RING button on the control panel. Verify that the Channel 1 light flashes.

STEP 4 Push the AVOW Channel 1 button. Verify that the AVOW Channel 1 button is ON steadily.

STEP 5 Push the AVOW Channel 1 button. Verify that the AVOW Channel 1 button is OFF.

4.7.14.6 DCM error rate test. (See 3.7.14.6)

STEP 1 Connect the DSVT Simulator as shown in Figure 41.

STEP 2 Measure and record the DCM error rate over a 30-second period. Verify that no errors are generated.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature.

4.8.1.1 High temperature. (See 3.8.1.1) The unit shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. The temperature for Step 7 shall be 62.7°C (145°F). The operational test specified in 4.7.14 shall be conducted before, during, and after the temperature test as specified in Method 501.1.

4.8.1.2 Low temperature. (See 3.8.1.2) The unit shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature for Step 2 shall be -57°C (-70°F) maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature for Step 4 shall be -32°C (-25°F). The operational test specified in 4.7.14 shall be conducted before, during, and after the temperature test as specified in Method 502.1.

4.8.2 Humidity. (See 3.8.2) The unit shall be subjected to the test of Method 507.1, Procedure II of MIL-STD-810. In Step 6 of the procedure, measurements shall be taken during the last 5 hours of the last cycle. The operational test specified in 4.7.14 shall be conducted before, during, and after the humidity test as specified in Method 507.1.

4.8.3 Altitude. (See 3.8.3) The unit shall be subjected to the test of Method 500.1, Procedure I of MIL-STD-810, except the operating and non-operating altitudes in Step 2 shall be 10,000 feet and 40,000 feet, respectively. In Steps 2 and 3, the rate of altitude change shall not exceed 2500 ±250 feet per minute. The operational test specified in 4.7.14 shall be conducted before, during, and after the altitude test specified in Method 500.1.

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4.8.4 Dust. (See 3.8.4) The unit shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810, except that the air velocity in Steps 1, 2, and 3 shall be 200 ±100 feet per minute. Step 3 may proceed immediately after stabilization in Step 2, if desired. The operational test specified in 4.7.14 shall be conducted before and after the dust exposure. In addition, a visual inspection shall be performed. Deterioration, corrosion, binding, or clogging of moving parts which in any way prevent the unit from meeting operational, service, or maintenance requirements shall provide reason to consider the unit defective.

4.8.5 Salt fog. (See 3.8.5) The unit in its operating configuration shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810. Electronic operation of the equipment shall not be required and failure criteria shall be limited to the following:

a. Corrosion of finishes and metals only. Such corrosion shall be defined as any degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface or exfoliation (see 6.4.2) of metal.

b. Clogging or binding when an attempt is made to operate any mechanical part through its full range of movement.

4.8.6 Fungus. (See 3.8.6) The unit shall be subjected to the fungus test specified in Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (see 6.4.3) on 50 percent or more of the area of the control items after 14 and 28 days. No cleaning of the equipment model is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. After the fungus test, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.4.4) as evidenced by growth colonization (see 6.4.3), which includes branching (see 6.4.1) and sporulation (see 6.4.5) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.4.6) shall not be included in this evaluation.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. (See 3.8.7.1)

a. Operate the test item as specified in 4.7.14 and record pre-performance electrical test measurements. Inspect the test item mechanically.

b. At the conclusion of 4.8.7.1.2 visually inspect the test item for any evidence of mechanical damage and operate the test item as specified in 4.7.14.

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4.8.7.1.1 Mounting techniques. The equipment in its combat transportation mode shall be attached to the vibrator exciter table by its normal mounting means of a rigid fixture capable of transmitting the vibration conditions as specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup. Whenever possible the test load shall be distributed uniformly on the vibrator exciter table in order to reduce to the smallest amount the effect of unbalanced loads. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect test item input control or response. The input control sensing device(s) shall be rigidly attached to the vibration table or to the intermediate structure, if used, at or as near as possible to the attachment point(s) of the test item.

4.8.7.1.2 Cycling. The test item shall be vibrated along each of its three mutually perpendicular axes in accordance with the test levels, frequency range, and time schedules outlined below. The frequency of the applied vibration shall be swept logarithmically over the specified frequency range.

a. Test level: Vibration frequency of 5.0 Hz to 200 Hz at an acceleration level of 1.5G's.

b. Frequency range: 5-200-5 Hz.

c. Time schedule: 5-1/2 hours per axis.

d. Sweep rate: 5-200-5 Hz in 12 minutes.

4.8.7.2 Shock. (See 3.8.7.2)

a. Operate the test item as specified in 4.7.14 and record pre-performance electrical test measurements. Inspect the test item mechanically.

b. At the conclusion of 4.8.7.2.1 and 4.8.7.2.2, visually inspect the test item for any evidence of mechanical damage and operate the test item as specified in 4.7.14.

4.8.7.2.1 Basic design test. The unit shall be subjected to 18 impact shocks of 15G's, consisting of three shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 +1 milliseconds. The shape of the shock pulses shall be a half sinewave. The "G" value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter, and maximum "G" value shall occur at approximately 5-1/2 milliseconds.

4.8.7.2.2 Bench handling test. The unit shall be subjected to the test of Method 516.2, Procedure V, of MIL-STD-810. The operational test specified in 4.7.14 shall be conducted before and after the bench handling test.

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4.9 EMI/EMC tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures ahll be in accordance with MIL-STD-462, Notice 3, and a Government-approved test plan. During susceptibility testinq, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI-EMC testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI/EMC test report.

4.10 Reliability testing. (See 3.10)

4.10.1 PATE reliability test. PATE reliability testing shall be in accordance with Test Plan V, Test Level B of MIL-STD-781.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan IVA, Test Level B of MIL-STD-781.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Duty cycle. Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

4.10.4 Reliability test method. The detailed test procedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor-prepared, government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.14.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or designed limits, it shall be considered a major defect.

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a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

4.13 Visual and mechanical inspection. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (systems safety engineering) which can be determined visually.

4.14 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.15 Markings. The equipment shall be inspected to verify that the markings conform to MIL-M-13231.

4.16 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item, except cords and cables.

4.17 Nuclear tests. (See 3.16) Tests for nuclear survivability shall be performed on the equipment in accordance with Appendix I to TT-B1-2102-0080.

4.18 TEMPEST tests. (See 3.17) TEMPEST tests shall be performed on the equipment in accordance with Appendix II to TT-B1-2102-0080.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or B as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Control, Orderwire C-10717()/TRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Control, Orderwire C-10717()/TRC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods or submethods prescribed in MIL-P-116 as specified herein, insuring compliance with the applicable requirements of that specification.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod IC-1.

5.1.1.4.2 Control, Orderwire C-10717()/TRC. Each item shall be unit packed method III as follows: Cushion each item on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, designed to protect the item and packaging components from physical damage during handling, transit and storage. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514 or any other weather-resistant cushioning material complying with Federal or Military specifications which will provide the protection specified. Place each cushioned item within a close-fitting fiberboard box conforming to PP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure shall be as specified in the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2b). Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents when tested in accordance with methods 5016 and 5017 of FED-STD-101.

5.2.1 Level A. A multiple quantity of Control, Orderwire C-10717()/TRC, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A. When a single unit is shipped to a single destination the unit shall be individually packed as specified above.

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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallex, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.2 Level B. A multiple quantity of Control, Orderwire C-10717()/TRC, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I and II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unix container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of Control, Orderwire C-10717()/TRC, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.3 Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The C-10717( )/TRC (OCU Type 2) is a shelter-mounted unit which provides an equipment assemblage operator access to, supervision of, and bridging facilities for analog and digital voice orderwires from other assemblages via radio or cable. The C-10717( )/TRC also monitors status signals of equipments in an assemblage, formats this status information and forwards the information along with information from interconnected assemblages to the Communications Nodal Control Element (CNCE) via data orderwire. The OCU Type 2 is intended for bottom and top of the hill assemblages.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or level B preservation and packing. (See Section 5).

c. When rough handling test is required.

d. Marking and shipping of samples.

e. Quantity of PATE units required.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

i. Group C failure requirement.

j. MIL-Q-9858 or MIL-I-45208.

k. Length of cables (if any are required).

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

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6.4.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies called "spores".

6.4.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular-shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Group C inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the government pending the decision from the contracting officer on the adequacy of corrective action.

6.5.1 Initial shipments. (See 4.5.3.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.6 Location of operational inspection. (See 4.7.14) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example; TD-1235W/TTC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN:
ARMY-CR

Preparing Activity
ARMY-CR
Project Nr. 5805-A289

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-49245 (CR) Control, Orderwire C-10717()/TRC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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